

Development of Multicellular Organisms

CHAPTER 21

Every multicellular organism, be it animal or plant, starts its life as a single cell—a fertilized egg, or **zygote**. During development, this cell divides repeatedly to produce many different kinds of cells, arranged in a final pattern of spectacular complexity and precision. The goal of developmental cell biology is to understand the cellular and molecular mechanisms that direct this amazing transformation (**Movie 21.1**).

Plants and animals have very different ways of life, and they use different developmental strategies. In this chapter, we focus mainly on animals. Three processes are fundamental to animal development: (1) cell proliferation, which produces many cells from one; (2) cell specialization, or **differentiation**, by which cells take on different characteristics and functions; and (3) morphogenesis, in which cells rearrange themselves to form structured tissues and organs (**Figure 21-1**).

Development of the zygote begins with multiple rounds of cell division, generating a large population of cells that can then be specialized for different functions. At each subsequent stage in its development, a cell is presented with a limited set of options, so that its developmental pathway branches repeatedly, reflecting a large set of sequential choices. Like the decisions we make in our own lives, the choices made by the cell are based on its internal state—which largely reflects its history—and on current influences from other cells, especially its close neighbors. To understand development, we need to know how each choice is controlled and how it depends on previous choices. Beyond that, we need to understand how the choices, once made, influence the cell's chemistry and behavior, and how cell behaviors act synergistically to determine the structure and function of the body.

As cells become specialized, they change not only their biochemistry but also their shape and their attachments to other cells and to the extracellular matrix. They move and rearrange themselves to create the complex architecture of the body, with all its tissues and organs, each structured precisely and defined in size. To understand this process of form generation, or **morphogenesis**, we will need to take account of the mechanical, as well as the biochemical, interactions between the cells.

At first glance, one would no more expect the worm, the flea, the eagle, and the giant squid to be generated by the same developmental mechanisms than one would suppose that the same methods were used to make a shoe and an airplane. Remarkably, however, research has revealed that much of the basic machinery of development is essentially the same in all animals—not just in all vertebrates, but in all invertebrates too. Recognizably similar, evolutionarily related molecules

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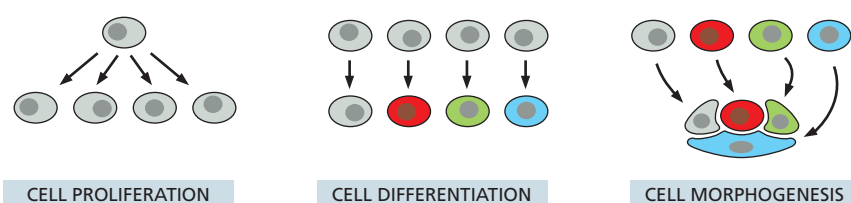


Figure 21-1 The three essential cell processes that allow a multicellular organism to develop.

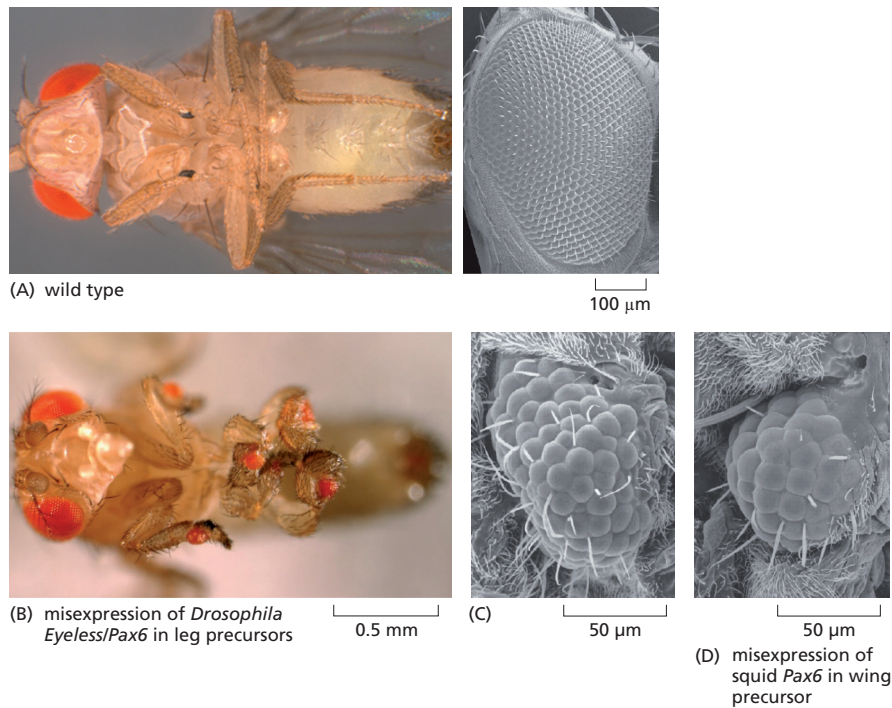


Figure 21-2 Homologous proteins often have conserved functions.

(A) The *Eyeless* protein (also called *Pax6*) controls eye development in *Drosophila*. In a wild-type fly (top row), the *Eyeless/Pax6* gene is expressed in the developing eye and directs formation of the eye structure shown in the scanning electron micrograph at right. Misexpression of *Eyeless/Pax6* in the legs (bottom row) results in the formation of eye tissue on leg precursor (B) (see Figure 7–38) or wing precursor (C). The homologous *Pax6* from a squid, when misexpressed in developing *Drosophila* legs, has the same effect (D). (Fly images courtesy of Katy Ong and Justin Kumar; electron micrographs from S.I. Tomarev et al., *Proc. Natl. Acad. Sci. USA* 94:2421–2426. Copyright 1997 National Academy of Sciences, USA. With permission from National Academy of Sciences.)

define the specialized animal cell types, mark the differences between body regions, and help create the animal body pattern. Homologous proteins often perform the same role in different species and are functionally interchangeable. Thus, a mouse or squid protein produced artificially in a fly, for example, can perform the same function as the fly's own version of that protein (Figure 21-2). Thanks to an underlying unity of mechanisms, developmental biologists have been making great strides toward a coherent understanding of animal development.

We begin this chapter with an overview of some of the basic mechanisms that operate in animal development, focusing on those that are best understood and generate differences between cells. We then discuss, in sequence, how cells in the embryo diversify to form patterns in space, how the timing of developmental events is controlled, how changing cell behaviors drive morphogenesis, and how the size of an animal is regulated.

OVERVIEW OF DEVELOPMENT

Animals live by eating other organisms. Thus, despite their remarkable diversity, animals as different as worms, mollusks, insects, and vertebrates share anatomical features that are fundamental to this way of life. Epidermal cells form a protective outer layer; gut cells absorb nutrients from ingested food; muscle cells allow movement toward food sources; and neurons and sensory cells control behavior. These various cell types are organized into tissues and organs, forming a sheet of skin covering the exterior, a mouth for feeding, and an internal gut tube for digestion—with muscles, nerves, and other tissues arranged in the space between the skin and the gut. Many animals have clearly defined axes—an anteroposterior axis, with mouth and brain anterior and anus posterior; a dorsoventral axis, with back dorsal and belly ventral; and a left–right axis. In this section, we discuss some fundamental mechanisms underlying how the basic animal body plan is established and how this diversity of cell types is created.

Conserved Mechanisms Establish the Core Tissues of Animals

The shared anatomical features of animals develop through conserved mechanisms. After fertilization, the zygote usually divides rapidly, or **cleaves**, to form many smaller cells called **blastomeres**. During this cleavage phase, the embryo, which cannot yet feed, does not grow. This step of development is initially driven

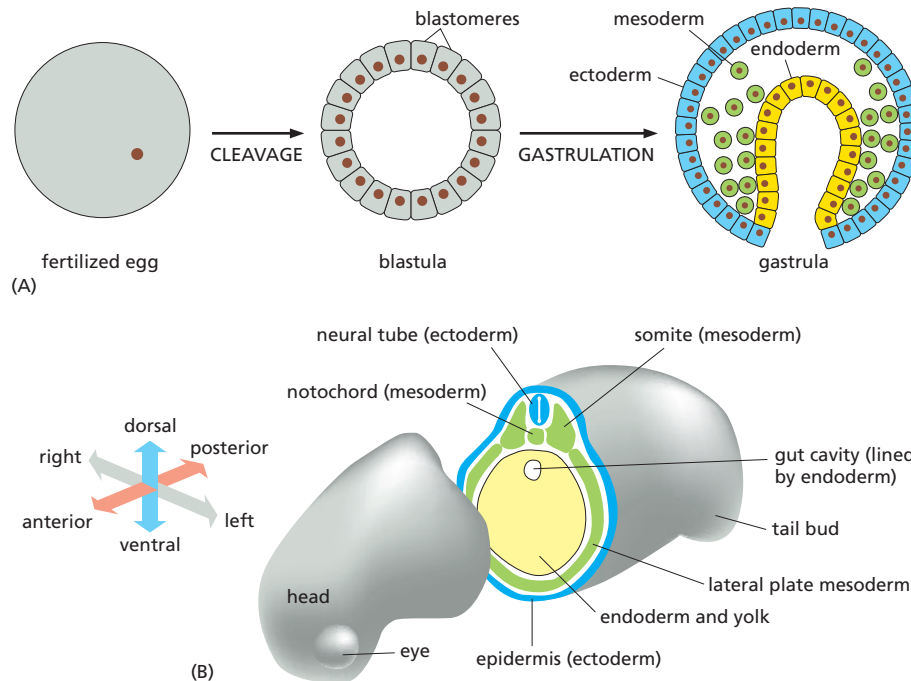


Figure 21-3 The early stages of development, as exemplified by a frog. (A) A fertilized egg divides to produce many blastomere cells that form an epithelial sheet often surrounding a cavity. During gastrulation, some of the cells tuck into the interior to form the mesoderm (green) and endoderm (yellow). Ectodermal cells (blue) remain on the outside. (B) A cross section through the trunk of an amphibian embryo shows the basic animal body plan, with a sheet of ectoderm on the outside, a tube of endoderm on the inside, and mesoderm sandwiched between them. The endoderm forms the epithelial lining of the gut, from the mouth to the anus. It gives rise not only to the pharynx, esophagus, stomach, and intestines, but also to many associated structures. The salivary glands, liver, pancreas, trachea, and lungs, for example, all develop from the wall of the digestive tract and grow to become systems of branching tubes that open into the gut or pharynx. The endoderm forms only the epithelial components of these structures—the lining of the gut and the secretory cells of the pancreas, for example. The supporting muscular and fibrous elements arise from the mesoderm.

and controlled entirely by the material deposited in the egg by the mother. The embryonic genome remains largely inactive until a point is reached when maternal mRNAs and proteins are used up or abruptly degraded. The embryo's genome is then activated—the maternal-zygotic transition that we will discuss later—and the cells cohere to form a **blastula**, typically a solid or a hollow fluid-filled ball of cells.

Even in these early stages of embryogenesis, genetic programs are under way that give rise to the basic tissue types. Complex cell rearrangements called **gastrulation** (from the Greek *gaster*, meaning “belly”) soon transform the blastula into a multilayered structure containing a rudimentary internal gut (**Figure 21-3**). Some cells of the blastula remain external, constituting the **ectoderm**, which will give rise to the epidermis and the nervous system; other cells invaginate, forming the **endoderm**, which will give rise to the gut tube and its appendages, such as lung, pancreas, and liver. In most animals, another group of cells moves into the space between ectoderm and endoderm and forms the **mesoderm**, which will give rise to muscles, connective tissues, blood, kidneys, and various other components. Further cell movements and accompanying cell differentiation create and refine the embryo's architecture.

The ectoderm, mesoderm, and endoderm formed during gastrulation constitute the three **germ layers** of the early embryo. This initial subdivision is the first step in defining the multitude of cell fates that will emerge in development. Many later developmental transformations will produce the elaborately structured organs. But the basic body plan and axes set up in miniature during gastrulation are preserved into adult life, when the organism may be billions of times larger (**Movie 21.2**).

The Developmental Potential of Cells Becomes Progressively Restricted

Concomitant with the refinement of the body plan, the individual cells within a *lineage*—that is, the progeny of a particular proliferating mother cell—become more and more restricted in their developmental potential. During the blastula stages, cells are often **totipotent** or **pluripotent**—they have the potential to give rise to all or almost all of the cell types of the adult body. The pluripotency is lost as gastrulation proceeds: a cell located in the endodermal germ layer, for

The mesoderm gives rise to the connective tissues—at first, to the loose mesh of cells in the embryo known as *mesenchyme*, and ultimately to cartilage, bone, and fibrous tissue, including the dermis (the inner layer of the skin). The mesoderm also forms the muscles, the entire vascular system—including the heart, blood vessels, and blood cells—and the tubules, ducts, and supporting tissues of the kidneys and gonads. The notochord forms from the mesoderm and serves as the core of the future backbone and the source of signals that coordinate the development of surrounding tissues.

The ectoderm will form the epidermis (the outer, epithelial layer of the skin) and epidermal appendages such as hair, sweat glands, and mammary glands. It will also give rise to the whole of the nervous system, central and peripheral, including not only neurons and glia but also the sensory cells of the nose, the ear, the eye, and other sense organs. (B, after T. Mohun et al., *Cell* 22:9–15, 1980.)

example, can give rise to the cell types that will line the gut or form gut-derived organs such as the liver or pancreas, but it no longer has the potential to form mesoderm-derived structures such as skeleton, heart, or kidney. Such a cell is said to be *determined* for an endodermal fate. Thus, **cell determination** starts early and progressively narrows the options as the cell steps through a programmed series of intermediate states—guided at each step by its genome, its history, and its interactions with neighbors. The process reaches its limit when a cell undergoes **terminal differentiation** to form one of the highly specialized cell types of the adult body (Figure 21–4). Some cell types in the adult, which maintain the ability to divide, also retain a degree of pluripotency, albeit with a generally narrow range of options. These adult stem cells are discussed in Chapter 22.

Cell Memory Underlies Cell Decision-Making

Underlying the richness and astonishingly complex outcomes of development is **cell memory** (see p. 435). Both the genes a cell expresses and the way it behaves depend on the cell's past, as well as on its present circumstances. The cells of our body—the muscle cells, the neurons, the skin cells, the gut cells, and so on—maintain their specialized characters largely because they retain a record of the extracellular signals their ancestors received during development, rather than because they continually receive such instructions from their surroundings. Despite their radically different phenotypes, almost all cells retain the same complete genome that was present in the zygote. Their differences arise instead from differential gene expression, which can lead to the stable inheritance of a particular cellular program. We have discussed the molecular mechanisms of gene regulation, cell memory, cell division, cell signaling, and cell movement in previous chapters. In this chapter, we shall see how these basic processes are collectively deployed as the developing animal self-assembles.

Several Model Organisms Have Been Crucial for Understanding Development

The anatomical features that animals share have undergone many extreme modifications in the course of evolution. As a result, the differences between species are usually more striking to our human eye than the similarities. But at the level of the underlying molecular mechanisms and the particular macromolecules involved, the reverse is true: the similarities among all animals are profound and extensive. Through more than half a billion years of evolutionary divergence, all animals have retained unmistakably similar sets of genes and proteins that are responsible for generating their body plans and for forming their specialized cells and organs.

This astonishing degree of evolutionary conservation was discovered not by broad surveys of animal diversity, but through intensive study of a small number of experimentally convenient species—the model organisms discussed in Chapter 1. For animal developmental biology, the most important have been the fly *Drosophila melanogaster*, the frog *Xenopus laevis*, the roundworm *Caenorhabditis elegans*, the mouse *Mus musculus*, and the zebrafish *Danio rerio*. In discussing the mechanisms of development, we shall draw our examples mainly from these few species, keeping in mind that they represent only a portion of the tree of animal life.

Regulatory DNA Seems Largely Responsible for the Differences Between Animal Species

Although many developmental mechanisms are conserved, they nevertheless produce animals that can be quite different. These differences arise primarily from variation in the activity of key development-controlling genes. As discussed in Chapter 7, each gene in a multicellular organism is associated with many thousands of nucleotides of noncoding DNA that contains regulatory elements. These

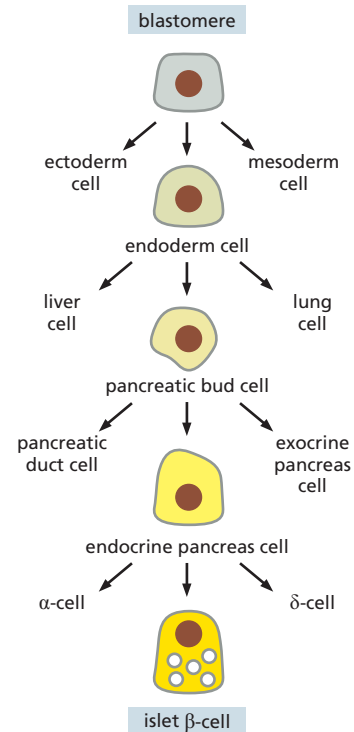


Figure 21–4 The lineage from blastomere to a terminally differentiated cell type.

As development proceeds, cells become more and more specialized. Blastomeres have the potential to give rise to most or all cell types. Under the influence of signaling molecules and gene regulatory factors, cells acquire more restricted fates until they differentiate into highly specialized cell types, such as the pancreatic β cells that secrete the hormone insulin.

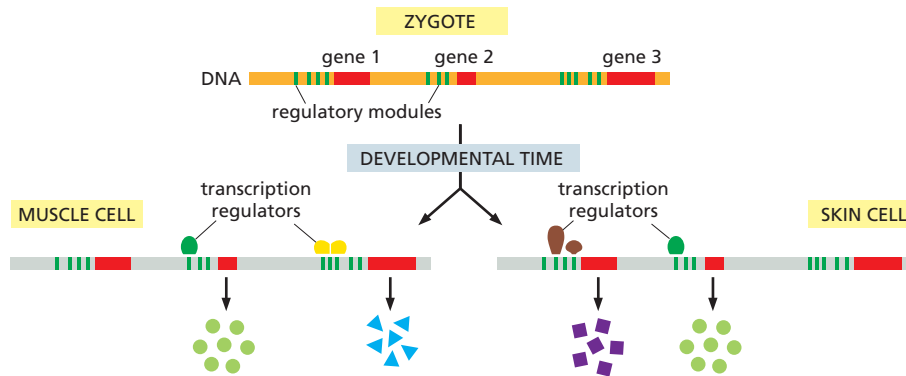


Figure 21-5 Regulatory DNA defines the gene expression patterns in development. The genome is the same in a muscle cell as in a skin cell, but different genes are active because these cells express different transcription regulators that bind to gene regulatory elements. For example, transcription regulators in skin cells recognize a regulatory element in gene 1, leading to its activation, whereas a different set of regulators is present in muscle cells, binding to and activating gene 3. Transcriptional regulators that activate the expression of gene 2 are present in both cell types.

regulatory elements determine when, where, and how strongly the gene product is expressed, according to the transcription regulators and chromatin structures that are present in the particular cell (Figure 21-5). Consequently, a change in sequence of the regulatory DNA that arises in evolution, even without any change in the coding DNA, can alter the logic of the gene regulatory network and change the outcome of development.

As discussed in Chapter 4, when we compare the genomes of different animal species, we find that evolution has altered the coding and regulatory DNA to different extents. The coding DNA can be quite conserved, but the noncoding regulatory DNA is usually much less so. It seems that changes in regulatory DNA are largely responsible for the dramatic differences between one class of animals and another (see p. 239). We can view the protein products of the conserved coding sequences as a kit of common molecular parts and the regulatory DNA as instructions for assembly: with different instructions, the same kit of parts can be used to make a whole variety of different body structures. We will return to this important concept later.

Small Numbers of Conserved Cell–Cell Signaling Pathways Coordinate Spatial Patterning

Spatial patterning of a developing animal requires that cells become different according to their positions in the embryo, which means that cells must respond to extracellular signals produced by other cells, especially their neighbors. In the most common mode of spatial patterning, a group of cells starts out with the same developmental potential, and a signal from cells outside the group then induces one or more members of the group to change their character. This process is called *inductive signaling*. Generally, the inductive signal is limited in time and space so that only a subset of the cells capable of responding—the cells close to the source of the signal—take on the induced character (Figure 21-6). Some inductive signals depend on cell–cell contact; others act over a longer range and are mediated by molecules that diffuse through the extracellular medium or are transported in the bloodstream (see Figure 15-2).

Most of the known inductive events in animal development are governed by a small number of highly conserved signaling pathways, including transforming growth factor- β (TGF β), Wnt, Hedgehog, Notch, and receptor tyrosine kinase (RTK) pathways (discussed in Chapter 15). The discovery of the limited vocabulary that developing cells use for intercellular communication has emerged as one of the great simplifying features of developmental biology.

Through Combinatorial Control and Cell Memory, Simple Signals Can Generate Complex Patterns

How can a small number of signaling pathways generate the huge diversity of cells and patterns? Several kinds of mechanisms are responsible. First, the effect of activating a signaling pathway depends on the previous experiences of the

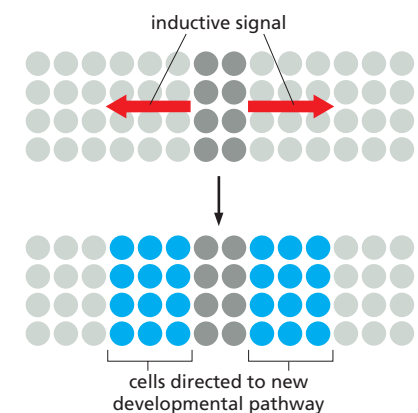


Figure 21-6 Inductive signaling. Cells (gray) expressing an extracellular signaling molecule direct a new cell fate in nearby neighbors (blue).

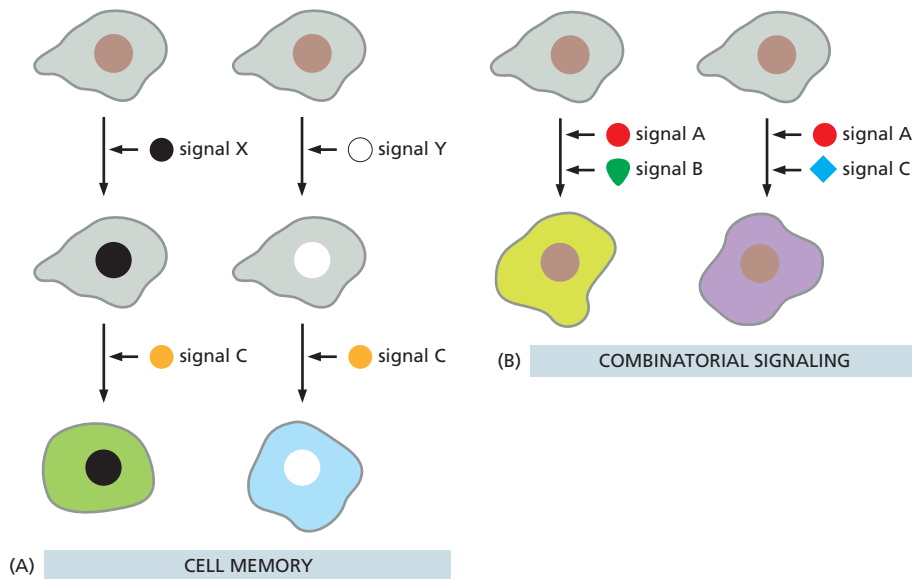


Figure 21-7 Two mechanisms for generating different responses to the same inductive signal. (A) Through cell memory, previous signals (or other events) can leave a lasting trace that alters the response to the current signal (see Figure 7-56). The memory trace is represented here in the coloring of the cell nucleus. (B) In combinatorial signaling, the effect of a signal depends on the presence of other signals received at the same time.

responding cell: past influences leave a lasting mark, registered in the state of the cell's chromatin and the selection of transcription regulators and RNA molecules that the cell contains. This cell memory enables cells with different histories to respond to the same signals differently (Figure 21-7A). Second, the response of a cell to a given signal depends on the other signals that the cell is receiving concurrently (Figure 21-7B). As a result, different combinations of signals can generate a large variety of different responses.

Alongside these two mechanisms, some components of a signaling pathway, such as ligands or receptors, are encoded by genes that have undergone duplication followed by functional divergence during evolution. These closely homologous genes can then be expressed in distinct cell types, where they can direct different signaling outcomes. Notch signaling, for example, may be mediated by Notch1 in one tissue, but by Notch4 in another, with each homolog inducing transcription of different target genes. Thus, the same few signaling pathways can be used repeatedly at different times and places with different outcomes, so as to generate patterns of unlimited complexity.

Morphogens Are Diffusible Inductive Signals That Exert Graded Effects

Signal molecules often govern simple yes-no choices—one outcome when their concentration is high, another when it is low or absent. In other cases, however, the responses are more finely graded: a high concentration of a signal molecule may, for example, direct cells into one developmental pathway, an intermediate concentration into another, and a low concentration into yet another.

One common way to generate such different concentrations of a signal molecule is for the molecule to diffuse out from a localized signaling source, creating a concentration gradient. Cells at different distances from the source are driven to behave in a variety of different ways, according to the signal concentration that they experience (Figure 21-8). A signal molecule that imposes a pattern on a whole field of cells in this way is called a **morphogen**. In the simplest case, a specialized group of cells produces a morphogen at a steady rate, and the morphogen is then degraded as it diffuses away from this source. The speed of diffusion and

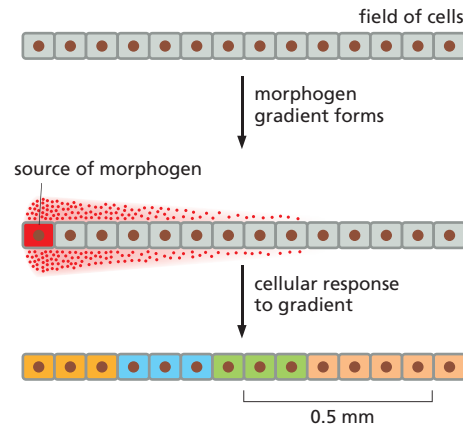


Figure 21–8 Gradient formation and interpretation. A gradient forms by the localized production of an inducer—a morphogen—that diffuses away from its source. Different concentrations of morphogen (or different durations of exposure) induce different gene expression patterns and cell fates in responding cells. Diffusive transport can generate steep gradients only over short distances, and morphogens generally act over distances of 1 mm or less.

the half-life of the morphogen will together determine the range and steepness of its resulting gradient (**Figure 21–9**).

Nearby cells then interpret their distance from the morphogen source according to how much signal they are exposed to, detected by binding to cell-surface receptors. After signal transduction from these receptors, target genes will become transcribed only if the morphogen concentration exceeds a specific threshold; low concentrations that fall beneath this threshold will not activate the target. In this manner, a graded signal can be converted into multiple discrete, on-or-off changes in gene activity. Thus, a single secreted protein, in combination with the physical properties of diffusion and the ability of cells to interpret the information, can generate several distinct fates within a field of cells.

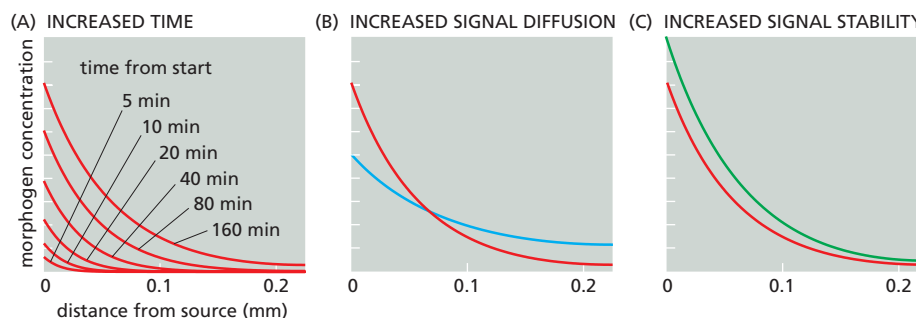
This simple mechanism can be modified in various ways. For example, cell-surface receptors may trap the diffusing morphogen and cause it to be endocytosed and degraded, shortening its effective half-life. Alternatively, the morphogen may bind to molecules in the extracellular matrix such as heparan sulfate proteoglycan (discussed in Chapter 19), thereby greatly reducing its diffusion rate.

Lateral Inhibition Can Generate Patterns of Different Cell Types

Morphogen gradients, and other types of inductive signals, exploit an existing asymmetry in the embryo to create further asymmetries and differences between cells: already, at the outset, some cells are specialized to produce the morphogen and thereby impose a pattern on another class of cells that are sensitive to it. But what if there is no clear initial asymmetry? Can a regular pattern arise spontaneously within a set of cells that are initially all alike?

The answer is yes. The fundamental principle underlying such *de novo* pattern formation is positive feedback: cells can exchange signals in such a way that any small initial discrepancy between cells at different sites becomes self-amplifying, driving the cells toward different fates. This is most clearly illustrated in the phenomenon of *lateral inhibition*, a form of cell–cell interaction that forces close neighbors to become different and thereby generates fine-grained patterns of different cell types.

Figure 21–9 Setting up a signal gradient by diffusion. Each graph shows the concentration of a morphogen signal molecule that is produced at a steady rate at the origin. In all cases, the molecule undergoes degradation as it diffuses away from the source, and the graphs are calculated on the assumption that diffusion is occurring along two axes in space (for example, radially from a source in an epithelial sheet). (A) The pattern of the morphogen assuming that production starts at time 0, the molecule has a half-life of 170 minutes, and that it diffuses with an effective diffusion constant of $D = 1 \mu\text{m}^2 \text{sec}^{-1}$, typical of a small protein molecule in extracellular tissues. *Red lines* show six successive stages in the buildup of the morphogen, which falls off exponentially with distance from the source. Panels B and C show how simple changes in the properties of the system alter the gradient at the 160-minute time point. (B) A threefold increase in the diffusion constant of the morphogen extends its range (*blue line*) but lowers its concentration next to the source. (C) A threefold increase in morphogen half-life (*green line*) increases its concentration throughout the tissue. Effects of the morphogen will depend not just on its concentration at some critical moment, but also on how each target cell integrates its response over time. (Courtesy of Patrick Müller.)



Consider a pair of adjacent cells that start off in a similar state. Each of these cells can both produce and respond to a certain signal molecule X, with the added rule that the stronger the signal a cell receives, the weaker the signal it generates itself (Figure 21-10). If one cell produces more X, the other is thus forced to produce less, for instance by reducing transcription of the X-encoding gene. This gives rise to a positive feedback loop that tends to amplify any initial difference between the two adjacent cells. Such a difference may arise from spontaneous random fluctuations, or “noise”—an inevitable feature of the genetic control circuitry in cells (discussed in Chapter 7). In either case, lateral inhibition means that if cell 1 makes a little more of X, it will thereby cause cell 2 to make less; and because cell 2 makes less X, it delivers less inhibition to cell 1 and so allows the production of X in cell 1 to rise higher still; and so on, until a stable state is reached where cell 1 produces a lot of X and cell 2 produces very little. The result is that the two cells are driven along different pathways of differentiation.

In almost all tissues, a balanced mixture and distribution of different cell types are required. Lateral inhibition provides a common way to generate the mixture. As we shall see, lateral inhibition is very often mediated by exchange of signals at cell-cell contacts via the Notch signaling pathway, driving cell diversification by enabling individual cells that express one set of genes to direct their immediate neighbors to express a different set, in exactly the way we have described (see also Figure 15-60).

Asymmetric Cell Division Can Also Generate Diversity

Cell diversification does not always depend on extracellular signals: in some cases, cells are born different as a result of an asymmetric cell division, in which some important molecule or molecules present in the mother cell are distributed unequally between the two daughters. This asymmetric inheritance during mitosis ensures that the two daughter cells develop differently (Figure 21-11). The mechanism here is intrinsic to the dividing cell, although daughter cells can also become asymmetric after division as a result of the extrinsic, inductive signals discussed earlier. Intrinsically asymmetric division is a common feature of early development. The fertilized egg may already possess an internal pattern, and cleavage of this large cell segregates different fate determinants into separate blastomeres. We shall see later that asymmetric division also plays a part in later developmental processes, as well as in stem cells (see Chapter 22).

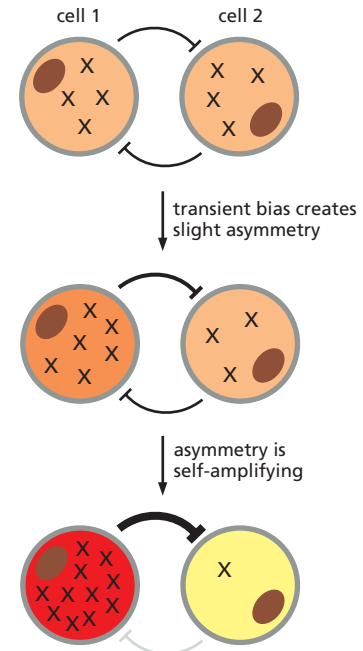


Figure 21-10 Genesis of asymmetry through lateral inhibition and positive feedback. In this example, two cells interact, each producing a substance X that acts on the other cell to inhibit its production of X, an effect known as lateral inhibition. An increase of X in one of the cells leads to a positive feedback that tends to increase X in that cell still further, while decreasing X in its neighbor. This can create an accelerating instability, making the two cells become radically different. Ultimately, the system comes to rest in one or the other of two opposite stable states. The final choice of state represents a form of memory: the small influence that initially directed the choice is no longer required to maintain it.

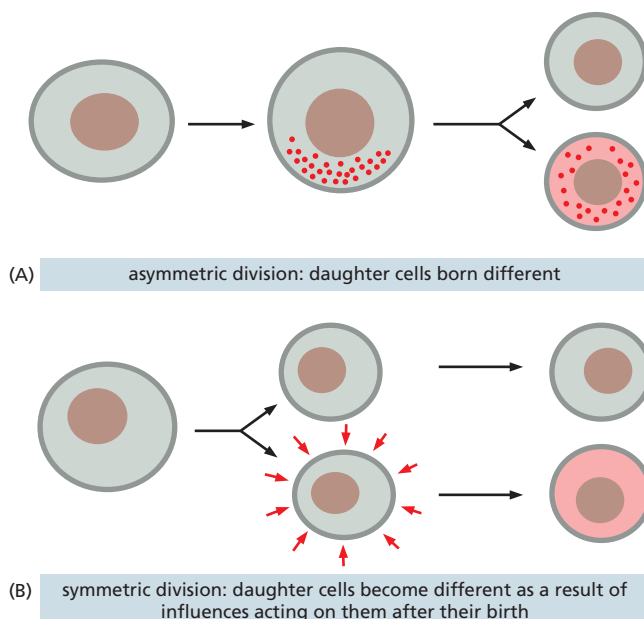


Figure 21-11 Two ways of making daughter cells different. Daughters can assume different fates either through (A) an intrinsically asymmetric division in which there is differential inheritance of cytoplasmic molecules or through (B) a symmetric division followed by signaling to only one daughter.

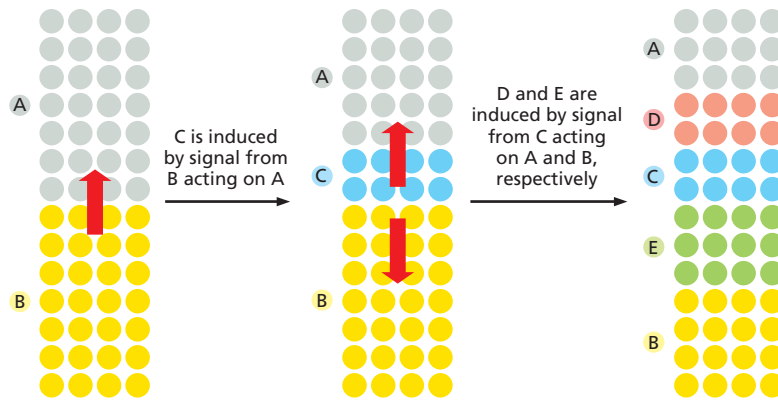


Figure 21-12 Patterning by sequential induction. A series of inductive interactions can generate many types of cells, starting from only a few.

Initial Patterns Are Established in Small Fields of Cells and Refined by Sequential Induction as the Embryo Grows

The signals that organize the spatial pattern of cells in an embryo generally act over short distances and govern relatively simple choices. A morphogen, for example, typically acts over a distance of less than 1 mm—an effective range for diffusion—and directs choices between several developmental options for the cells on which it acts. Yet the organs that eventually develop are much larger and more complex than this.

As the organ grows via cell proliferation, the refinement of its initial pattern is explained by a series of successive interactions that add increasing levels of detail to an initially simple sketch. For example, as soon as two types of cells are present in a developing tissue, one of them can produce a signal that induces a subset of neighboring cells to specialize in a third way. The third cell type can in turn signal back to the other two cell types nearby, generating a fourth and a fifth cell type, and so on (**Figure 21-12**).

This strategy for generating a progressively more complicated pattern is called **sequential induction**. It is chiefly through sequential inductions that the body plan of a developing animal, after being first roughed out in miniature, becomes elaborated with finer and finer details as development proceeds.

Developmental Biology Provides Insights into Disease and Tissue Maintenance

The rapid progress in understanding animal development has been one of the great success stories in biology, and it has important practical implications. Some 2–5% of all human babies are born with anatomical abnormalities, such as heart malformations, truncated limbs, cleft palate, or spina bifida. Advances in developmental biology help us understand how these defects arise, even if we cannot yet prevent or cure most of them.

Less obvious, but even more important from a practical point of view, is that developmental biology also provides insights into the workings of cells and tissues in the adult body. Developmental processes do not halt at birth; they continue throughout life, as tissues are maintained and repaired. The fundamental mechanisms of cell growth and division, cell–cell signaling, cell memory, cell adhesion, and cell movement are all involved in adult tissue maintenance and repair—just as they are in embryo development. These are also the main mechanisms perturbed in tumor cells, as we saw in Chapter 20.

Embryos are simpler than adults, and they allow us to analyze such basic processes more easily. Studies of the early *Drosophila* embryo, for example, were crucial to the discovery of several conserved signaling pathways, including the Wnt, Hedgehog, and Notch pathways. They also provided the key to understanding the central role of these pathways in the maintenance of normal adult human tissues and laid the foundation for manipulating the pathways in the cause of regenerative medicine. Finally, they have identified both targets and rational mechanisms for therapies to fight cancer and other diseases.

In Chapter 22, we shall consider how these and other core developmental mechanisms operate in the normal adult body, especially in tissues that are continually renewed by means of stem cells—including the gut, skin, and the hematopoietic system. But now, we must look more closely at the way in which an early embryo generates its spatial pattern of specialized cells, beginning with the transformations that create the adult body plan.

Summary

Animal development is an amazing self-assembly process, in which the initially similar cells of the embryo become different from one another and organize themselves into increasingly complex structures. The process begins with a single large cell, the fertilized egg, which cleaves to produce a blastula that undergoes gastrulation to generate the three germ layers of the embryo—ectoderm, mesoderm, and endoderm. As development continues, the cells become more and more narrowly specialized according to their locations and their interactions with one another, eventually forming one of the differentiated cell types of the adult body.

Differences between developing cells arise in various ways and have to be properly coordinated in space. In one common strategy, initially similar cells within a group become different by exposure to different levels of an inductive signal or morphogen emanating from a source outside the group. Neighboring cells can also become different by lateral inhibition, in which a cell signals to its neighbors not to follow the same fate. These cell–cell interactions are mediated by a small number of highly conserved signaling pathways, which are used repeatedly in different organisms and at different times during development. Not all cell diversification arises by cell–cell interactions, however: daughter cells can be born different as a result of asymmetric cell division.

Regulators of transcription and chromatin structure bind to regulatory DNA and determine the fate of each cell. Differences of body plan seem to arise to a large extent from differences in the regulatory DNA associated with each gene. This DNA has a central role in defining the sequential program of development, calling genes into action at specific times and places according to the pattern of gene expression that was present in each cell at the previous developmental stage.

Development has been most thoroughly studied in a handful of model organisms. But most of the genes and mechanisms thereby identified are used in all animals and repeatedly at different stages of development. Thus, insights from worms, flies, fish, frogs, and mice deeply inform our understanding of embryology and adult tissue maintenance in humans, as well as how aberrations in these processes cause birth defects and cancer.

MECHANISMS OF PATTERN FORMATION

A developing multicellular organism has to create different cell fates in fields of cells that were nearly indistinguishable and accomplish this task in a spatially ordered manner so that functional tissues are formed. Some of the early microscopists imagined the entire shape and structure of the human body to be already present in the sperm as a “homunculus,” a miniature human; after fertilization, the homunculus would simply grow and generate a full-sized person. We now know that this view is incorrect, and that development is a progression from simple to complex, through a gradual refinement of an animal’s anatomy. To see how the whole sequence of events of spatial patterning and cell determination is set in motion, we must return to the egg and the early embryo.

Different Animals Use Different Mechanisms to Establish Their Primary Axes of Polarization

Surprisingly, the earliest steps of animal development are among the most variable, even within a phylum. A frog, a chicken, and a mammal, for example, even though they develop in similar ways later, make eggs that differ radically in size and structure, and they begin their development with different sequences of cell

divisions and cell specializations. Likewise, there is great variation in the time and manner in which the primary axes of the body become marked out. However, this *polarization* of the embryo usually becomes discernible very early, before gastrulation begins—it is the first step of spatial patterning.

Two axes have to be established. The *anteroposterior* (A-P) axis specifies the locations of future head and tail. The *dorsoventral* (D-V) axis specifies the future back and belly. For species that show bilateral asymmetry, creation of an additional *left-right* (L-R) axis is also important. Finally, the eggs of many animals have an *animal-vegetal* (A-V) axis that is not evident in the adult, but defines which parts are to become internal (through the movements of gastrulation) and which are to remain external. (The bizarre name dates from a century ago and has nothing to do with vegetables.)

At one extreme within animal diversity, the egg is spherically symmetrical, and the axes only become defined during embryogenesis. The mouse is one example, with little obvious sign of polarity in the egg. Correspondingly, the blastomeres produced by the first few cell divisions all seem to be alike and are remarkably adaptable. If the early mouse embryo is split in two, a pair of identical twins can be produced—two complete, normal individuals from a single cell. Similarly, if one of the cells in a two-cell mouse embryo is destroyed by pricking it with a needle and the resulting “half-embryo” is placed in the uterus of a foster mother to develop, in many cases a perfectly normal mouse will emerge.

At the opposite extreme, an egg is built with an asymmetric structure that itself defines the future axes of the body. This is the case for most species, including insects such as *Drosophila*, as we shall see shortly. Other organisms lie somewhere in between. The egg of the frog *Xenopus*, for example, has a clearly defined A-V axis before fertilization: the pronucleus near the top defines the animal pole, while the mass of yolk (the embryo’s food supply, destined to be incorporated in the gut) toward the bottom defines the vegetal pole. Several types of mRNA molecules are already localized in the vegetal cytoplasm of the egg, where they produce their protein products. After fertilization, these mRNAs and proteins act in and on the cells in the lower and middle part of the embryo, giving the cells their specialized characters.

Formation of the D-V axis of the *Xenopus* embryo is triggered by fertilization. At the site where the sperm enters, which is only permitted in the region of the animal pole, the sperm centrosome nucleates a microtubule aster. This reorganization of the microtubule cytoskeleton causes the outer cortex of the egg to rotate relative to the central core of the egg cytoplasm, so that the animal pole of the cortex becomes slightly shifted to one side (**Figure 21-13**). Cortical rotation

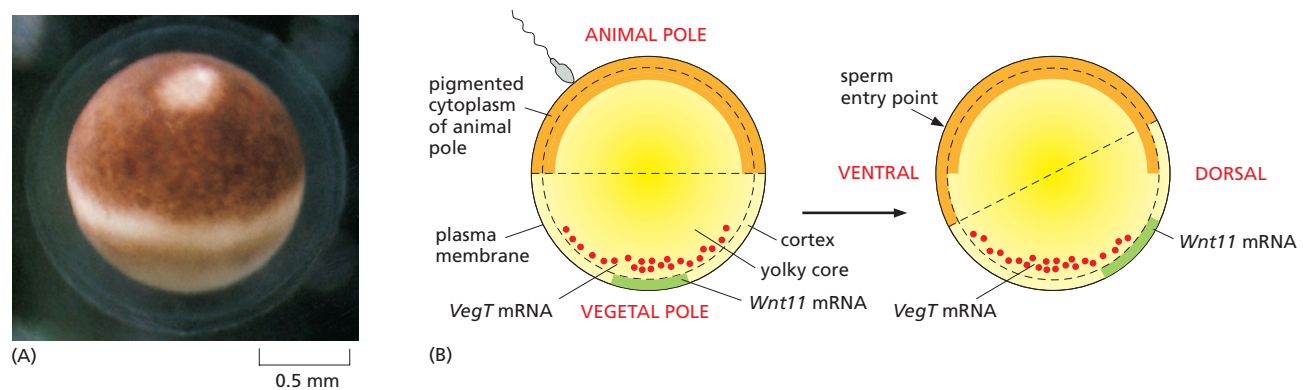


Figure 21-13 The frog egg and its asymmetries. (A) Side view of a *Xenopus* egg photographed just before fertilization. (B) The asymmetric distribution of molecules inside the egg along the animal-vegetal axis, and how fertilization activates dorsoventral asymmetry. Vegetally localized *VegT* in the unfertilized egg defines the vegetal source of signals that will induce endoderm and mesoderm. Sperm entry initiates a reorganization of the microtubule cytoskeleton that triggers a rotation of the egg cortex (a layer a few micrometers deep) through about 30° relative to the core of the egg. Cortical rotation relocates Wnt signaling components including *Wnt11* mRNA to the future dorsal side, which sets up the dorsoventral axis of the embryo. (A, courtesy of Tony Mills.)

relocalizes signaling molecules and initiates a cascade of events that will organize the dorsoventral axis of the body. (The A-P axis of the embryo will only become clear later, in the process of gastrulation.) This general process, by which a cue creates a new axis in the embryo, is known as *symmetry breaking*.

Although different animal species use a variety of different mechanisms to specify their overall organization, the outcome has been relatively well conserved in evolution: head is distinguished from tail, back from belly, and gut from skin. It seems that it does not much matter what tricks the embryo uses, it always manages to break the initial symmetry and set up its basic body plan.

Studies in *Drosophila* Have Revealed Many Genetic Control Mechanisms Underlying Development

It is the fly *Drosophila*, more than any other organism, that has provided the key to our present understanding of how genes govern early development. Decades of study culminated in a large-scale genetic screen, focusing on the early embryo and searching for mutations that disrupt its pattern. This revealed a group of key developmental genes that act in a relatively small set of regulatory pathways. The discovery of these genes and the subsequent analysis of their functions is a famous *tour de force* that has had a revolutionary impact on all of biology, earning its discoverers a Nobel Prize. Some parts of the machinery revealed in this way are conserved between flies and vertebrates, some parts not. But the logic of the experimental approach and the general strategies of genetic control that it revealed have transformed our understanding of multicellular development in general.

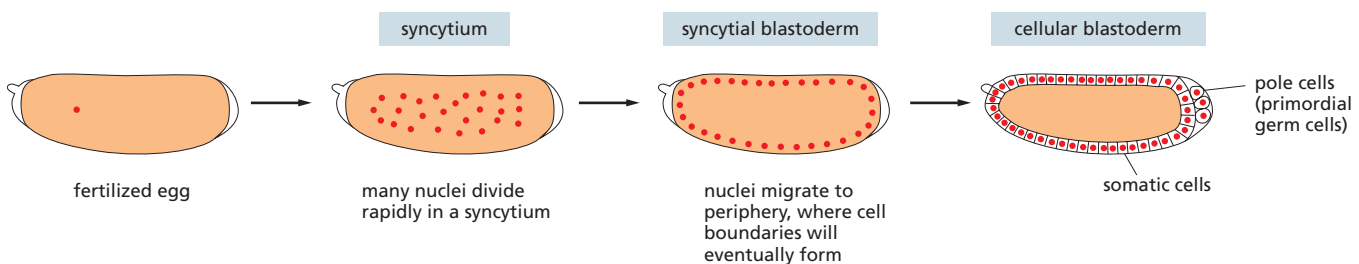
To understand how the early developmental machinery operates in *Drosophila*, it is important to note a peculiarity of fly development. Like the eggs of other insects, but unlike most vertebrates, the *Drosophila* egg—shaped like a cucumber—begins its development with an extraordinarily rapid series of nuclear divisions without cell division, producing multiple nuclei in a common cytoplasm—a **syncytium**. The nuclei then migrate to the cell cortex, forming a structure called the *syncytial blastoderm*. After about 6000 nuclei have been produced, the plasma membrane folds inward between them and partitions them into separate cells, converting the syncytial blastoderm into the *cellular blastoderm* (Figure 21–14).

We shall see that the initial patterning of the *Drosophila* embryo depends on molecules that diffuse through the cytoplasm at the syncytial stage and exert their actions on genes in the rapidly dividing nuclei, before the partitioning of the egg into separate cells. Here, there is no need for the usual forms of cell–cell signaling; neighboring regions of the syncytial blastoderm can communicate by means of transcription regulators that move through the cytoplasm of the giant multinuclear cell.

Gene Products Deposited in the Egg Organize the Axes of the Early *Drosophila* Embryo

As in most insects, the main axes of the future body of *Drosophila* are defined before fertilization by a complex exchange of signals between the developing egg, or *oocyte*, and the mother's somatic cells that surround it in the ovary.

Figure 21–14 Development of the *Drosophila* egg from fertilization to the cellular blastoderm stage.



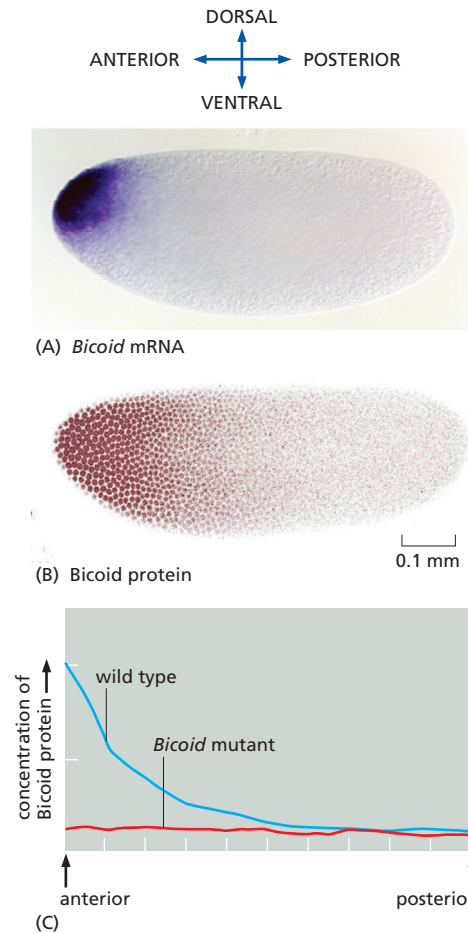


Figure 21–15 The Bicoid protein gradient. (A) *Bicoid* mRNA is deposited at the anterior pole during oogenesis. (B) Local translation followed by diffusion generates the Bicoid protein gradient. (C) Absence of the Bicoid protein gradient in embryos from *Bicoid* homozygous mutant mothers. (A and B, courtesy of Stephen Small.)

These signals can be described as **maternal effect**, because it is the genome of the mother rather than the zygote that produces them. Before fertilization, the anteroposterior and dorsoventral axes of the future embryo become defined by systems of **egg-polarity genes** that create landmarks—either mRNA or protein—in the oocyte. After fertilization, each landmark serves as a beacon, providing a signal that organizes the developmental process in its neighborhood.

The nature of the egg-polarity genes emerged from studies of mutants in which the patterning of the embryo was altered. Some of these mutations gave embryos with disrupted polarity; for example, one caused tail-end structures at both ends of the body, with no head-end structures. This particular mutation allowed the identification of the landmark that organizes the anterior end of the embryo, called *Bicoid*. A deposit of *Bicoid* mRNA molecules is localized, before fertilization, at the anterior end of the egg. Upon fertilization, the mRNA is translated to produce Bicoid protein. This protein is an intracellular morphogen and transcription regulator that diffuses away from its source to form a concentration gradient within the syncytial cytoplasm, with its maximum at the head end of the embryo (**Figure 21–15**). The different concentrations of Bicoid along the A-P axis help determine different cell fates by directly regulating the transcription of genes in the nuclei of the syncytial blastoderm (discussed in Chapter 7).

There are three other egg-polarity gene systems that pattern the syncytial nuclei; two act along the A-P axis and one acts along the D-V axis. Together with the *Bicoid* group of genes, and acting in a broadly similar way, their gene products mark out three fundamental partitions of body regions—head versus rear,

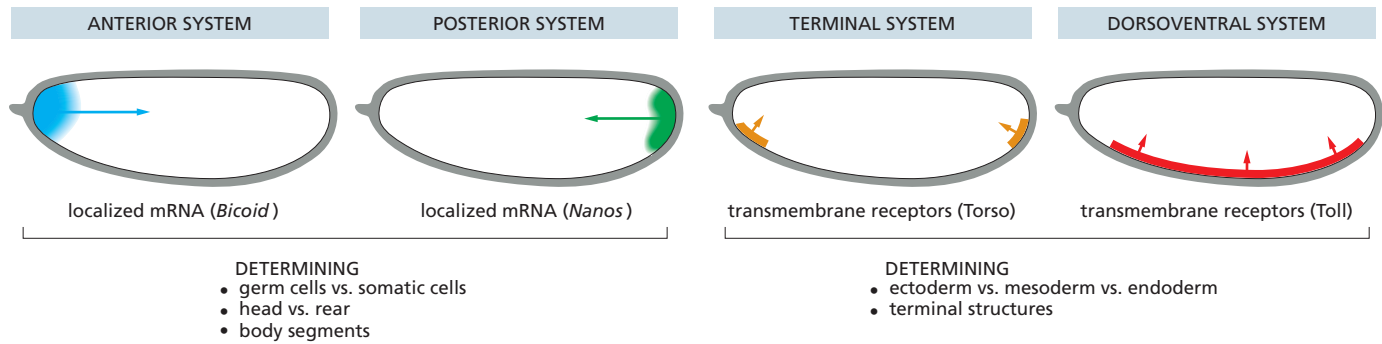


Figure 21-16 The organization of the four egg-polarity gradient systems in *Drosophila*. *Bicoid* mRNA encodes a transcriptional activator that determines the head and thoracic regions. *Nanos* is a translational repressor that governs the formation of the abdomen. Localized *Nanos* mRNA is also incorporated into the germ cells as they form at the posterior of the embryo, and *Nanos* protein is necessary for germ-line development. Toll and Torso are receptor proteins that are distributed all over the membrane but are activated only at the sites indicated by the coloring, through localized exposure to the extracellular ligands Spaetzle (the ligand for Toll) and Trunk (the ligand for Torso). Toll activity determines the mesoderm and Torso activity determines the formation of terminal structures at the head and tail.

dorsal versus ventral, and endoderm versus mesoderm and ectoderm—as well as a fourth partition, no less fundamental to the body plan of animals: the distinction between germ cells and somatic cells (Figure 21-16).

The egg-polarity genes act first in a hierarchy of gene systems that define a progressively more detailed pattern of body parts. In the next few pages, we begin with the molecular mechanisms that pattern the developing *Drosophila* embryo and larva along the A-P axis, before considering the patterning along the D-V axis.

Three Groups of Genes Control *Drosophila* Segmentation Along the A-P Axis

The body of an insect is divided along its A-P axis into a series of **segments**. The segments are repetitions of a theme with variations: each segment forms highly specialized structures, all built according to a similar fundamental plan (Figure 21-17). The gradients of transcription regulators set up along the A-P axis in the early embryo by the egg-polarity genes are the prelude to the creation of the segments. These regulators initiate the orderly transcription of *segmentation genes*, which refine the pattern of gene expression to define the boundaries and ground plan of the individual segments. Segmentation genes are expressed by subsets of cells in the embryo, and their products are among the first components that the embryo's own genome contributes to embryonic development; they are therefore called **zygotic-effect genes**, to distinguish them from the earlier-acting maternal-effect genes. Mutations in segmentation genes can alter either the number of segments or their basic internal organization.

The **segmentation genes** fall into three groups according to their mutant phenotypes (Figure 21-18). It is convenient to think of the three groups as acting in sequence, although in reality their functions overlap in time. First to be expressed is a set of at least six **gap genes**, whose products mark out coarse A-P subdivisions of the embryo. Mutations in a gap gene eliminate one or more groups of adjacent segments: in the mutant *Krüppel*, for example, the larva lacks eight segments. Next comes a set of eight **pair-rule genes**. Mutations in these genes

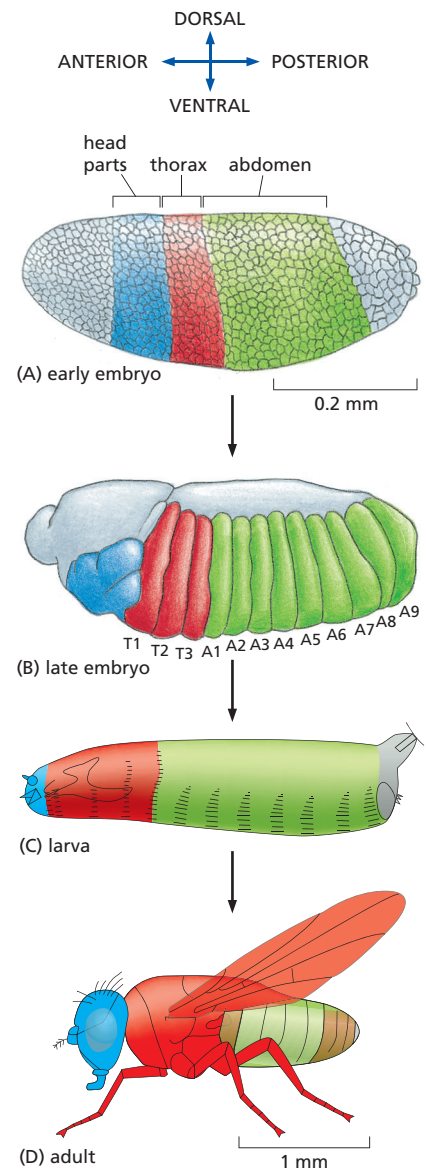


Figure 21-17 The origins of the *Drosophila* body segments. (A) At 3 hours, the embryo (shown in side view) is at the blastoderm stage and no segmentation is visible, although a fate map can be drawn showing the future segmented regions (color). (B) At 10 hours, all the segments are clearly defined (T1: first thoracic segment; A1: first abdominal segment). See [Movie 21.3](#). (C) The segments of the *Drosophila* larva and their correspondence with regions in the embryo. (D) The segments of the *Drosophila* adult and their correspondence with regions in the embryo.

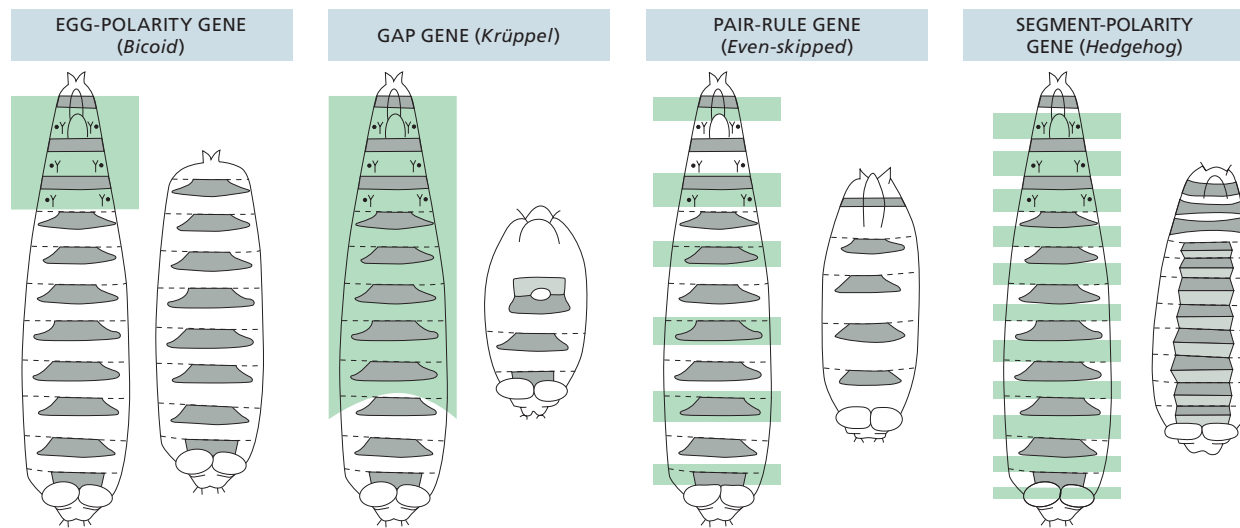


Figure 21-18 Examples of the phenotypes of mutations affecting egg-polarity genes and the three types of segmentation genes. In each case, the areas shaded in green on the normal larva (left) are deleted in the mutant (right) or are replaced by mirror-image duplicates of the unaffected regions. (Modified from C. Nüsslein-Volhard and E. Wieschaus, *Nature* 287:795–801, 1980.)

cause a series of deletions affecting alternate segments, leaving the embryo with only half as many segments as usual; although all the mutants display this two-segment periodicity, they differ in the precise pattern. Finally, there are at least 10 **segment-polarity genes**, in which mutations produce a normal number of segments but with a part of each segment deleted and replaced by a mirror-image duplicate of all or part of the rest of the segment.

The phenotypes of the various segmentation mutants suggest that the segmentation genes form a coordinated system that subdivides the embryo progressively into smaller and smaller domains along the A-P axis, each distinguished by a different pattern of gene expression. Molecular genetics has helped to reveal how this system works.

A Hierarchy of Gene Regulatory Interactions Subdivides the *Drosophila* Embryo

Like *Bicoid*, most of the segmentation genes encode transcription regulators. Their control by the egg-polarity genes and their actions on one another and on still other genes can be deciphered by comparing gene expression in normal and mutant embryos. By using appropriate probes to detect RNA transcripts or their protein products, one can observe genes switch on and off in changing patterns. These patterns reveal the wealth of spatial information created within the morphologically uniform embryo by the egg-polarity gene network. By comparing these patterns in different mutants, one can begin to discern the logic of the entire gene control system.

The products of the egg-polarity genes provide the global positional signals in the early embryo (see Figure 21-16). The Bicoid protein, as we have seen, acts as a morphogen and activates different sets of genes at different positions along the A-P axis: some gap genes are only activated in regions with high levels of Bicoid, others only where levels of Bicoid are lower. There are only six gap genes, but a combination of overlapping expression as well as different levels within their domains provides each cell along the A-P axis with a rich variety of positional identities. After the gap-gene products refine their positions by repressing each other's expression, they provide a second tier of positional signals that act more locally to regulate finer details of patterning. They control the expression of the pair-rule genes, through combinatorial effects as discussed in Chapter 7 for the pair-rule gene *Even-skipped* (see pp. 423–424). The pair-rule genes demarcate

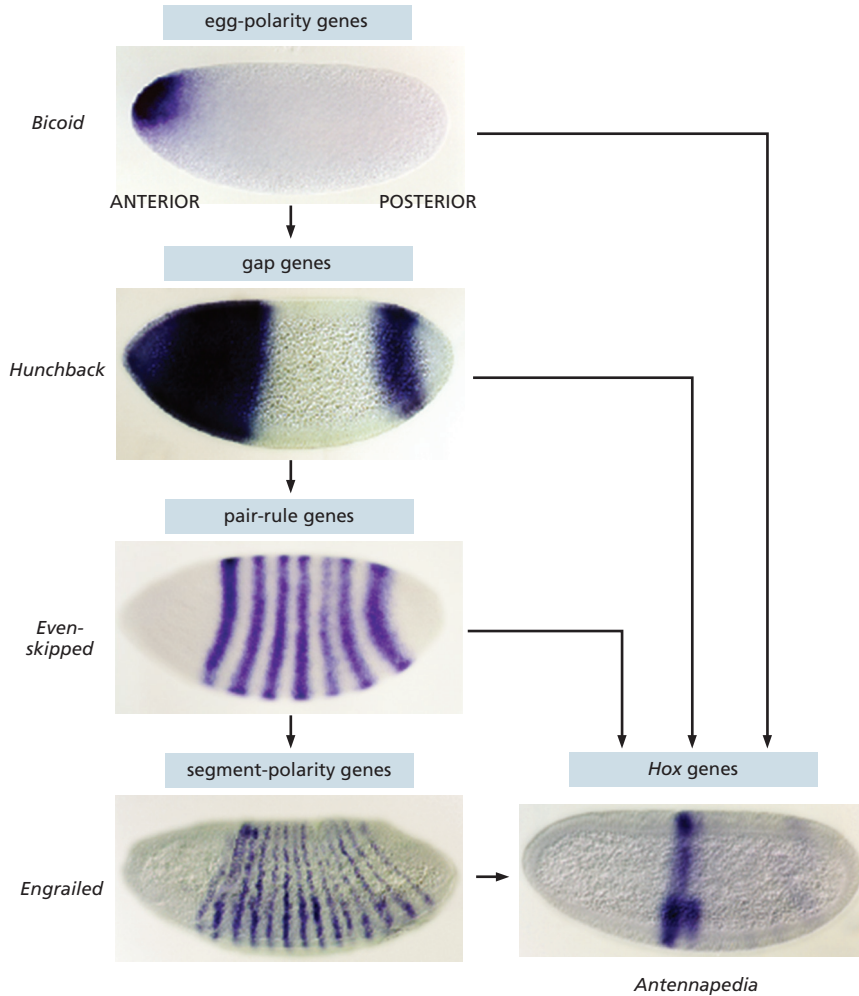


Figure 21-19 The regulatory hierarchy of A-P patterning in the *Drosophila* embryo. Egg-polarity genes define the A-P axis and also initiate expression of three groups of genes (gap, pair-rule, and segment polarity) that create segments. The identity of each segment is specified by *Hox* genes (discussed shortly), whose expression is controlled by input from both egg-polarity and segmentation genes. The photographs show mRNA expression patterns of representative examples of genes of each type. (Courtesy of Stephen Small.)

the repeated groups of cells that will later become segments and, in turn, collaborate with one another and with the gap genes to set up a regular, periodic pattern of expression of the segment-polarity genes, which define the internal pattern of each individual segment (Figure 21-19).

A large subset of the segment-polarity genes codes for components of two signaling pathways—the Wnt pathway and the Hedgehog pathway, including the secreted signal proteins Wingless (the first-named member of the Wnt family) and Hedgehog. (The Hedgehog pathway was first discovered through study of *Drosophila* segmentation, and it takes its name from the prickly appearance of the surface of the *Hedgehog* mutant embryo.) Wingless and Hedgehog are synthesized in different bands of cells that serve as signaling centers within each segment. The two proteins mutually maintain each other's expression while regulating the expression of genes such as *Engrailed* in neighboring cells (Figure 21-20). In such a manner, a series of sequential inductions creates a fine-grained pattern of gene expression within each segment.

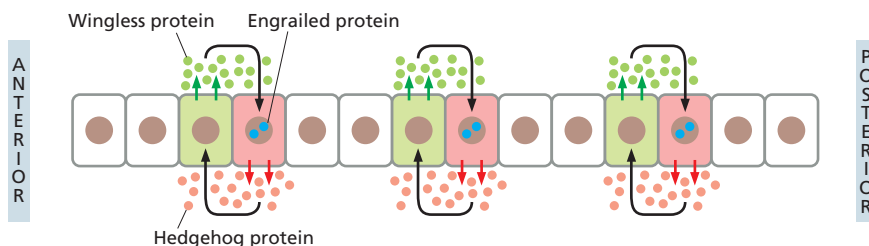


Figure 21-20 Mutual maintenance of *Hedgehog* and *Wingless* expression. *Engrailed* is a transcription regulator (blue) that drives the expression of *Hedgehog*. *Hedgehog* encodes a secreted protein (red) that activates a signaling pathway in neighboring cells and thereby drives them to express the *Wingless* gene. In turn, *Wingless* encodes a secreted protein (green) that acts back on neighbors of the *Wingless*-expressing cell to maintain their expression of *Engrailed*. *Engrailed* then maintains *Hedgehog* expression to complete the loop. As indicated, the same network repeats along the A-P axis of the fly. (Based on S. DiNardo et al., *Curr. Opin. Genet. Dev.* 4:529–534, 1994.)

Egg-Polarity, Gap, and Pair-Rule Genes Create a Transient Pattern That Is Remembered by Segment-Polarity and *Hox* Genes

The gap genes and pair-rule genes are activated within the first few hours after fertilization. Their mRNA products initially appear in patterns that only approximate the final picture; then, within a short time, this fuzzy initial pattern resolves itself into a regular, crisply defined system of stripes. But this pattern itself is unstable and transient: as the embryo proceeds through gastrulation and beyond, the pattern disintegrates. The genes' actions, however, have passed on an enduring memory of their patterns of expression by inducing the expression of certain segment-polarity genes along with another class of genes called *Hox* genes (discussed shortly). After a period of pattern refinement mediated by cell-cell interactions, the expression patterns of these new groups of patterning genes are stabilized to provide positional labels that serve to maintain the segmental organization of the larva and adult fly.

The segment-polarity gene *Engrailed* provides a good example. Its RNA transcripts form a series of 14 bands in the cellular blastoderm, each approximately one cell wide. These stripes lie immediately anterior to similar stripes of expression of another segment-polarity gene, *Wingless*. As the cells in the developing embryo continue to divide and move, signaling between the *Wingless*-expressing cells and the *Engrailed*-expressing cells maintains narrow stripes of their expression (see Figure 21-20). This interaction triggers a stable *Engrailed* expression pattern that will last throughout the life of the fly, long after the signals that induced and refined it have disappeared. The segment borders in embryo, larva, and adult will all form at the posterior edge of each such *Engrailed* stripe (Figure 21-21).

In addition to regulating the segment-polarity genes, the products of pair-rule genes collaborate with those of gap genes to induce the precisely localized activation of a further set of genes—the *Hox* genes (see Figure 21-19). It is the *Hox* genes that first define and then permanently distinguish one segment from another. In the next section, we examine these important genes in detail; we shall see that this role is critical in a wide range of animals, including ourselves.

Hox Genes Permanently Pattern the A-P Axis

As animal development proceeds, the body becomes more and more complex. But again and again, in every species and at every level of organization, we find that complex structures are made by repeating a few elementary themes, with variations. Thus, a subset of basic differentiated cell types, such as muscle cells or fibroblasts, recur at different sites and are organized into tissues such as muscle or tendon. Subtle variations in how and where patterning mechanisms are deployed determines how structures such as teeth or digits are built, giving rise to molars and incisors, fingers and thumbs and toes.

Wherever we find this phenomenon of *modulated repetition*, we can break down the developmental biologist's problem into two kinds of questions: What is the basic construction mechanism common to all the objects of the given class, and how is this mechanism modified to give the observed variations in different animals? The segments of the insect body provide a good example. We have thus far sketched the way in which the rudiment of a single body segment is constructed and how cells within each segment become different from one another. We now consider how one segment becomes determined, or *specified*, to be different from another.

The first glimpse of the answer to this problem came more than 80 years ago, with the discovery of a set of mutations in *Drosophila* that cause bizarre disturbances in the organization of the adult fly. In the *Antennapedia* mutant, for example, legs sprout from the head in place of antennae, whereas in the *Bithorax* mutant, portions of an extra pair of wings appear where normally there should be the much smaller appendages called halteres (Figure 21-22). These mutations transform parts of the body into structures appropriate to other positions, and

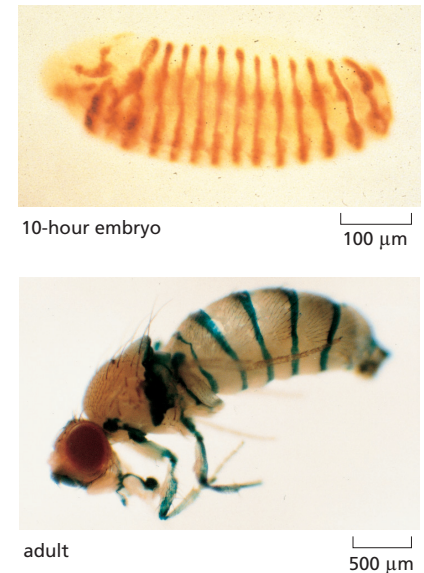


Figure 21-21 The pattern of expression of *Engrailed*, a segment-polarity gene. The *Engrailed* pattern is shown in a 10-hour embryo and an adult (whose wings have been removed in this preparation). The pattern is revealed by constructing a strain of *Drosophila* containing the control sequences of the *Engrailed* gene coupled to the coding sequence of the reporter *LacZ*, whose product is detected histochemically through the brown product generated by immunohistochemistry against *LacZ* itself (10-hour embryo) or through the blue product generated by a reaction that *LacZ* catalyzes (adult). Note that the *Engrailed* pattern marks segment boundaries and, once established, is preserved throughout the animal's life. (Courtesy of Tom Kornberg.)

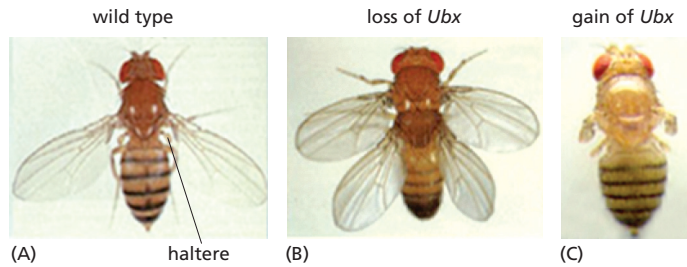


Figure 21–22 Homeotic mutations.

Ultrabithorax, or *Ubx*, is one of three genes in the *Bithorax* gene complex (a *Hox* gene cluster). *Ubx* is responsible for all of the differences between the second (wing-bearing) and third (haltere-bearing) thoracic segments. (A and B) *Ubx* loss-of-function mutations transform the haltere-bearing segment (A) into a wing-bearing segment, resulting in four-winged flies (B). (C) *Ubx* gain-of-function in the second thoracic segment transforms this wing-bearing segment into a haltere-bearing segment, resulting in wingless flies.

(A, courtesy of the Archives, California Institute of Technology; C, courtesy of L.S. Shashidhara.)

they are called *homeotic* mutations (from the Greek *homoios*, meaning “similar”) because the transformation is between structures of a recognizably similar general type, changing one kind of limb or one kind of segment into another. It was eventually discovered that a whole set of genes, the **homeotic selector genes** or **Hox genes**, serve to permanently specify the A-P characters of the whole set of animal segments. These genes are all related to one another as members of a multigene family.

There are eight *Hox* genes in the fly, and they all lie in one or the other of two gene clusters known as the ***Bithorax* complex** and the ***Antennapedia* complex**. The genes in the *Bithorax* complex control the differences among the abdominal and thoracic segments of the body, while those in the *Antennapedia* complex control the differences among thoracic and head segments. Comparisons with other species show that the same genes are present in essentially all animals, including humans. These comparisons also reveal that the *Antennapedia* and *Bithorax* complexes are two halves of a single entity, called the **Hox complex**, that has become split in the course of the fly’s evolution, and whose members operate in a coordinated way to exert their control over the head-to-tail pattern of the body.

The products of the *Hox* genes, the **Hox proteins**, are transcription regulators, all of which possess a highly conserved, 60-amino-acid-long DNA-binding *homeodomain* (see p. 404). The homeodomain-encoding DNA sequence is called a “homeobox,” from which, by abbreviation, the *Hox* complex takes its name. There are many homeobox-containing genes, but only those located in a *Hox* complex are *Hox* genes.

Hox Proteins Give Each Segment Its Individuality

The Hox proteins can be viewed as molecular address labels possessed by the cells of each segment: these labels give the cells in each region a **positional value**; that is, an intrinsic character that differs according to a cell’s location. If the address labels in a developing *Drosophila* segment are changed, the segment behaves as though it were located somewhere else; if all the *Hox* genes in an embryo are deleted, the body segments in the larva will all be alike.

To a first approximation, each *Hox* gene is normally expressed in those regions that develop abnormally when that gene is mutated or absent. How does each Hox protein give a segment its permanent identity? Recall that the Hox proteins are transcription regulators, which can bind to gene regulatory DNA; each Hox protein targets a different set of genes for activation or repression. Hundreds of genes are under this type of Hox-modulated control, including genes that control cell-cell signaling, transcriptional regulation, cell polarity, cell adhesion, cytoskeletal function, cell growth, and cell death, all conspiring to give each segment its distinctive Hox-dependent character.

Hox Genes Are Expressed According to Their Order in the Hox Complex

How, then, is the expression of the *Hox* genes themselves regulated? The coding sequences of the eight *Hox* genes in *Drosophila* are interspersed amid a much larger quantity of regulatory DNA. This DNA includes binding sites for the

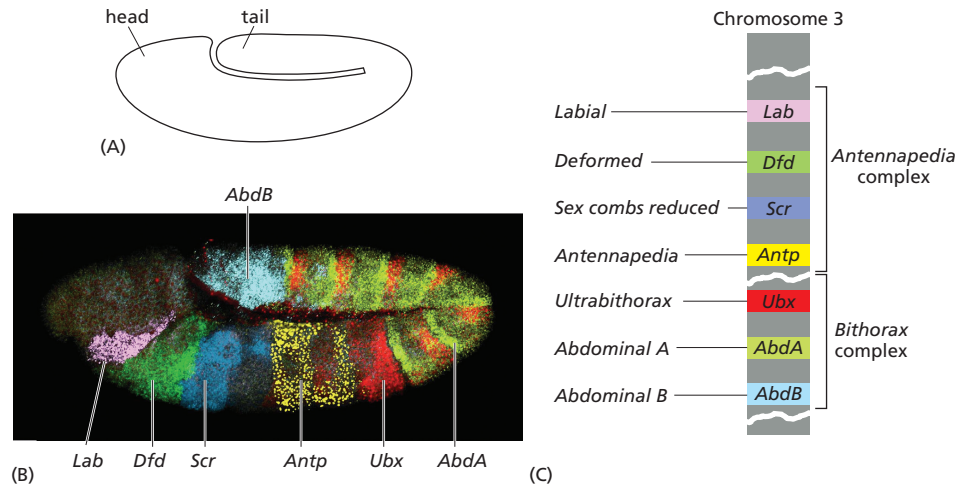


Figure 21-23 The patterns of expression compared to the chromosomal locations of the genes of the *Hox* complex.

(A) Diagram of a *Drosophila* embryo at the so-called germ band retraction stage, about 10 hours after fertilization when the developing body axis is folded over on itself. (B) An embryo at this stage has been stained by *in situ* hybridization using differently labeled probes to detect the mRNA products of different *Hox* genes in different colors. (C) The spatial pattern in the photograph corresponds, with minor deviations, to the sequence of genes in each of the two subdivisions of the chromosomal complex. (B, courtesy of William McGinnis, adapted from D. Kosman et al., *Science* 305:846, 2004. With permission from AAAS.)

products of the egg-polarity and segmentation genes, thereby serving as an interpreter of the detailed spatial information supplied to it by all of these transcription regulators. The net result is that a particular set of *Hox* genes is transcribed in a specific region along the A-P body axis.

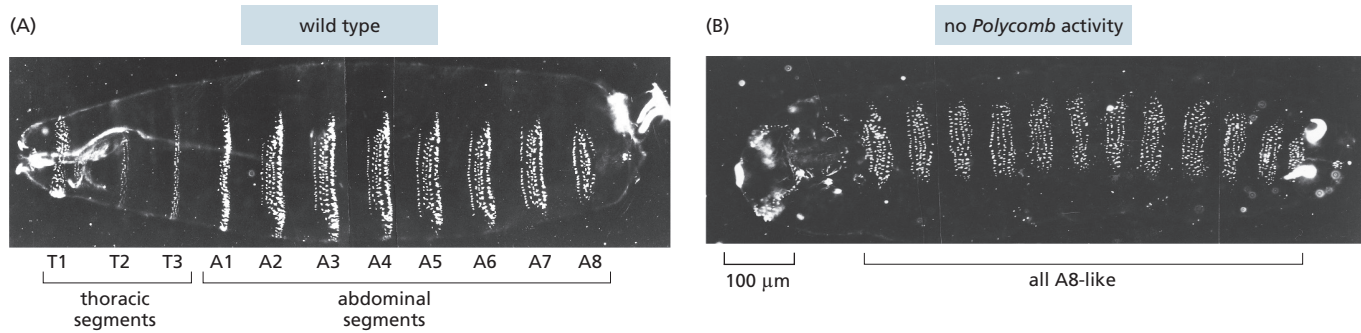
The pattern of *Hox* gene expression exhibits a remarkable regularity that suggests an additional form of control. The sequence in which the genes are ordered along the chromosome, in both the *Antennapedia* and the *Bithorax* complexes, corresponds almost exactly to the order in which they are expressed along the A-P axis of the body (Figure 21-23). This hints at some process of gene activation, perhaps dependent on chromatin structures that propagate along the *Hox* complexes, switching on one *Hox* gene after another according to their order along the chromosome. The most “posterior” of the *Hox* genes that are expressed in a cell generally dominates, driving down expression and activity of the “anterior” genes and dictating the character of the segment. The gene regulatory mechanisms underlying these phenomena are still not well understood, but their consequences are profound. We shall see that the serial organization of gene expression in the *Hox* complex is a fundamental feature that has been highly conserved in the course of animal evolution.

Trithorax and Polycomb Group Proteins Regulate *Hox* Expression to Maintain a Permanent Record of Positional Information

The spatial pattern of expression of the genes in the *Hox* complex is set up by signals acting early in development, but the effects are long lasting. Although the pattern of expression undergoes complex adjustments as development proceeds, the *Hox* pattern stamps each cell and all of its progeny with a permanent record of the A-P position that the cell occupied in the early embryo. In this way, the cells of each segment maintain a memory of their location along the A-P axis of the body, which governs the segment-specific identity not only of the larval segments but also of the structures of the adult fly.

Two molecular mechanisms ensure that a cell remembers its positional information. One is from the *Hox* genes themselves: many of the *Hox* proteins autoactivate the transcription of their own genes, thereby helping to keep the genes on indefinitely. Another crucial input is from two large, complementary sets of proteins, called the **Trithorax group** and the **Polycomb group**, which imprint the chromatin of the *Hox* complex with a heritable record of its embryonic state of activation or repression. These are key general regulators of chromatin structure that are critical for cell memory: if genes of the *Trithorax* or *Polycomb* group are defective, the pattern of expression of the *Hox* genes is set up correctly at first, but it is not correctly maintained as cells divide and the embryo grows older.

The two sets of regulators act in opposite ways. Trithorax group proteins are needed to maintain the transcription of *Hox* genes in cells where their



transcription has already been switched on. In contrast, Polycomb group proteins form stable complexes that bind to the chromatin of the *Hox* complex and maintain the repressed state in cells where *Hox* genes have not yet been activated (Figure 21–24). Although first discovered because of their influence on *Hox* genes in flies, Polycomb and Trithorax group proteins are general regulators of chromatin structure that control many genes in plants as well as animals. How such changes in chromatin can store developmental cell memory is discussed in Chapters 4 and 7.

The D-V Signaling Genes Create a Gradient of the Transcription Regulator Dorsal

We now turn to patterning of the second major axis of the *Drosophila* embryo. As with the patterning along the A-P axis just discussed, the patterning along the dorsoventral (D-V) axis begins with maternal gene products that define this axis in the egg (see Figure 21–16) and then progresses through zygotic gene products that further subdivide the D-V axis in the embryo.

Initially, a protein that is produced by the mother's somatic cells underneath the future ventral region of the embryo leads to the localized activation of a transmembrane receptor called **Toll** on the ventral side of the egg membrane. (Curiously, *Drosophila* Toll and vertebrate Toll-like proteins also operate in innate immune responses, as discussed in Chapter 24.) The localized activation of Toll controls the distribution of **Dorsal**, a transcription regulator of the NFκB family discussed in Chapter 15. The Toll-regulated activity of Dorsal, like that of NFκB, depends on the translocation of Dorsal protein from the cytosol, where it is held in an inactive form, to the nucleus, where it regulates gene expression (see Figure 15–63). In the newly laid egg, both Dorsal mRNA and protein are distributed uniformly in the cytosol. After the nuclei in the syncytial blastoderm have migrated to the surface of the embryo, but before cellularization (see Figure 21–14), Toll receptor activation on the ventral side induces a remarkable redistribution of the Dorsal protein. On the dorsal side, the protein remains in the cytosol, but ventrally it becomes concentrated in the nuclei, with a smooth gradient of nuclear localization between these two extremes (Figure 21–25).

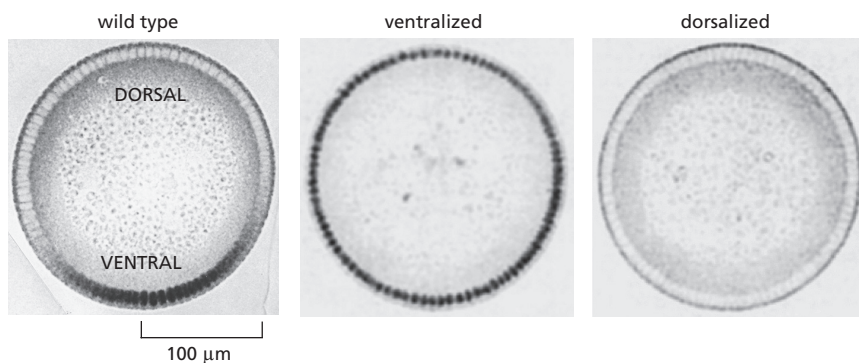


Figure 21–24 The role of genes of the Polycomb group. (A) Photograph of a wild-type *Drosophila* embryo, imaged by dark-field microscopy. (B) Photograph of a mutant embryo defective for the gene *Extra sex combs (Esc)* and derived from a mother also lacking this gene. The gene belongs to the Polycomb group. Essentially all segments have been transformed to resemble the most posterior abdominal segment, A8. In the mutant, the pattern of expression of the homeotic selector genes, which is roughly normal initially, is unstable in such a way that all these genes soon become switched on all along the body axis. (From G. Struhl, *Nature* 293:36–41, published 1981 by Nature Publishing Group. Reproduced with permission of SNCSC.)

Figure 21–25 The concentration gradient of Dorsal protein in the nuclei of the blastoderm. In wild-type *Drosophila* embryos, the protein is present in the dorsal cytoplasm and absent from the dorsal nuclei; ventrally, it is depleted in the cytoplasm and concentrated in the nuclei. In a mutant in which the Toll pathway is activated everywhere and not just ventrally, Dorsal protein is everywhere concentrated in the nuclei; the result is a ventralized embryo. Conversely, in a mutant in which the Toll signaling pathway is inactivated, Dorsal protein everywhere remains in the cytoplasm and is absent from the nuclei; the result is a dorsalized embryo. (From S. Roth et al., *Cell* 59:1189–1202, 1989. With permission from Elsevier.)

Similar to Bicoid along the A-P axis, Dorsal acts as a morphogen along the D-V axis. Once inside the nucleus, the Dorsal protein turns on or off the expression of different sets of genes depending on Dorsal's concentration. The expression of each responding gene depends on its regulatory DNA—specifically, on the number and affinity of the binding sites that this DNA contains for Dorsal and other transcription regulators. In this way, the regulatory DNA interprets the positional signal provided by the nuclear Dorsal protein gradient, so as to define a distinct D-V series of territories—complementary bands of cells that run the length of the embryo. Most ventrally—where the nuclear concentration of Dorsal protein is highest—it switches on, for example, the expression of a gene called *Twist*, which directs mesodermal fate. Most dorsally, where the nuclear concentration of Dorsal protein is lowest, the cells switch on a gene called *Decapentaplegic* (*Dpp*). And in an intermediate region, where the nuclear concentration of Dorsal protein is high enough to repress *Dpp* but too low to activate *Twist*, the cells switch on another set of genes, including one called *Short gastrulation* (*Sog*) (Figure 21–26A).

Products of the genes directly regulated by the Dorsal protein generate in turn more local signals, which define finer subdivisions along the D-V axis. These signals act after cellularization and take the form of conventional extracellular diffusible proteins. In particular, *Dpp* codes for a secreted TGF β family protein, which forms a local morphogen gradient in the dorsal part of the embryo. *Sog*, produced ventrally to *Dpp*, encodes another secreted protein that acts as an antagonist of Dpp protein, by binding to it and preventing Dpp from activating its receptor. The opposing diffusion gradients of these two signal proteins create a steep gradient of Dpp activity: the highest Dpp activity levels, in combination with certain other factors, cause development of the most dorsal tissue of all—an

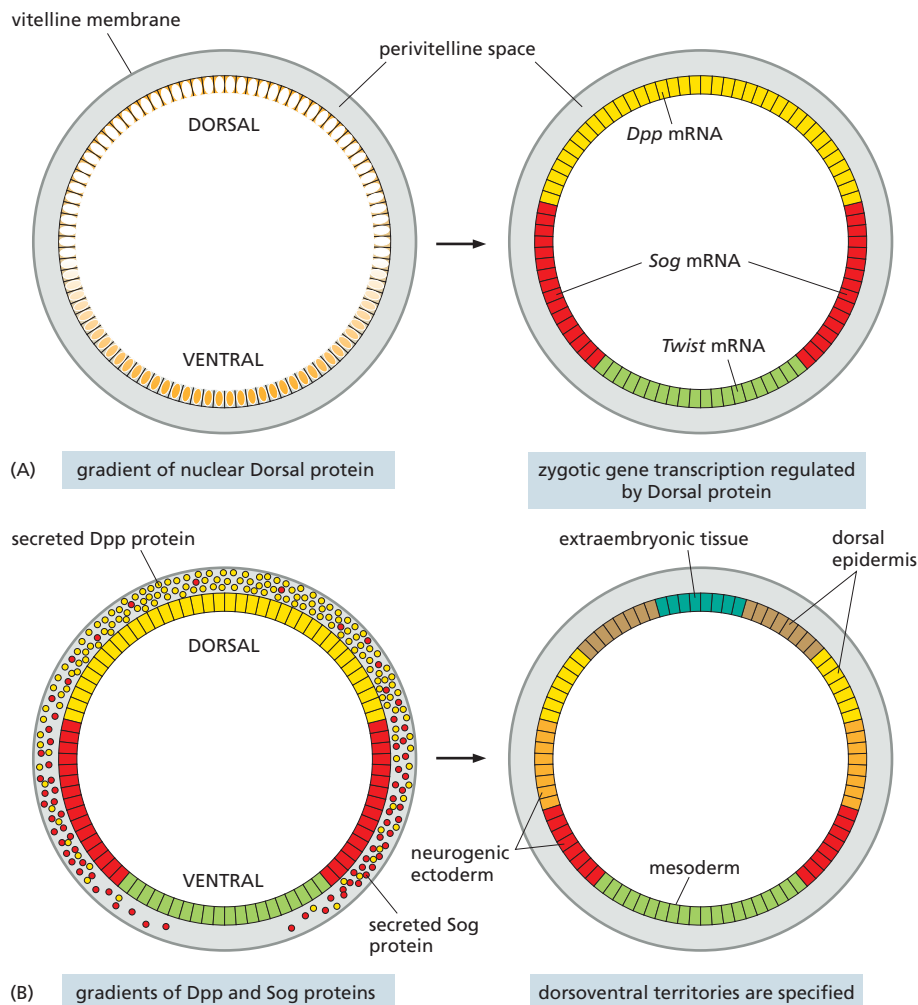


Figure 21–26 How morphogen gradients guide a patterning process along the dorsoventral axis of the *Drosophila* embryo. (A) Initially, a gradient of Dorsal protein defines three broad territories of gene expression, marked here by the expression of three representative genes: *Dpp*, *Sog*, and *Twist*. (B) Slightly later, the cells expressing *Dpp* and *Sog* secrete, respectively, the signal proteins Dpp (a TGF β family member) and Sog (an antagonist of Dpp). These two proteins then diffuse and interact with one another (and with certain other factors) to create the dorsoventral (D-V) territories shown.

extraembryonic membrane. Intermediate levels cause development of dorsal epidermis; and the absence of Dpp activity in cells expressing Sog allows the development of neurogenic ectoderm, which will give rise to the nervous system (Figure 21-26B).

A Hierarchy of Inductive Interactions Subdivides the Vertebrate Embryo

The molecular genetic analysis of *Drosophila* development has uncovered how a cascade of transcription regulators and signaling pathways sequentially subdivides the embryo. The same principle of progressive pattern refinement is used during the development of all animal embryos, including vertebrates. Remarkably, conservation is not restricted to the general strategy of pattern formation, but also extends to many of the molecules involved.

As mentioned previously, the earliest phases of vertebrate development are surprisingly variable, even between closely related species, and it is even hard to say precisely how the A-P and D-V axes of an early fly embryo correspond to those of an early frog or mouse embryo. Nevertheless, we shall see that amid this display of evolutionary plasticity, some features of early development turn out to be highly conserved. The same is true of later developmental stages also, often to an astonishing degree. From our own anatomy, it is obvious that we are cousins to birds and fish. But looking at molecular mechanisms, we see that we are related to flies and worms too.

In the following pages, we discuss how vertebrate embryos are patterned by the interplay of signaling molecules and transcription regulators. We begin by discussing the formation and patterning of the embryonic axes in amphibians, taking the frog *Xenopus* as our example. We have already broached this topic earlier in the chapter. Here, we pick up the thread and draw comparisons with the fly.

As noted earlier, the origins of the embryonic axes and the three germ layers in the frog can be traced back to the blastula (see Figure 21-3A). By labeling individual blastomeres, we can track cells through all their divisions, transformations, and migrations and see what they become and where they come from. The precursors of ectoderm, mesoderm, and endoderm are arranged in order along the animal-vegetal axis of the blastula: the endoderm derives from the most vegetal blastomeres, the ectoderm from the most animal, and the mesoderm from a middle set. Within each of these territories, the cells have diverse fates according to their positions along the D-V axis of the later embryo. For ectoderm, epidermal precursors are located ventrally, and future neurons are found dorsally; for mesoderm, precursors for notochord, muscle, kidney, and blood are arranged from dorsal to ventral. All this can be represented by a **fate map** that shows which later cell types derive from which regions of the early embryo (Figure 21-27). The fate map confronts us with the central question: How are the cells in different positions driven toward their different fates? We have already explained how maternal factors deposited in the developing frog egg define its animal-vegetal axis, and how cortical rotation triggered by fertilization defines the orientation of the dorsoventral axis (see Figure 21-13). But how does the establishment of axes lead on to the subdivision of the embryo into the future body parts?

The answer is that the maternal gene products lead to the formation of *signaling centers* on the vegetal and dorsal sides of the embryo. The dorsal signaling center in particular has a special place in the history of developmental biology. Experiments in the early twentieth century identified it as a small cluster of cells with an extraordinary property: when the cells were transplanted to an opposite site, they could trigger a radical reorganization of the neighboring tissue, causing it to form a second whole-body axis (Figure 21-28). The discovery of this signaling center, called the **Organizer**, led the way to a pioneering analysis of the chain of inductive interactions that establish the framework of the vertebrate body.

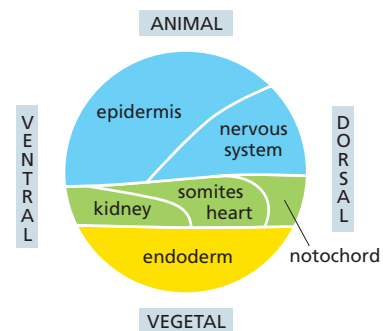


Figure 21-27 Blastula fate map in a frog embryo. The endoderm derives from the most vegetal blastomeres (yellow), the ectoderm from the most animal (blue), and the mesoderm from a middle set (green) that contributes also to endoderm and ectoderm. Different cell types derive from different positions along the dorsoventral axis.

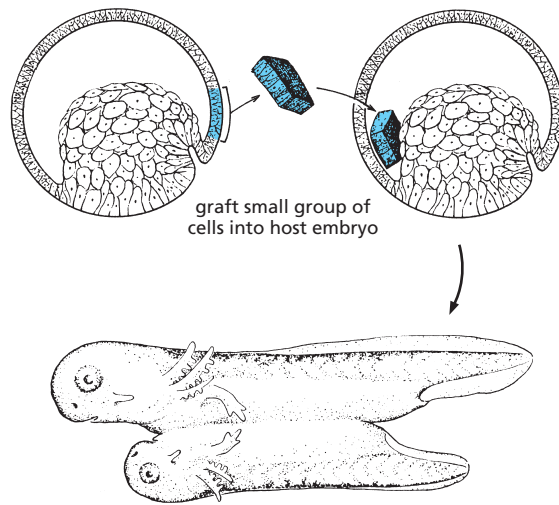


Figure 21–28 Induction of a secondary axis by the Organizer. An amphibian embryo receives a graft of a small cluster of cells taken from a specific site, called the Organizer region, on the dorsal side of another embryo at the same stage. Signals from the graft organize the behavior of neighboring cells of the host embryo, causing development of a pair of conjoined (Siamese) twins. See **Movie 21.4**. [After J. Holtfreter and V. Hamburger, in *Analysis of Development* (B.H. Willier, P.A. Weiss, and V. Hamburger, eds.), pp. 230–296. Philadelphia: Saunders, 1955.]

In contrast to the *Drosophila* syncytial embryo, the fertilized frog egg undergoes conventional cleavage divisions that result in an embryo consisting of thousands of cells. Patterning must therefore be mediated by extracellular signal molecules that diffuse through the embryo from cell to cell, not by transcription regulators that move through the cytoplasm of a syncytium. Not surprisingly, the Organizer is now known to be a major source of secreted signals. As we shall see, this includes not only ligands that bind and activate transmembrane receptors (see Chapter 15), but also secreted proteins that inhibit the activity of these ligands.

A Competition Between Secreted Signaling Proteins Patterns the Vertebrate Embryonic Axes

The signal molecules that pattern the frog embryo along the animal-vegetal (A-V) axis belong to the TGF β family: they are secreted by a signaling center at the vegetal pole and form concentration gradients along the A-V axis. These *Nodal* proteins act over a relatively short range: cells closest to the vegetal pole are exposed to high levels and respond by switching on genes that promote the development of endoderm; cells farther away are exposed to lower levels and activate genes that promote the formation of mesoderm. The cells at the vegetal pole that produce *Nodal* also produce a second, more rapidly diffusing protein called *Lefty*, which antagonizes *Nodal*. The high ratio of *Lefty* to *Nodal* at the animal pole allows *Lefty* to block *Nodal* signaling; this causes the cells there to develop as ectoderm (**Figure 21–29A**). Thus, a mid-range activation by *Nodal*, combined with a long-range inhibition by *Lefty*, sets up the pattern of progenitors along the A-V axis for the three germ layers—endoderm, mesoderm, and ectoderm.

The frog uses a somewhat related strategy to subdivide the germ-layer territories along the D-V axis of the embryo. It relies on patterned inhibition of otherwise uniform signaling by the *bone morphogenetic proteins* (BMPs; members of yet another subclass of the TGF β family), which are secreted throughout the embryo. The dorsal signaling system exerts its influence by secreting several proteins, including *Chordin* and *Noggin*, that block BMP signaling when their own concentrations are high. In this way, *Chordin* and *Noggin* create a dorsal-to-ventral gradient of BMP, with low activity on the dorsal side and high activity on the ventral side (**Figure 21–29B**). Ectodermal cells that experience high levels of BMP signaling are driven to epidermal fates, whereas cells that experience little or no BMP signaling remain neural. We can note that this strategy for patterning the D-V axis by opposing gradients of BMP family signals and diffusible inhibitors is similar to that used in *Drosophila*, and indeed the particular molecules used are homologous.

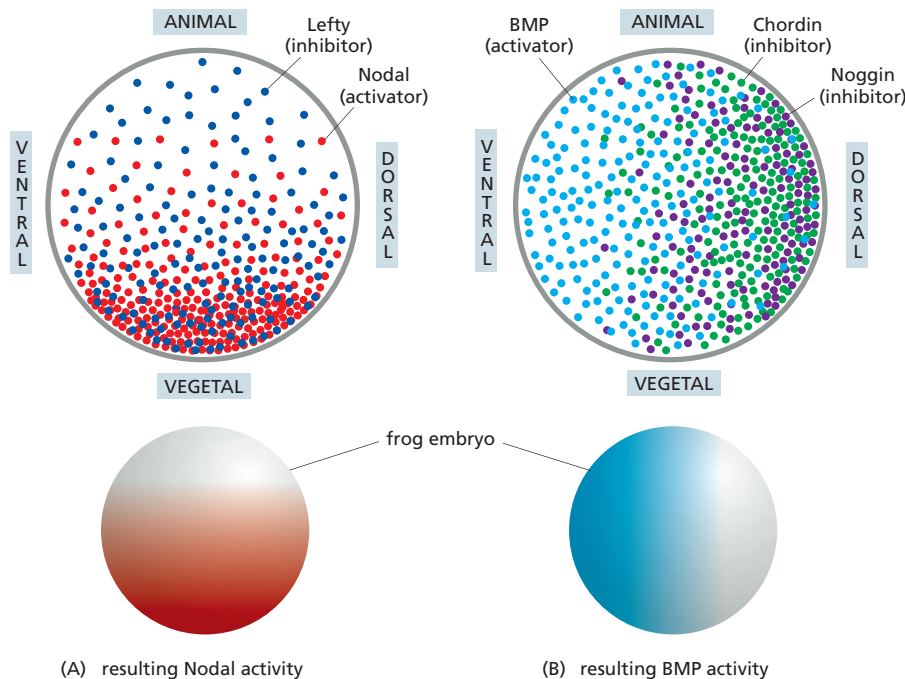


Figure 21–29 How Nodal and bone morphogenetic protein (BMP) signaling pattern the embryonic axes. Nodal and its antagonist Lefty pattern the animal–vegetal axis, while BMP and its antagonists Chordin and Noggin pattern the dorsoventral axis. (A) In the animal-pole region, where Nodal levels are low relative to Lefty, Lefty blocks Nodal from binding to its receptors. In the vegetal region, there is an excess of Nodal, resulting in Nodal pathway activation. (B) Along the dorsoventral axis, BMP is widely present, but Chordin and Noggin are concentrated at the dorsal side: there, they bind to BMP and block its binding to receptors. The resulting patterns of Nodal and BMP activity are illustrated at the bottom of the figure.

Knowing the signals that specify the three germ layers and various tissue types of the vertebrate body, one can reproduce this specification in a culture dish. Frog cells taken from the animal-pole region of the embryo, for example, will differentiate into blood (a ventral mesodermal tissue) when diverted from their original fate by exposure to intermediate concentrations of Nodal and high concentrations of BMP. Similarly, mouse or human embryonic stem cells can be coaxed into generating specific cell types by exposing them in culture to appropriate combinations of signal molecules. In this way, the insights gained through studies of animal development can be used to generate the cell types needed for regenerative medicine, as we discuss in the next chapter.

Hox Genes Control the Vertebrate A-P Axis

The conservation of developmental mechanisms between *Drosophila* and vertebrates extends far beyond the D-V signaling system. *Hox* genes are found in almost every animal species studied, where they are often grouped in complexes similar to the insect *Hox* complex. In mice and humans, for example, there are four such complexes—called the *HoxA*, *HoxB*, *HoxC*, and *HoxD* complexes—each on a different chromosome. Individual genes in each complex can be recognized by their sequences as counterparts of specific members of the *Drosophila* set. Indeed, mammalian *Hox* genes can function in *Drosophila* as partial replacements for the corresponding *Drosophila Hox* genes. It appears that each of the four mammalian *Hox* complexes is, roughly speaking, the equivalent of one complete insect *Hox* complex (that is, an *Antennapedia* complex plus a *Bithorax* complex) (Figure 21–30).

The ordering of the genes within each vertebrate *Hox* complex is essentially the same as in the insect *Hox* complex, suggesting that all four vertebrate complexes originated by duplications of a single primordial complex present in the common ancestor of vertebrates and insects, and have preserved its basic organization. Most tellingly, the members of each vertebrate *Hox* complex are expressed in a head-to-tail series along the axis of the embryo, just as they are in *Drosophila*. As in *Drosophila*, vertebrate *Hox* gene expression patterns are often aligned with vertebrate segments. This alignment is especially clear in the hindbrain (see Figure 21–30), where the segments are called *rhombomeres*.

The products of the vertebrate *Hox* genes, the Hox proteins, specify positional values that control the A-P pattern of parts in the hindbrain, neck, and trunk (as well as some other parts of the body). As in *Drosophila*, when a posterior *Hox*

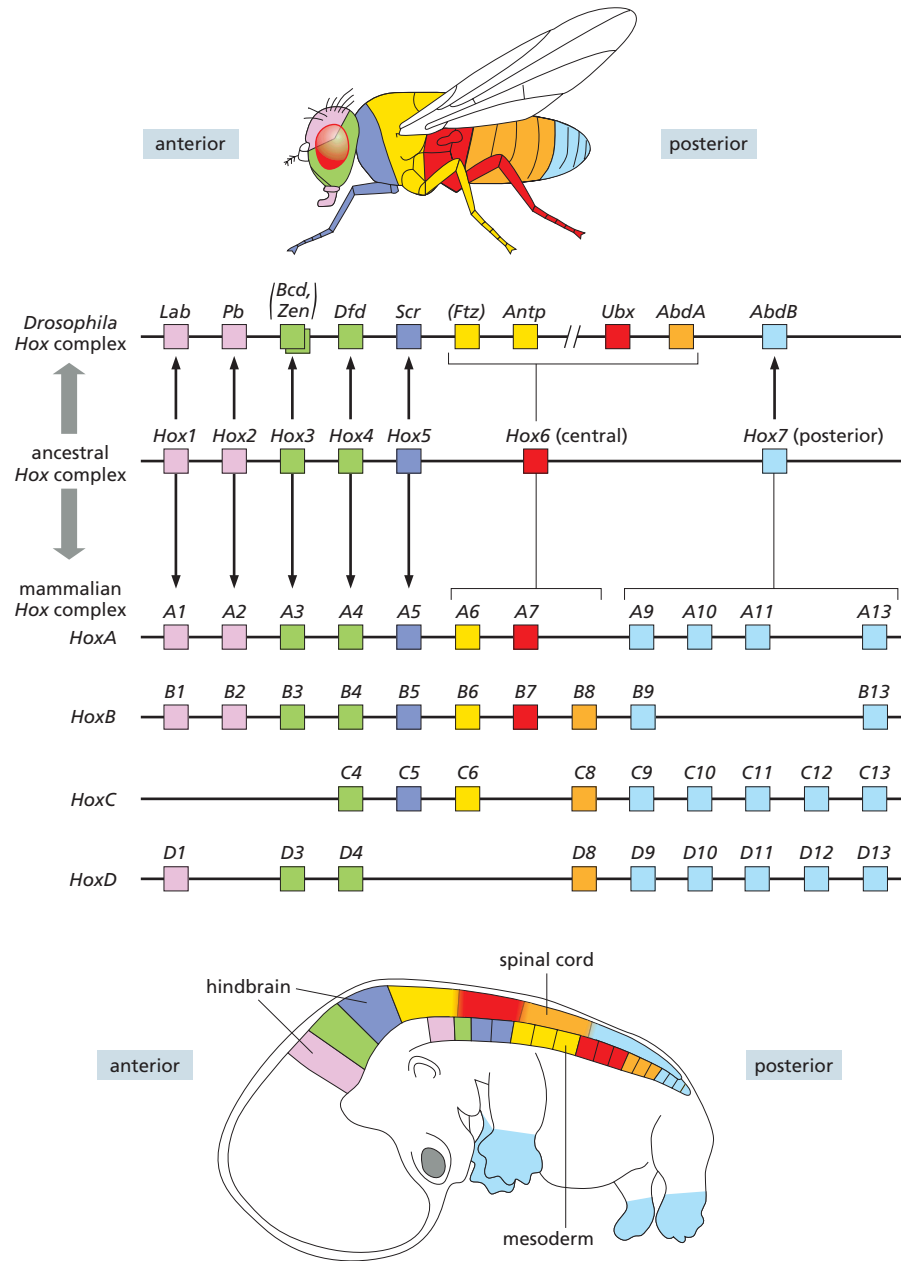


Figure 21–30 The *Hox* complexes of an insect and a mammal, compared and related to body regions. The genes of the *Antennapedia* and *Bithorax* complexes of *Drosophila* are shown in their chromosomal order in the top line (see Figure 21–23). The corresponding genes of the four mammalian *Hox* complexes are shown below, also in chromosomal order. The gene expression domains in fly and mammal are indicated in a simplified form by color in the diagrams of animals above and below. There is a remarkable parallelism. However, the details of the patterns depend on developmental stage and vary somewhat from one mammalian *Hox* complex to another. Also, in many cases, genes shown here as expressed in an anterior domain are also expressed more posteriorly, overlapping the domains of more posterior *Hox* genes.

The complexes are thought to have evolved as follows: first, in some common ancestor of worms, flies, and vertebrates, a single primordial homeotic selector gene underwent repeated duplication to form a series of such genes in tandem—the ancestral *Hox* complex. In the *Drosophila* sublineage, this single complex became split into separate *Antennapedia* and *Bithorax* complexes. Meanwhile, in the lineage leading to the mammals, the whole complex was repeatedly duplicated to give four *Hox* complexes. The parallelism is not perfect because apparently some individual genes have been duplicated and others lost. Still others have been co-opted for different purposes (genes in parentheses in the top line) over the time that has elapsed since the complexes diverged. (Based on a diagram courtesy of William McGinnis.)

gene is artificially expressed in an anterior region, it can convert the anterior tissue to a posterior character. Conversely, loss of posterior *Hox* genes allows the posterior tissue where they are normally expressed to adopt an anterior character (Figure 21–31). Because of overlapping functions between genes in the four *Hox* gene clusters, the transformations observed in mouse *Hox* mutants are not always so straightforward as those in the fly, and they are often incomplete. Nonetheless, it seems clear that the fly and the mouse use essentially the same molecular machinery to impart individual characteristics to successive regions along at least a part of the A-P axis.

Some Transcription Regulators Can Activate a Program That Defines a Cell Type or Creates an Entire Organ

Just as there are single genes that specify the identity of a particular region of the body, there are also genes whose products act as triggers for the development of a specific cell type or even a specific organ, initiating and coordinating the whole complex program of gene expression that is required. An example is the *MyoD*

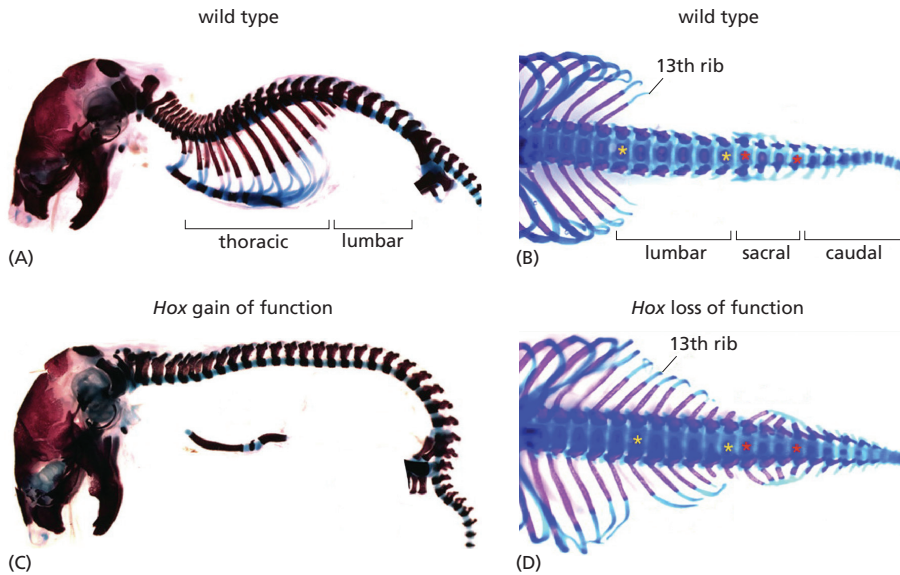


Figure 21-31 Control of anteroposterior pattern by *Hox* genes in the mouse. (A and B) A normal mouse (wild type) has about 65 vertebrae, differing in structure according to their position along the body axis: 7 cervical (neck), 13 thoracic (with ribs), 6 lumbar (bracketed by yellow asterisks in panel B), 4 sacral (bracketed by red asterisks in panel B), and about 35 caudal (tail). Panel A shows a side view, and panel B shows a dorsal view; for clarity, the limbs have been removed in each picture. (C) The *HoxA10* gene is normally expressed in the lumbar region (together with its paralogs *HoxC10* and *HoxD10*); here it has been artificially expressed in the developing vertebral tissue all along the body axis. As a result, the cervical and thoracic vertebrae are all converted to a ribless, lumbar character. (D) Conversely, when *HoxA10* is removed along with *HoxC10* and *HoxD10*, vertebrae that should normally have a lumbar or sacral character take on a thoracic, rib-bearing character instead. (A and C, from M. Carapuço et al., *Genes Dev.* 19:2116–2121, 2005. With permission from Cold Spring Harbor Laboratory Press; B and D, from D.M. Wellik and M.R. Capecchi, *Science* 301:363–367, 2003. With permission of AAAS.)

family of transcription regulators that we encountered in Chapter 7. These proteins drive cells to differentiate into muscle, expressing muscle-specific actins and myosins and all the other specialized cytoskeletal, metabolic, and membrane proteins that a muscle cell needs. Analogously, members of the Achaete/Scute family of transcription regulators drive cells to become neural progenitors. In both these examples, the proteins belong to the helix-loop-helix (HLH) class of transcription regulators (see p. 405), and the same is true for several other proteins that induce the differentiation of particular cell types. These *master transcription regulators* exert their powerful differentiation-inducing activity by binding to many different regulatory sites in the genome and thereby controlling the expression of large numbers of downstream target genes. In one well-studied case, that of an Achaete/Scute family member called Atonal homolog 1 (*Atoh1*), the number of direct target genes in the mouse genome is more than 600. It is important to note, however, that even such powerful drivers of cell differentiation can have radically different effects according to the context and history of the cells in which they act: *Atoh1*, for example, drives differentiation of certain classes of neurons in the brain, of sensory hair cells in the inner ear, and of secretory cells in the lining of the gut.

Other genes encoding transcription regulators can drive the formation and assembly of the multiple cell types that constitute an entire organ. A famous example is the transcription regulator *Eyeless* (see Figure 21-2). When it is artificially expressed in a patch of cells in the leg precursors of *Drosophila*, a well-organized eye-like organ develops on the leg, with the various eye cell types correctly arranged; conversely, loss of the *Eyeless* gene results in flies that lack eyes. Moreover, loss of the *Eyeless* homolog *Pax6* in vertebrates likewise leads to loss of eye structures; humans with *Pax6* mutations display a condition called congenital aniridia (Figure 21-32). Similar organ-selector proteins are known for foregut, heart, pancreas, and other organs. They are all master transcription regulators that directly regulate hundreds of target genes, the products of which then specify and construct the different elements of the appropriate organ. However, as in the example of *Atoh1*, they usually exert their specific effect only in combination with the right partners, which are only expressed in cells that were appropriately primed during their earlier development.

Notch-mediated Lateral Inhibition Refines Cellular Spacing Patterns

After the establishment of the basic body plan and the generation of organ precursors, many further steps of pattern refinement are required to achieve the adult pattern of terminally differentiated cells in tissues and organs. Lateral

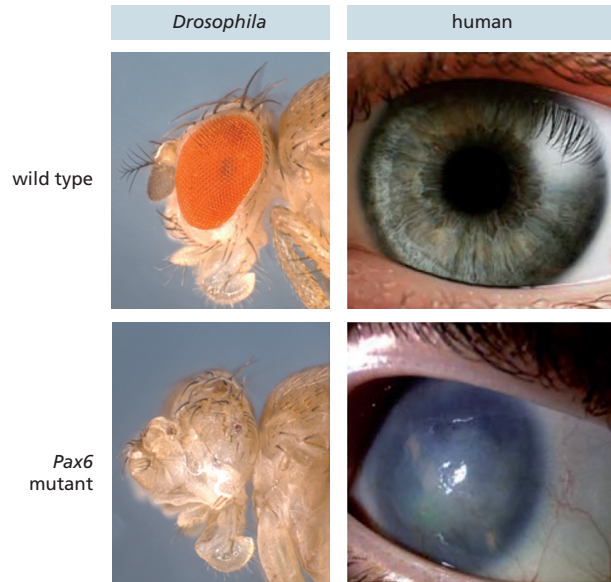


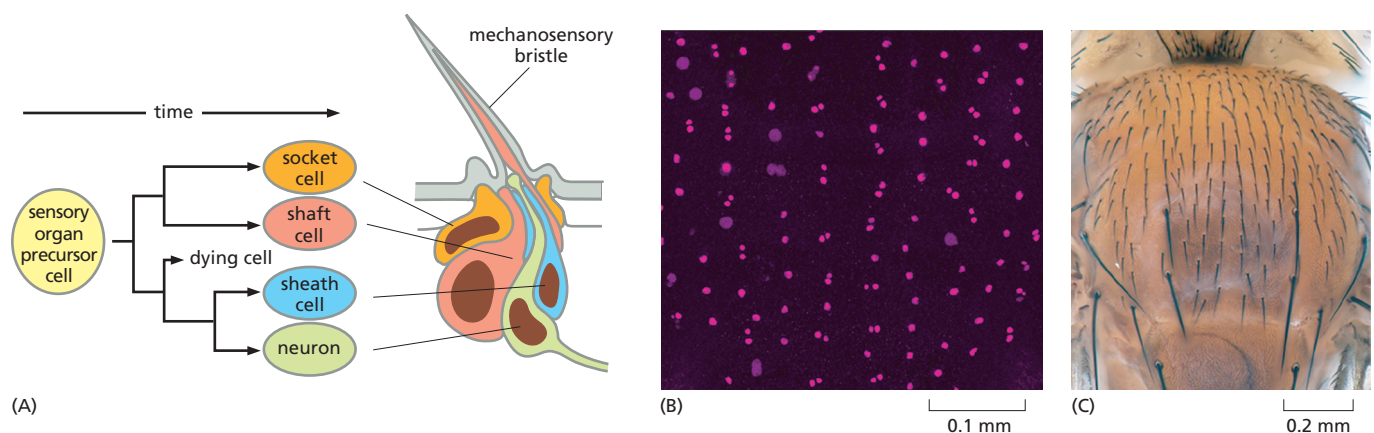
Figure 21-32 The Pax6 protein (also called Eyeless in *Drosophila*) controls development of light-sensing organs in many animal species. Flies or humans carrying mutations in the *Pax6* gene lack certain eye structures present in normal animals. [Fly pictures courtesy of Katy Ong; human pictures from N.L. Washington et al., *PLoS Biol.* 7(11):e1000247, 2009.]

inhibition (see Figure 21-10) mediated by Notch signaling is crucial for both cell diversification and fine-grained patterning in an enormous variety of tissues in all animals.

Certain tissues require the evenly spaced distribution of a specialized cell type, such as sensory cells, secretory cells, or stem cells, across their expanse. A good example is the development of **sensory bristles** in *Drosophila*, most easily seen on the fly's back, but also present on most of its other exposed surfaces. Each of these is a miniature sense organ, consisting of a sensory neuron and a small set of supporting cells. Some bristles respond to chemical stimuli and others to mechanical stimuli, but they are all constructed in a similar way (Figure 21-33).

The proneural genes *Achaete* and *Scute* mentioned earlier mark the clusters of epidermal cells within which bristles will form, but exactly which cells form a bristle depends on competitive interactions among them. Signaling within each cluster selects a single cell—called the *sensory organ precursor cell*—to serve as the bristle progenitor. The progenitor becomes distinct from surrounding cells through lateral inhibition (see Figure 21-10), which is mediated by the Notch signaling pathway. Binding of the transmembrane receptor *Notch* on the surface of a cell in the cluster by its transmembrane ligand *Delta* on a neighboring cell generates signals that inhibit differentiation to form a sensory organ precursor and also inhibit production of Delta in that same cell. At first, all cells in the cluster express both Notch and Delta and inhibit one another from differentiating.

Figure 21-33 *Drosophila* mechanosensory bristles. (A) The lineage of the four cells of the bristle—all descendants of a single sensory organ precursor cell—is shown on the left. The sensory organ precursor, once it is specified, generates this set of cells through a short program of division cycles. In each generation of the progeny, lateral inhibition operates again to drive the newborn cells toward different fates: one of the ultimate progeny cells will become the neuron; another, the shaft of the bristle; others, supporting cells of various sorts. (B) The distribution of sensory organ precursors in the pupal epidermis, seen using a fluorescent reporter for the *senseless* gene (*pink*). (C) The pattern of bristles on the thorax of an adult fly. (B and C, courtesy of François Schweisguth.)



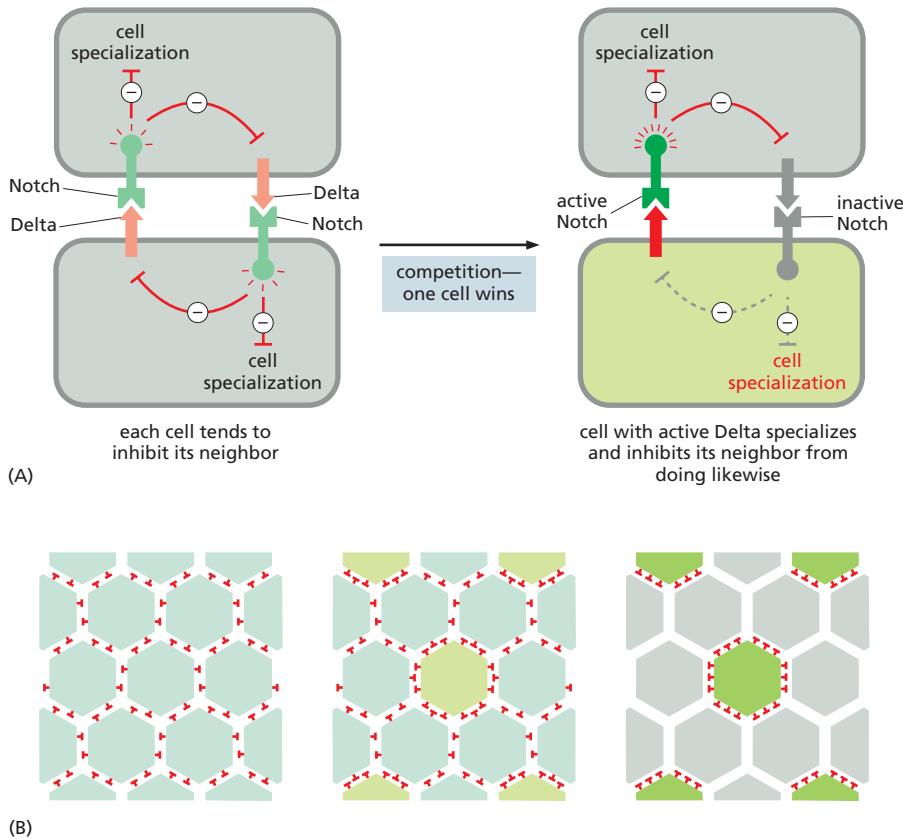


Figure 21–34 Lateral inhibition. (A) The basic mechanism of Notch-mediated competitive lateral inhibition, illustrated for just two interacting cells. Proteins or effector lines shown in gray indicate inactivity. (B) The outcome of the same process operating in a larger patch of cells. Initially, all cells in the patch are equivalent and signal through Delta–Notch interactions to discourage their neighbors from specializing as a sensory organ precursor. This balanced inhibition is disrupted when an individual cell generates a slightly stronger Delta signal than that of its neighbors: it then inhibits its neighbors more strongly while itself becoming more committed to the sensory organ precursor fate. As neighboring cells lose their capacity to differentiate, they also lose their capacity to inhibit other cells from doing so. Lateral inhibition thus makes adjacent cells follow different fates. Although the interaction is thought to be dependent primarily on cell–cell contacts, the sensory organ precursor can also deliver an inhibitory signal to cells that are more than one cell diameter away; for example, by sending out long protrusions to touch them.

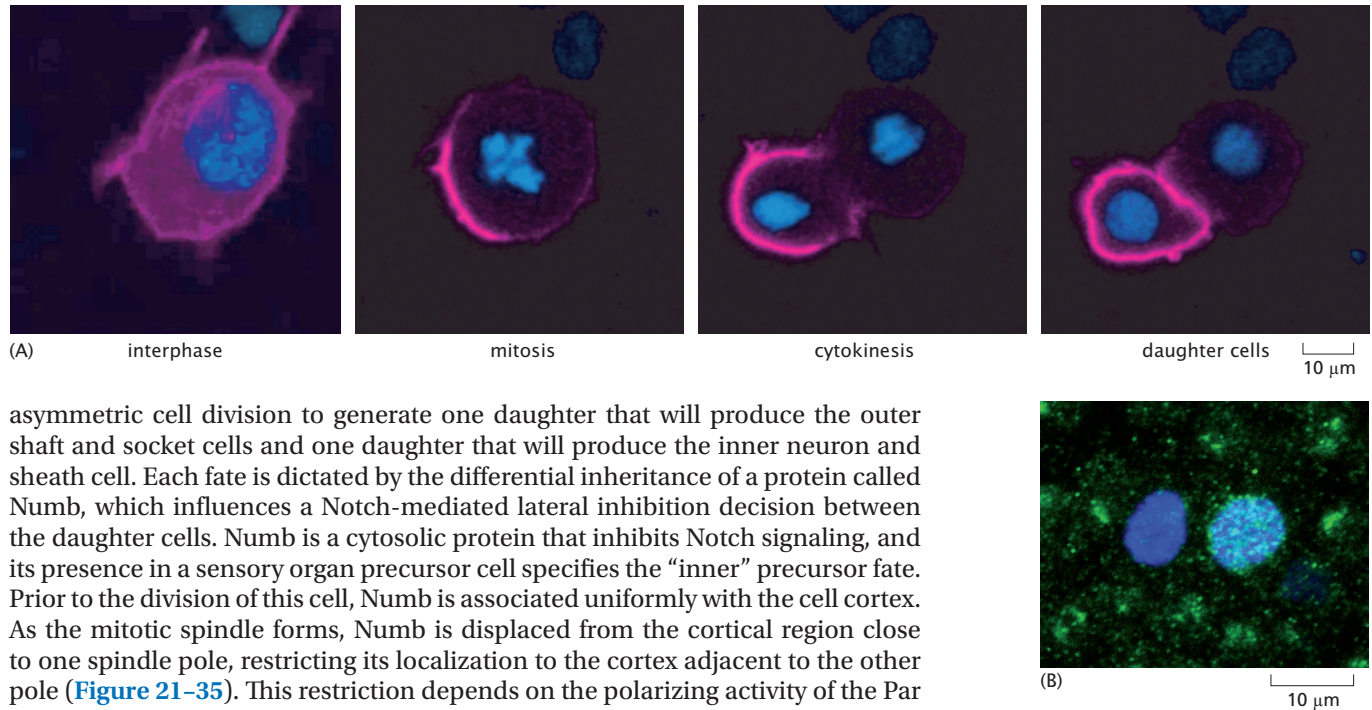
Gradually, however, small signaling differences lead to a positive feedback loop in which a cell producing a slightly stronger Delta signal both discourages its neighbors from differentiating and reduces their production of Delta. The result is a competition, from which a single sensory organ precursor cell eventually emerges as winner, sending a strong inhibitory signal to its immediate neighbors but receiving no such signal in return (Figure 21–34).

The sensory organ precursor goes through a short program of further divisions to generate the set of cells that form the final bristle (see Figure 21–33). Notch signaling acts repeatedly at successive stages in this program to drive the descendants of the sensory organ precursor along different pathways and assign them to their various specialized fates. However, it does so in conjunction with additional mechanisms that bias the outcome of the competition mediated by lateral inhibition. Determinants that are asymmetrically localized inside the dividing cells have this role in sensory bristle development, as we now discuss.

Cell-fate Determinants Can Be Asymmetrically Inherited

Cell diversification does not always depend on extracellular signals: in some cases, daughter cells are born different as a result of an intrinsically asymmetric cell division, during which some significant set of molecules is divided unequally between them. This asymmetrically segregated molecule (or set of molecules) then acts as a determinant for one of the cell fates by directly or indirectly altering the pattern of gene expression within the daughter cell that receives it (see Figure 21–11). A good example of the asymmetric segregation of molecules occurs in the early frog embryo: *VegT* RNA is localized specifically in the vegetal region of the fertilized egg. After the cleavage divisions, only vegetal cells will inherit *VegT* RNA.

Asymmetric divisions often occur at the beginning of development, but they are also encountered at some later stages. To return to our example of the adult *Drosophila* bristle (see Figure 21–33), the sensory organ precursor undergoes an



asymmetric cell division to generate one daughter that will produce the outer shaft and socket cells and one daughter that will produce the inner neuron and sheath cell. Each fate is dictated by the differential inheritance of a protein called Numb, which influences a Notch-mediated lateral inhibition decision between the daughter cells. Numb is a cytosolic protein that inhibits Notch signaling, and its presence in a sensory organ precursor cell specifies the “inner” precursor fate. Prior to the division of this cell, Numb is associated uniformly with the cell cortex. As the mitotic spindle forms, Numb is displaced from the cortical region close to one spindle pole, restricting its localization to the cortex adjacent to the other pole (**Figure 21–35**). This restriction depends on the polarizing activity of the Par protein complex described in Chapter 16 (and see Figure 16–76). When cytokinesis is completed, Numb is present in only one of the two daughters, where it inhibits Notch signaling within that cell—thereby acting as an asymmetrically segregated fate determinant.

Numb operates in an analogous manner during asymmetric divisions of the *Drosophila* neural stem cell lineage to generate the large numbers and balanced proportions of neurons and glia. Asymmetric division of neural progenitors also drives neurogenesis in the cortex of the vertebrate brain. In both cases, Notch signaling determines the difference between a more progenitor-like and a more differentiated fate.

Evolution of Regulatory DNA Explains Many Morphological Differences

In the preceding sections, we have seen that animals contain the same essential cell types, have a similar collection of genes, and share many of the molecular mechanisms of pattern formation. But how can we reconcile these ideas with the radical differences that we see in the body structures of animals as diverse as a worm, a fly, a frog, and a mouse? We asserted earlier, in a general way, that these differences usually seem to reflect differences in the regulatory DNA that controls the conserved basic set of developmental regulatory proteins. We now examine the evidence more closely.

When we compare animal species with similar basic body plans—different vertebrates, for example, such as fish, birds, and mammals—we find that corresponding genes can have similar sets of regulatory elements that are conserved and recognizably homologous in the different animals. The same is true if we compare different species of nematode worms or insects. But, when we compare vertebrate regulatory regions with those of worms or flies, it is hard to see any such resemblance. The protein-coding sequences are unmistakably similar, but the corresponding regulatory DNA sequences appear very different, suggesting that the differences in body plans mainly reflect differences in regulatory DNA. Although variations in the proteins themselves also contribute, differences in regulatory DNA would be enough to generate extremely different tissues and body structures even if the proteins were the same.

It is not yet possible to trace all of the genetic steps that have led to the spectacular diversity of animals. Their lineages have diverged over hundreds of millions

Figure 21–35 Asymmetric cell division creates different daughter-cell fates in the *Drosophila* bristle lineage. (A) Images of a dividing sensory organ precursor cell in the *Drosophila* pupa. Initially, the Numb protein (pink) is distributed uniformly around the cell cortex, but it partitions to one side of the cell during metaphase. Nuclear DNA is stained blue. Division results in the segregation of Numb into only one of the two daughter cells. (B) In this pair of daughter cells, asymmetrically inherited Numb (not shown) prevents Notch signaling in the daughter cell on the left; Notch activity is restricted to the daughter cell on the right, as shown by a reporter of the Notch target gene *Enhancer of Split m8* (green), which appears as speckles in the cell nucleus (stained blue). (A, from J.A. Knoblich, *Nat. Rev. Mol. Cell Biol.* 11:849–860, 2010; B, courtesy of François Schweisguth.)

of years, and in most cases too many changes have occurred for us to be able to say that this or that feature results from this or that mutation. The picture is clearer, however, for more recent evolutionary events. Studies of closely related animal populations and plant populations whose members have different morphologies have revealed that dramatic developmental effects can result from specific changes in regulatory DNA.

A well-studied example is the morphological diversity found in stickleback fish. Marine sticklebacks extend sharp spines from their pelvic skeleton. These spines are thought to help protect the fish from soft-mouthed fish predators that live in the ocean. After the last ice age ended about 10,000 years ago, some marine sticklebacks colonized newly formed freshwater streams and lakes, which often lacked such predators. Many of these populations of freshwater sticklebacks evolved to lose their pelvic spines. The development of this different morphology reflects differences in control of the expression of a transcription regulator called *Pitx1*. Whereas marine sticklebacks express the *Pitx1* gene in the hindlimb bud cells that will give rise to pelvic spikes, freshwater sticklebacks have lost this expression as a result of a change at the *Pitx1* locus. These changes do not lie in the coding sequence. Instead, each is a small deletion of a block of regulatory DNA that controls *Pitx1* expression specifically in the pelvic cells (Figure 21–36).

The *Pitx1* protein has important functions elsewhere in the fish body, for example in the jaw and pituitary gland, and these functions must be retained. The DNA sequences that encode the *Pitx1* protein, as well as the regulatory DNA responsible for tissue-specific *Pitx1* expression at these other body sites, is preserved in both marine and freshwater fish. The evolution of pelvic development

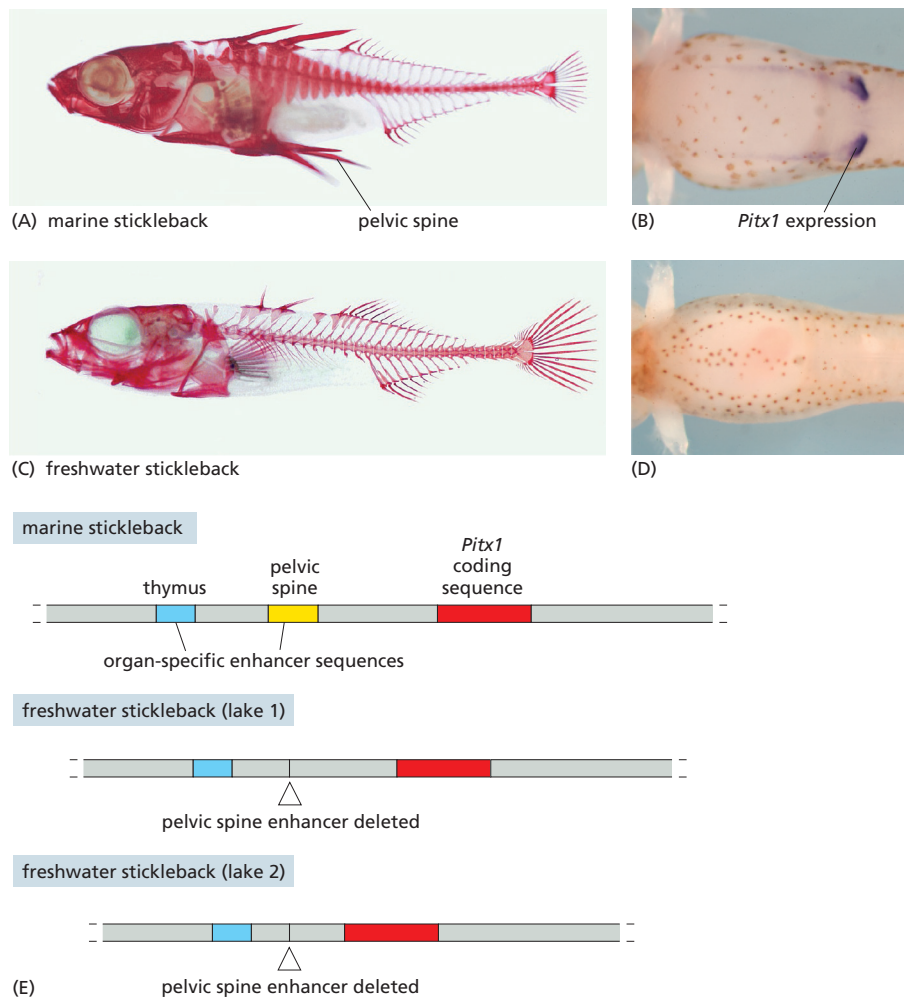


Figure 21–36 Morphological diversity in stickleback fish is caused by changes in regulatory elements. (A–D) Pelvic spines are present in marine (A) but missing in some freshwater (C) populations. Correspondingly, *Pitx1* is expressed in the pelvic area in marine (B) but not in freshwater (D) fish. (E) The lack of expression in the pelvic area of freshwater populations is caused by regulatory deletions that removed pelvic-specific enhancer sequences; these deletions occurred independently in different populations and are of different sizes. The coding sequence and other enhancers and sites of expression for *Pitx1* are conserved in marine and freshwater sticklebacks. (A–D, courtesy of Michael D. Shapiro.)

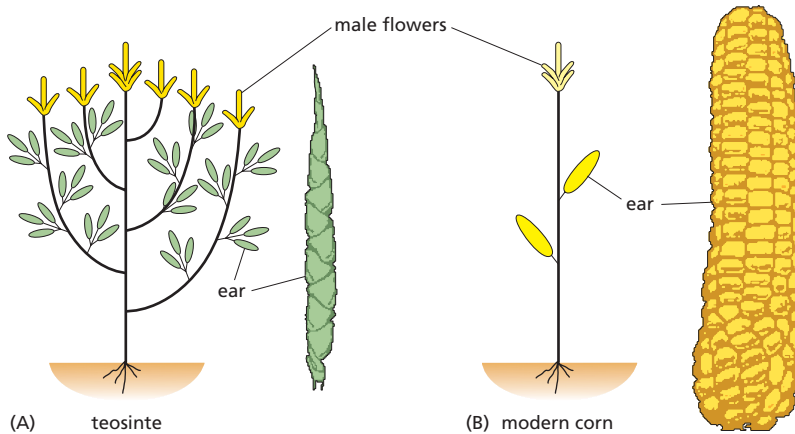


Figure 21-37 The insertion of a mobile genetic element helped produce modern corn. Today's corn plants were originally bred from a wild plant called teosinte (A). This wild ancestor produced numerous ears that contained small, hard seeds. (B) Modern corn, by contrast, produces fewer ears—but they contain numerous, plump, sweet seeds (kernels). The insertion of a mobile genetic element in the regulatory region of a gene involved in seed development helped drive the change. Here, the two plants are drawn to the same scale; for simplicity, the leaves are not shown.

in sticklebacks shows how the modular nature of regulatory DNA elements that we encountered in Chapter 7 (see Figure 7-31) allows independent modification of the different parts of the body, even when formation of multiple body parts depends on the same proteins.

In the recent evolution of plants, changes in their structure can be traced in a similar way to changes in regulatory DNA. In an important example for agriculture, such changes account for a large part of the dramatic difference between the wild teosinte plant and its modern descendant, maize, through some 10,000 years of selection upon natural genetic variation by Native Americans (Figure 21-37).

Summary

Drosophila has been the foremost model organism for the study of the genetics of animal development. Its embryonic pattern is initiated by the products of maternal-effect genes called egg-polarity genes, which operate by setting up graded distributions of transcription regulators in the egg and early embryo. The gradient of Bicoid protein along the anteroposterior (A-P) axis, for example, helps initiate the orderly expression of gap genes, pair-rule genes, and segment-polarity genes. These three classes of segmentation genes, through a hierarchy of interactions, become expressed in some regions of the embryo and not others, progressively subdividing an initially uniform field of cells along the A-P axis into a regular series of repeating modular units called segments.

Superimposed on the pattern of gene expression that repeats itself in every segment, there is a serial pattern of expression of Hox genes that confers on each segment a different identity. These genes are grouped in complexes and are arranged in a sequence that matches their sequence of expression along the A-P axis of the body.

Although Hox gene expression is initiated in the embryo, it is subsequently maintained by the action of chromatin-binding proteins of the Polycomb and Trithorax group, which stamp the chromatin of the Hox complex with a heritable record of its embryonic state of repression or activation, respectively. Hox complexes homologous to that of *Drosophila* are found in virtually every type of animal, where they help pattern the A-P axis of the body.

Signaling gradients are also set up along the dorsoventral (D-V) axis. Initially, Toll signaling generates a nuclear gradient of Dorsal protein, which induces an extracellular signaling gradient of the TGF β family protein Dpp and its antagonist, Sog. This creates a gradient of Dpp activity that helps refine the assignment of different characters to cells at different positions along the D-V axis.

In *Xenopus*, the polarity of the egg and the site of sperm entry set up the embryonic axes. A gradient generated by the TGF β family protein Nodal induces different fates along the animal-vegetal axis, whereas BMP and Chordin—proteins homologous to *Drosophila* Dpp and Sog, respectively—control the patterning of the D-V axis.

Transcription regulators control the formation of specific cell types. Members of the MyoD family drive the process of muscle-cell determination, coordinating the many components required, whereas Achaete/Scute transcription regulators control neural fate. Other genes encoding such master transcriptional regulators can regulate the formation of entire organs. Eyeless, for example, is both necessary and sufficient to generate eye structures in Drosophila.

To refine the anatomical pattern within such an organ, the cells interact locally, both by diffusible inductive signals and by short-range mechanisms. Often, the cells compete with one another by lateral inhibition. This process results in activation of the Notch signaling pathway in one cell and inhibition in its neighbors, generating two different cell types. Asymmetric cell divisions, in which daughter cells inherit different molecular determinants from the mother cell, provide an additional way to organize a fine-grained diversity of cell types.

Evidence from recent evolutionary events indicates that anatomical changes are mostly driven by changes in regulatory DNA sequences that determine when and where developmental genes are expressed. How the striking diversity in body structures has evolved over longer times remains largely unknown, although it seems likely that similar principles apply.

DEVELOPMENTAL TIMING

Developmental events unfold over minutes, hours, days, weeks, months, or even years, with each organism following its own strict timetable. The cascades of inductive interactions and transcriptional regulatory events described earlier take time, as signals are transmitted and transcription regulators are synthesized and then bind to DNA to activate or repress their target genes. So too do the changes in cell size, shape, and organization that result from changes in gene expression. All of these events require correct parameters of timing, including order (the sequence of events), interval (the time between events), and, in some cases, rhythm (repeated cycles of an event). Each developmental process must thus occur at an appropriate rate, tuned by evolution to fit with the timing of other processes in the embryo or in the environment. The control of timing is a compelling problem in developmental biology, but one of the least understood.

Molecular Lifetimes Play a Critical Part in Developmental Timing

Developmental processes are complex, but they are built up from simple steps. A first challenge is to understand the timing of these steps. How long does it take, for example, to switch the expression of a gene on or off? This is not like throwing a light switch: it involves delays. First, it takes time to make an mRNA molecule: the RNA polymerase must travel the length of the gene, the primary RNA transcript must be spliced and otherwise processed, and the resulting mRNA must be exported from the nucleus and delivered to the site where it will be translated. This adds up to what one might call the *gestation time* of the individual molecule. Second, it takes time for the individual mRNA molecules to accumulate to their fully effective concentration; as explained in Chapter 15, this *accumulation time* is dictated by the average lifetime of the molecules—the longer they last, the higher their ultimate concentration, and the longer the time taken to attain it. Similar delays occur at the next step, where the mRNA is translated into protein: synthesis of each individual protein molecule involves a gestation delay, and attainment of an effective concentration of protein molecules involves an accumulation delay that depends on the time it takes for each protein to fold into its three-dimensional structure and its lifetime once produced. The time for the whole gene switching process is the sum of the gestation delays and the accumulation delays (basically, the molecular lifetimes) for both the mRNA and the protein molecules. Somewhat counterintuitively, it is the combined length of these delays, rather than the rate of molecular synthesis (the number of molecules synthesized per second), that chiefly determines the switching time.

The same additive principle applies to long cascades of gene switching, where gene *A* activates gene *B*, and gene *B* activates gene *C*, and so on. It also applies in other circumstances, such as in signaling pathways where one protein directly regulates the activation of the next. In all these cases, molecular lifetimes, along with gestation delays, play a key part in determining the pace of development. The lifetimes of mRNA and protein molecules are enormously variable, from a few minutes or hours to days or more, explaining much of the tempo of developmental events.

Gene switching delays, however, are not the be-all and end-all of developmental timing. Development involves many other kinds of delay that contribute to timing. Chromatin structure takes time to remodel. Inductive signals take time to diffuse across a field of cells (see Figure 21-9). Cells take time to move and rearrange themselves in space. Nevertheless, the timing of gene switching plays a fundamental part in developmental timing, as illustrated in an especially clear and striking way by a gene expression oscillator that controls the segmentation of the vertebrate body axis, as we now explain.

A Gene Expression Oscillator Acts as a Clock to Control Vertebrate Segmentation

The main body axis of all vertebrates has a repetitive, periodic structure, seen in the series of vertebrae, ribs, and segmental muscles of the neck, trunk, and tail. These segmental structures originate from the mesoderm that lies as a long slab on either side of the embryonic midline. This slab becomes broken up into a regular repetitive series of separate blocks, or **somites**—cohesive groups of cells, separated by clefts (Figure 21-38A). In contrast to the segments of a *Drosophila* embryo that appear simultaneously, as we discussed previously, somites in the vertebrate are formed sequentially. The somites form (as bilateral pairs) one after another, in a regular rhythm, starting in the region of the head and ending in the tail. Depending on the species, the final number of somites ranges from fewer than 40 (in a frog or a zebrafish) to more than 300 (in a snake).

The cells that will form somites are generated in the posterior, most immature part of the mesodermal slab, called the *presomitic mesoderm*. This is a proliferative zone, and as new cells are generated, the zone is pushed tailward, extending

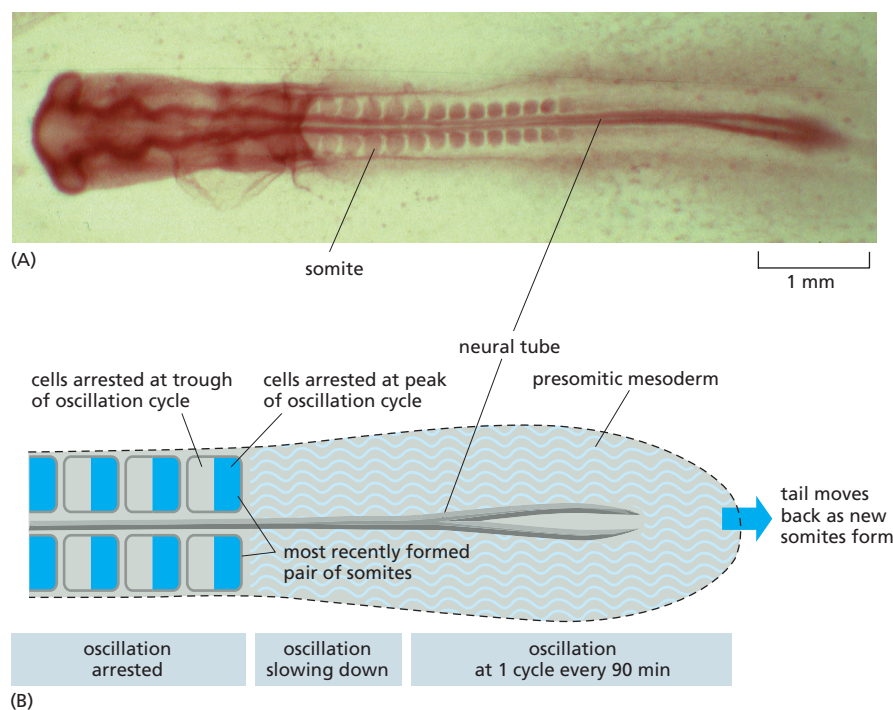


Figure 21-38 Somite formation in the chick embryo. (A) Brightfield image of a chick embryo at 40 hours of incubation. (B) How the temporal oscillation of gene expression in the presomitic mesoderm becomes converted into a spatial alternating pattern of gene expression in the formed somites. In the posterior part of the presomitic mesoderm, each cell oscillates with a cycle time of 90 minutes. As cells mature and emerge from the presomitic region, their oscillation is gradually slowed down and finally brought to a halt, leaving them in a state that depends on the phase of the cycle they happen to be in at the critical moment. In this way, a temporal oscillation of gene expression traces out an alternating spatial pattern. (Movie 21.5.) (A, from Y.J. Jiang et al., *Curr. Biol.* 8:R868–R871, 1998. With permission from Elsevier.)

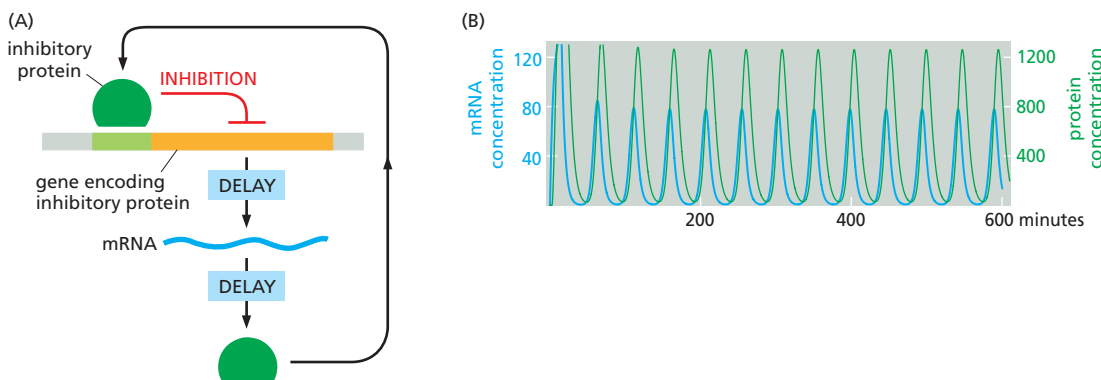
the embryo (**Figure 21-38B**). As the presomitic mesoderm moves tailward, it deposits a trail of somites formed from cells that group together into blocks as they emerge from the anterior end of the presomitic region. The special character of the presomitic mesoderm is maintained by a combination of fibroblast growth factor (FGF) and Wnt signals, produced by a signaling center at the tail end of the embryo, and the range of these signals seems to define the region from which somites can be formed. The somites emerge with clocklike timing, but what determines the rhythm of the process?

In the posterior part of the presomitic mesoderm that is destined to be segmented, the expression of certain genes oscillates in time. Such oscillations can be observed in time-lapse movies of embryos containing fluorescent reporters of individual genes. One new somite pair is formed in each oscillation cycle, and, in mutants where the oscillations fail to occur, somite segmentation is disrupted: the cells may still break up, belatedly, into separate clusters, but they do so in a haphazard, irregular way. The gene expression oscillator controlling regular segmentation is called the **segmentation clock**. The length of one complete oscillation cycle depends on the species: it is 30 minutes in a zebrafish, 90 minutes in a chick, 120 minutes in a mouse, and 300 minutes in a human.

As cells emerge from the presomitic mesoderm to form somites—in other words, as they escape from the influence of the FGF and Wnt signals—their oscillating gene expression stops. Some become arrested in one state, some in another, according to the phase of the oscillation cycle at the time they leave the presomitic region. In this way, the temporal oscillation of gene expression in the presomitic mesoderm leaves its trace in a spatially periodic pattern of gene expression in the maturing mesoderm. This in turn dictates how the tissue will break up into physically separate blocks, through effects on the pattern of cell–cell adhesion.

How does the segmentation clock work? The first somite oscillator genes to be discovered were *Hes* genes, which code for inhibitory transcription regulators. As well as regulating other genes, the products of *Hes* genes can directly regulate their own expression, creating a simple negative feedback loop. Auto-regulation of *Hes* genes is thought to be the basic generator of the oscillations of the somite clock. Although the machinery has been modified in various ways in different species, the underlying principle seems to be conserved. When the key *Hes* gene is transcribed, the amount of Hes protein builds up until it is sufficient to block *Hes* gene transcription. Synthesis of the protein then ceases, and as the protein decays, transcription is permitted to begin again; and so on, cyclically (**Figure 21-39**). The period of oscillation, which determines the size of each somite, depends on the delay in the feedback loop. This delay equals the sum of the gestation delays and accumulation delays (that is, the molecular lifetimes) of the *Hes* mRNA and protein molecules, according to the principle discussed earlier. Mathematical modeling (see Chapter 8) allows us to relate these basic molecular parameters to the cycle time of the segmentation clock: to a first approximation, the cycle period is simply equal to twice the total delay in the negative feedback loop, and thus twice the sum of the delays occurring

Figure 21-39 Delayed negative feedback giving rise to oscillating gene expression. (A) A single gene, coding for a transcription regulator that inhibits its own expression, can behave as an oscillator. For oscillation to occur, there must be a delay (or several delays) in the feedback circuit, and the lifetimes of the mRNA and protein (which contribute to the delay) must be short compared with the total delay. The total delay determines the period of oscillation. It is thought that a feedback circuit like this, based on a pair of redundantly acting genes called *Her1* and *Her7* in the zebrafish—or their counterpart, *Hes7*, in the mouse—is the pacemaker of the segmentation clock governing somite formation. (B) The predicted oscillation of *Her1* and *Her7* mRNA and protein, computed using rough estimates of the feedback circuit parameters appropriate to this gene in the zebrafish. Concentrations are measured as numbers of molecules per cell. The predicted period is close to the observed period, which is 30 minutes per somite in the zebrafish.



at each step of the loop. Moreover, an experimental manipulation that changes one of these parameters (for example, altering the size of a *Hes* intron to perturb mRNA-processing time) results in the predicted effect on the period of somite formation.

The feedback loop just described is intracellular, and each cell in the presomitic mesoderm can generate oscillations on its own, even when cultured in a dish. But these oscillations at the single-cell level are somewhat erratic and imprecise, reflecting the fundamentally noisy, stochastic nature of the control of gene expression, as discussed in Chapter 7. A mechanism is needed to keep all the cells in the presomitic mesoderm that will form a particular somite oscillating in synchrony. This is achieved in part through cell-cell communication via the Notch signaling pathway. In this context, Notch signaling does not drive neighboring cells to be different, as in lateral inhibition, but does just the opposite: it keeps them in unison. In mutants where Notch signaling fails, including mutants defective in Delta or Notch itself, the cells drift out of synchrony, and somite segmentation is again disrupted. This leads to gross deformity of the vertebral column—an extraordinary display of the consequences of the noisy temporal control of gene expression at the single-cell level, writ large in the structure of the vertebrate body as a whole.

Cell-intrinsic Timing Mechanisms Can Lead to Different Cell Fates

Although signaling between cells plays an essential part in driving the progress of development, this does not mean that cells always require signals from other cells to prod them into changing their character as development proceeds. Some of these changes are intrinsic to the cell (like the ticking of a circadian clock, discussed in Chapter 15) and depend on *intracellular developmental programs* that can operate even when the cell is removed from its normal environment.

The best-understood example is the development of neuroblasts, the stem cells of the *Drosophila* central nervous system. These cells go through a succession of asymmetric cell divisions to generate distinct types of neurons and glial cells, always in the same sequence and with the same timing. Each neuroblast produces this wide range of daughter-cell fates using a time-dependent developmental program. As the neuroblast goes through its set schedule of divisions, it successively changes its internal state by expressing a series of different transcription regulators. For example, most embryonic neuroblasts sequentially express the transcription regulators Hunchback, Krüppel, Pdm, and Castor in a fixed order (**Figure 21-40**). When a neuroblast divides, the transcription regulators expressed at that moment are inherited by and maintained in its progeny; thus, the differentiated neural cells are endowed with different characters according to their time of birth.

When neuroblasts are taken from an embryo and maintained in culture, isolated from their normal surroundings, they step through much the same stereotyped developmental program as if they had been left in the embryo. Moreover, many of the neuroblast transitions occur even when cell division is blocked. The neuroblasts seem to have a built-in timer that determines when each of the transcription regulators is expressed, and this timer can continue to run in the absence of cell-cycle progression. The molecular basis of the timing is not well understood. Because cross-regulatory transcriptional cascades are involved, it likely depends on the time taken for gene switching, as described earlier, but it might also depend on slow progressive changes in chromatin structure and genome organization, which also can serve to measure the passage of time in the embryo.

Timing mechanisms can coordinate cell fates with the spatial organization of tissues. For example, during development of the mammalian cerebral cortex, the ordered expression of different transcription factors in the progenitor cells that divide to produce neurons and glial cells leads to sequential production of specific neural cell types. The newborn neurons migrate to their appropriate position

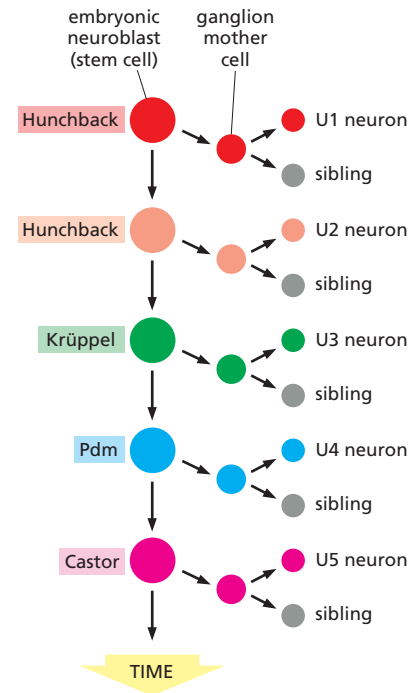


Figure 21–40 Temporal patterning of neuroblast fate in *Drosophila* embryos. Each neuroblast undergoes an asymmetric cell division to renew itself and to produce a daughter called a ganglion mother cell. The ganglion mother cell subsequently divides to produce a pair of terminally differentiated neurons or glia. The types of neurons and glia are determined by the expression of a transcription regulator in the parent neuroblast that is then inherited by its ganglion-mother-cell daughter. Each neuroblast will sequentially express the transcription regulators Hunchback, Krüppel, Pdm, and Castor as it progresses through its lineage. After dividing several times while expressing one of these regulators, the neuroblast switches its pattern of gene expression, expressing an updated set of genes that will be passed along to its daughter cells, which differentiate into specific cell types accordingly. (After B.J. Pearson and C.Q. Doe. *Nature* 425:624–628, 2003.)

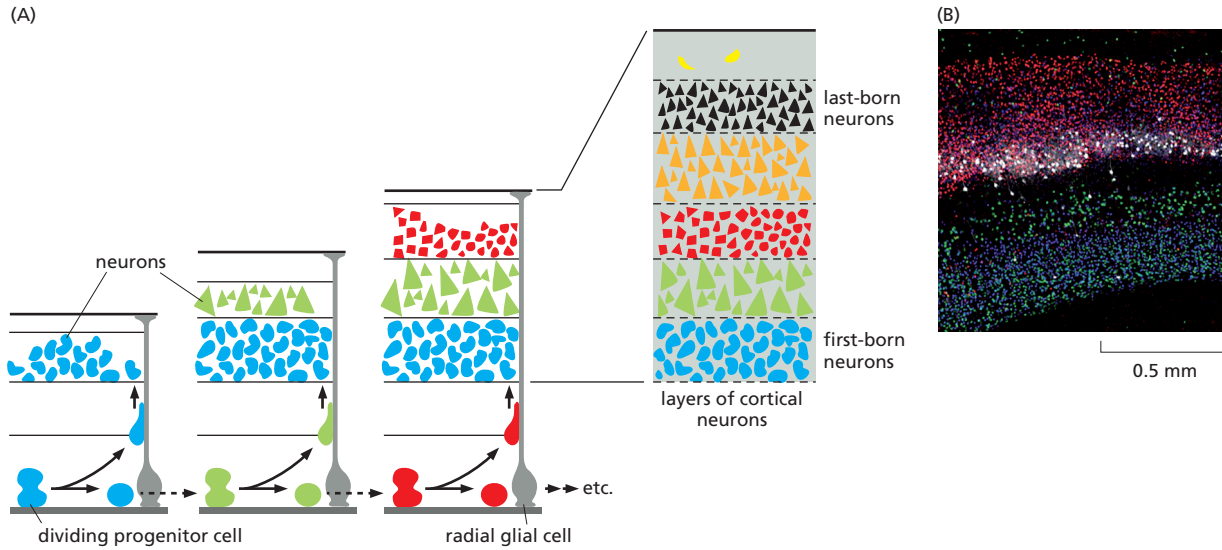


Figure 21-41 Timed expression of neural genes drives laminar organization of the brain. (A) Many neurons and glial cells in the mammalian cerebral cortex develop from a common progenitor, which proliferates on the inner surface of the cortical neuroepithelium to produce successive generations of neurons (colored here *blue*, *green*, *red*, *orange*, and *black*). The developing neurons migrate outward, crawling along processes that extend from radial glial cells, which also develop from the same progenitor cells. The first-born neurons settle closest to their birthplace, while neurons born later crawl past them to settle farther out. Successive generations of neurons thus occupy different layers in the cortex and have different intrinsic characters according to their birth dates. (B) Fluorescence micrograph showing the laminar organization of excitatory neurons labeled with antibodies to different transcription factors (*blue*, *green*, *white*, and *red*) in the mammalian cerebral cortex. Neural cell types are specified by the temporal sequence of transcription factor expression in their progenitors, so that birth order specifies both the fate and location of differentiated neurons. (From I. Holguera and C. Desplan, *Science* 362: 176–180, 2018.)

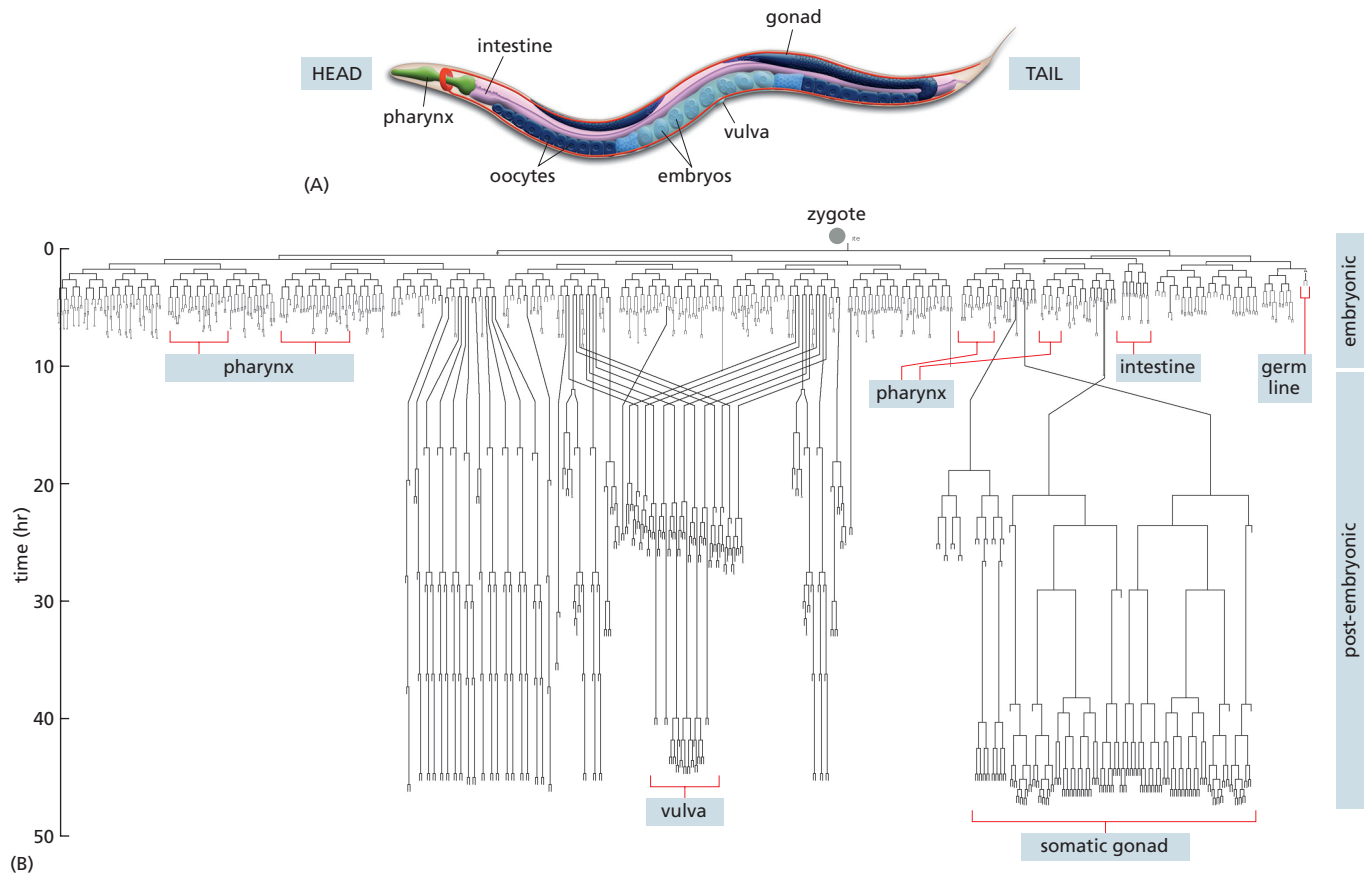
in the developing cortex, ultimately occupying different layers of the adult brain. Thus, all neurons that populate a layer form in the same time window and have common fates (Figure 21-41).

Cells Rarely Count Cell Divisions to Time Their Development

Many specialized cells develop from proliferating progenitor cells that stop dividing and terminally differentiate after a limited number of cell divisions. In these cases, it is tempting to speculate that the cell-division cycle serves as an intracellular timer to control the timing of cell differentiation. The cell cycle would be the ticking clock that sets the tempo of other developmental processes, with maturational changes in gene expression being dependent on cell-cycle progression. Most of the evidence, however, indicates that this appealing idea is wrong. Although there are examples where cells change their maturation state with each division and the change depends on cell division, this is not the general rule. As we just saw for neuroblasts in the *Drosophila* embryo, cells in developing animals often carry on with their normal timetable of maturation and differentiation even when cell division is artificially blocked; necessarily, some abnormalities occur, if only because a single undivided cell cannot differentiate in two ways at once. But it seems that most developing cells can change their developmental state without a requirement for cell division. Developmental control genes can switch the cell-division-cycle machinery on or off, and it is the dynamics of these genes, rather than the cell cycle, that sets the tempo of development.

MicroRNAs Can Regulate Developmental Transitions

Because genetic screens are useful for tracking down the genes involved in almost any biological process, they have been used to search for mutations that alter developmental timing. Such screens were performed in the nematode worm



Caenorhabditis elegans (Figure 21–42). This small creature follows an astonishingly precise and predictable developmental program that has been described in extraordinary detail, so that one can map out the exact lineage of every cell in the body. The time and place of each cell division, as well as the fate of each of its daughters, are invariant from one individual to the next.

A great advantage of the lineage diagram describing the normal development of *C. elegans* is that one can see exactly how the developmental program is altered in a mutant. Genetic screens in *C. elegans* revealed mutations that disrupt developmental timing in a particularly striking way: in these so-called **heterochronic** mutants, certain cells in a larva at one stage of development behave as though they were in a larva at a different stage of development or cells in the adult carry on dividing as though they belonged to a larva (Figure 21–43).

Genetic analyses showed that the products of the heterochronic genes act in series, forming regulatory cascades. Unexpectedly, two genes at the top of their respective cascades, called *Lin4* and *Let7*, were found to code not for protein but instead for **microRNAs (miRNAs)**—short, untranslated, regulatory RNA molecules, 21 or 22 nucleotides long. These act by binding to complementary sequences in the noncoding regions of mRNA molecules transcribed from other heterochronic genes, thereby repressing their translation and promoting their degradation, as discussed in Chapter 7. The *Lin4* miRNA binds to the 3' untranslated region of mRNA produced by the heterochronic gene *Lin14*, which itself promotes early-stage larva-cell behaviors. A developmentally regulated increase of *Lin4* levels thus governs the progression from early to late-stage larva-cell behaviors, by gradually reducing levels of Lin14 protein. Increasing levels of *Let7* miRNA govern the progression from late larva to adult in an analogous way by regulating a different target gene called *Lin41*. In fact, *Lin4* and *Let7* were the first miRNAs to be described in any animal: it was through developmental genetic studies in *C. elegans* that the importance of this whole class of molecules for gene regulation in animals was discovered.

Figure 21–42 Cell lineage in *Caenorhabditis elegans*. (A) Some of the major tissues in the adult worm (see Figure 1–42). (B) The complete cell lineage of the worm was determined by meticulous analysis of the fate of every daughter cell after each cell division, beginning with the single-celled zygote and ending in the 959 cells of the adult. Each vertical line represents a single cell, with branch points reflecting a cell division. Labels indicate the cells in the lineage that make up the tissues shown in panel A. (A and B, courtesy of D.H. Hall, Worm Atlas from <https://www.wormatlas.org/celllineages.html>.)

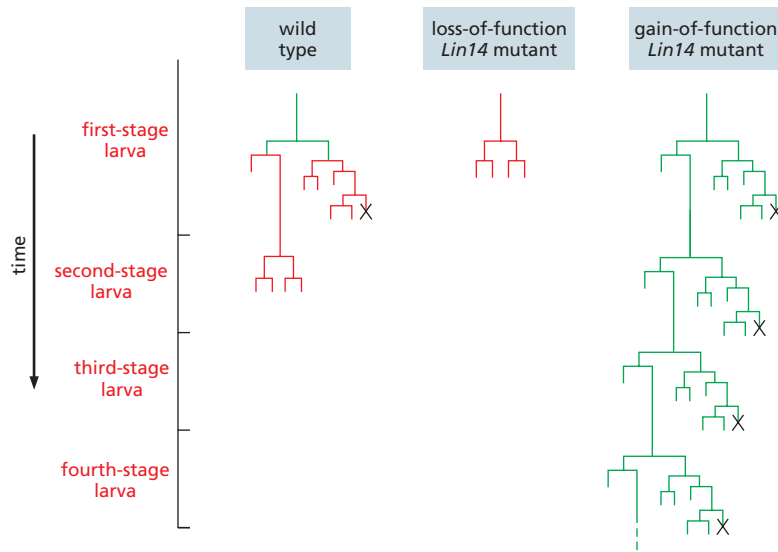


Figure 21-43 Heterochronic mutations in the *Lin14* gene of *C. elegans*. Only the effects on one of the many altered lineages are shown. A loss-of-function (recessive) mutation in *Lin14* causes premature occurrence of the pattern of cell division and differentiation characteristic of a late larva, so that the animal reaches its final state prematurely and with an abnormally small number of cells. The gain-of-function (dominant) mutation has the opposite effect, causing cells to reiterate patterns of cell divisions characteristic of the first larval stage, continuing through as many as five or six molt cycles. The *Lin14* gain-of-function mutations can result from altering the binding site for the *Lin4* miRNA, rendering *Lin14* protein immune from down-regulation. The *black X* denotes a programmed cell death. *Green lines* represent cells that contain *Lin14* (which binds to DNA), *red lines* those that do not. (Adapted from V. Ambros and H.R. Horvitz, *Science* 226:409–416, 1984; and P. Arasu et al., *Genes Dev.* 5:1825–1833, 1991.)

Cell and Nuclear Size Relationships Schedule the Onset of Zygotic Gene Expression

As we discussed earlier, many gene products are deposited in the egg by the mother, allowing the first steps of development to begin immediately upon fertilization. Maternal supplies eventually run out, however, and the zygote begins to produce its own gene products. This **maternal-zygotic transition (MZT)** occurs in most organisms after a substantial delay and marks the temporal window during which the embryo's own genome largely takes over control of development from maternal macromolecules (Figure 21-44).

The MZT occurs in most species shortly before gastrulation, when rapid and synchronous cleavage divisions give way to slower and more conventional cell cycles. During this transition, thousands of genes initiate transcription from the previously silent zygotic genome, and many maternal gene products are cleared from the embryo. In several species, including frogs, fish, and flies, some of the earliest zygotic genes to be transcribed include specific miRNAs that target maternal mRNAs for translational repression and degradation. Here again, miRNAs function to sharpen developmental transitions by blocking and removing mRNAs that define an earlier developmental stage.

How is the timing of the MZT controlled? One trigger appears to be the nuclear-to-cytoplasmic ratio. Many animal embryos do not grow or change volume during the earliest stages of development. Rather, rapid rounds of DNA replication and mitosis transform the fertilized egg into thousands of smaller cells. During these cleavage divisions, the total amount of cytoplasm in the embryo remains constant, but the number of cell nuclei in the embryo increases exponentially, thereby increasing the ratio of DNA to cytoplasm. Strikingly, haploid embryos containing

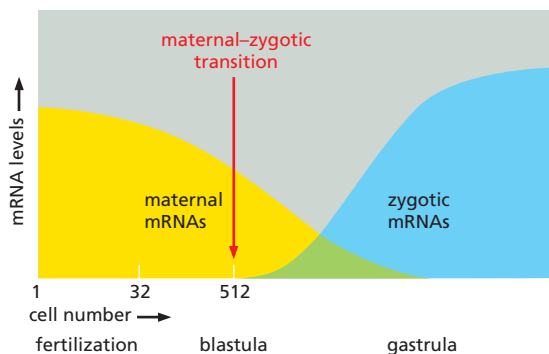


Figure 21-44 The maternal-zygotic transition in a zebrafish embryo. Maternal mRNAs are deposited by the mother into the egg and drive early development. These mRNAs are degraded during different stages of embryogenesis, including blastula and gastrula stages, but a relatively abrupt change occurs at the maternal-zygotic transition (MZT). Before this, the embryonic (zygotic) genome is transcriptionally inactive; afterward, zygotic genes start to be transcribed. In zebrafish embryos, the zygotic genome begins to be activated at the 512-cell stage.

half the amount of DNA per cell undergo the MZT one cell cycle later than normal diploid embryos. According to one model, the nuclear-to-cytoplasmic ratio might be measured through the titration of a maternally provided transcription repressor against the increasing amount of nuclear DNA. The total amount of repressor would stay constant during cleavage divisions, but the amount of repressor per genome would decrease by half with each round of DNA synthesis until a threshold is reached that allows the zygotic genome to become transcriptionally active. One candidate for the repressor is maternal histone proteins, whose concentration regulates MZT timing in several organisms. Other evidence suggests that certain maternally provided transcription regulators trigger the MZT after they accumulate to some threshold sufficient to activate their gene targets. Different species may rely on either or both mechanisms to time activation of the zygotic genome appropriately.

Hormonal Signals Coordinate the Timing of Developmental Transitions

We have so far emphasized timing mechanisms that operate locally and separately in the different parts of the embryo or in specific subsystems of the molecular control machinery. Evolution has tuned each of these largely independent processes to run at an appropriate rate, matched to the needs of the organism as a whole. For some purposes, however, this is not enough: a global coordinating signal is required. This is especially true where changes have to occur throughout the body in response to a cue that depends on the environment. For example, when an insect or amphibian undergoes *metamorphosis*—the transition from larva to adult—almost every part of the body is transformed. The timing of metamorphosis depends on external factors such as the supply of food, which determines when the animal reaches an appropriate size. All the bodily changes have to be triggered together at the right time, even though they are occurring in widely separated sites. The coordination in such cases is provided by **hormones**—signal molecules that spread throughout the body.

The metamorphosis of amphibians provides a spectacular example. During this developmental transition, amphibians switch from an aquatic to a terrestrial life. Larva-specific organs such as gills and tail disappear, and adult-specific organs such as legs form. This dramatic transformation is triggered by thyroid hormone, produced in the thyroid gland. If the gland is removed or if thyroid hormone action is blocked, metamorphosis does not occur, although growth continues, producing a giant tadpole. Conversely, a dose of thyroid hormone given to a tadpole by an experimenter can trigger metamorphosis prematurely.

Thyroid hormone is distributed throughout the body by the vascular system and induces changes in receiving cells by binding to intracellular nuclear hormone receptors, which regulate hundreds of genes. This does not mean, however, that target tissues all respond in the same way to the hormone: organs differ not only in their levels of thyroid hormone receptors and levels of extracellular proteins that locally regulate the amount of active hormone, but also in the sets of genes that respond. Thyroid hormone induces muscle in the limbs to grow and muscle in the tail to die. The timing of the responses also differs; for example, the legs form early in response to a very low concentration of circulating hormone, but it requires a high level of the hormone to induce resorption of the tail.

A surge of thyroid hormone triggers metamorphosis, but how is the timing of the surge controlled? One mechanism depends on coupling hormone synthesis to the size of the thyroid gland, which reflects the size of the tadpole. Only when the gland attains a certain size does it produce enough thyroid hormone to initiate metamorphosis. However, environmental cues other than nutrition also play a part: conditions such as temperature and light are sensed by the nervous system, which regulates the secretion of another tier of hormones (neurohormones) that stimulate the secretion of thyroid hormone. Thus, tadpole-intrinsic factors such as size combine with environmental factors to determine when metamorphosis begins.

Environmental Cues Determine the Time of Flowering

Another striking example of environmentally controlled developmental timing is the flowering of plants. Flowering involves a transformation of the behavior of the cells at the growing apex of the plant shoot—the *apical meristem*. During ordinary vegetative growth, these cells behave as stem cells, generating a steady succession of new leaves and new segments of stalk. In flowering, the meristem cells switch to making the components of a flower, with its sepals and petals, its stamens carrying pollen, and its ovary containing the female gametes.

To time the switch correctly, the plant has to take account of both past and present conditions. One important cue, for many plants, is day length. To sense this, the plant uses its circadian clock—an endogenous 24-hour rhythm of gene expression—to generate a signal for flowering only when there is light for the appropriate part of the day. The clock itself is influenced by light, and the plant in effect uses the clock to compare past to present lighting conditions. Important parts of the genetic circuitry underlying these phenomena have been identified, including the phytochromes and cryptochromes that act as light receptors (discussed in Chapter 15). The flowering signal that is carried from the leaves to the stem cells via the vasculature depends on the product of a gene called *Flowering locus T* (*Ft*).

But this signal will trigger flowering only if the plant is in a receptive condition from prior long-term cold exposure. Many plants need winter before they will flower—a process called *vernalization*. Cold over a period of weeks or months progressively reduces the level of expression of a remarkable gene called *Flowering locus C* (*Flc*). *Flc* encodes a transcriptional repressor that suppresses expression of the *Ft* flowering promoter.

How does vernalization shut down *Flc* so as to lift the block to flowering? The effect involves at least three long noncoding RNAs, including an antisense transcript called *Coolair* that overlaps with the *Flc* gene and is produced when the temperature is low (Figure 21–45). Together with cold-induced chromatin modifiers, including Polycomb-group proteins, *Coolair* coordinates the switching of *Flc* chromatin to a silent state (discussed in Chapters 4 and 7). The degree of silencing depends on the length of cold exposure, enabling the plants to distinguish the odd chilly night from the whole of winter.

The effect on the chromatin is long lasting, persisting through many rounds of cell division even as the weather grows warmer. Thus vernalization creates a

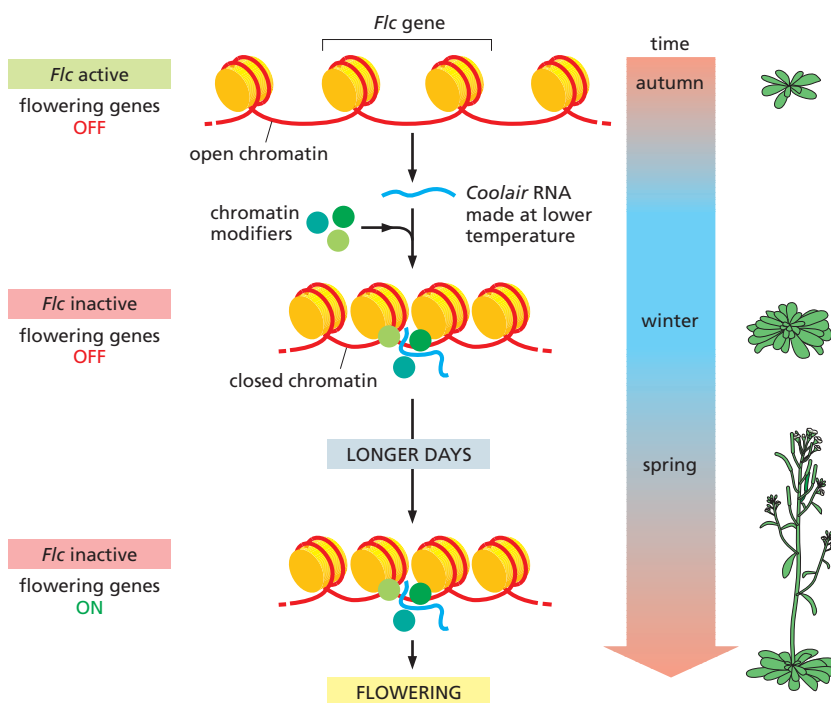


Figure 21–45 Temporal control of flowering in *Arabidopsis*. The *Flc* gene is active and blocks flowering when plants have been grown without exposure to winterlike temperatures. Exposure to a prolonged period of cold leads to the production of the noncoding RNA *Coolair*, which overlaps with the *Flc* gene. *Coolair* induces long-term chromatin changes that turn off *Flc*. These changes persist after the end of the cold period and allow the plant to flower when longer days promote flowering.

persistent block in production of *Flc*, enabling the *Ft* signal to be generated when day length is sufficiently long.

Mutations affecting the regulation of *Flc* expression alter the time of flowering and thus the ability of a plant to flourish in a given climate. The whole control system governing the switch to flowering is thus of vital importance for agriculture, especially in an era of rapid climate change.

The example of vernalization suggests a general point about the role of chromatin modification in developmental timing. The plant uses changes in chromatin to record its experience of prolonged cold. It may be that in other organisms—animals as well as plants—slow, progressive changes in chromatin structure provide long-term timers for those mysterious developmental processes that unfold slowly, over a period of days, weeks, months, or years. Such chromatin timers may be among the most important clocks in the embryo, but as yet we understand very little about them.

Summary

*Developmental timing is controlled at many levels. It takes time to switch a gene on or off, and this time delay depends on the lifetimes of the molecules involved, which can vary widely. Cascades of gene regulation involve cascades of delays. Feedback loops can give rise to temporal oscillations in gene expression, and these may serve to generate spatially periodic structures. During vertebrate segmentation, for example, expression of the *Hes* genes oscillates, and one new pair of somites is formed during each oscillation cycle. *Hes* genes encode transcription repressor proteins that can act back on expression of the *Hes* genes themselves. This negative feedback generates oscillations with a period that reflects the delay in the autoregulatory gene switching loop. The period of oscillation of this “segmentation clock” controls the sizes of the somites. Notch signaling between neighboring cells synchronizes their oscillations: when Notch signaling fails, the cells drift out of synchrony because of genetic noise in their individual clocks, and the segmental organization of the vertebral column is disrupted.*

*Timing does not always depend on cell–cell interactions; many developing animal cells have intrinsic developmental programs that play out even in isolated cells in culture. Neuroblasts in *Drosophila* embryos, for example, go through set programs of asymmetric divisions, generating different neural-cell types at each division with a predictable sequence and timing and through a cascade of gene switching events. Studies in both vertebrates and invertebrates show that such programs are rarely governed by the timing of cell division and can unfold even when cell division is blocked. MicroRNAs produced at critical moments sharpen developmental transitions by blocking the translation and promoting the degradation of specific sets of mRNAs. One such event is the maternal–zygotic transition, when transcription from the embryonic genome initiates. The onset of this event involves both titration of a maternal repressor by multiplying zygotic nuclei and the time-dependent accumulation of maternal transcriptional activators. Global coordination of developmental timing is achieved by hormones that disseminate and act on cells throughout the organism: as a tadpole grows, for example, thyroid hormone levels surge and trigger its metamorphosis into a frog. Environmental control of developmental timing is especially striking in plants and reveals the presence of molecular timers that act over the long term. In vernalization, for example, prolonged cold induces changes in chromatin that chart the passage through winter so as to allow flowering only in the spring. Slow, progressive changes in chromatin structure are likely to be important timers in the long-term programming of development in animals too.*

MORPHOGENESIS

The specialization of cells into distinct types at specific times is important, but it is only one aspect of animal development. Equally important are the movements and deformations that cells go through to assemble into tissues and organs

with specific shapes and sizes. These physical forms, which are a central output of cell-fate programs, are essential for each organ's physiological function. Like developmental timing, the process of *morphogenesis* ("form generation") is less well understood than the processes of differential gene expression and inductive signaling that lead to cell-type specialization. The cell movements can be readily described, and the underlying molecular mechanisms that coordinate the movements are now being deciphered.

In Chapter 19, we saw how cells cohere to form epithelial sheets or surround themselves with extracellular matrix to create connective tissues. We also discussed how the basic features of tissues, such as the polarity of epithelia, arise from the properties of individual cells. In this section, we consider how the dynamic rearrangements of cells during animal development give shape to the embryo and all the individual organs and appendages of the body.

A small number of cell behaviors are basic to morphogenesis. Cells can change shape, and coordinated shape changes can bend an epithelial sheet into a tube or a hollow ball. Cells can rearrange with respect to their neighbors, causing elongation, constriction, or thickening of a tissue. By stretching out while holding onto their companions, specialized sets of cells can form growing tubular networks such as the system of blood or lymph vessels. Individual cells can also extract themselves from their neighbors and form physically separate groups that then migrate through the embryo along defined tracks. Group migrations, as occur in gastrulation, can transform the entire topology of the embryo. Underlying all these processes are changes in physical forces that alter cell shape and cell contacts—either with other cells or with extracellular matrix. We begin by considering how these forces are generated.

Imbalance in Physical Forces Acting on Cells Drives Morphogenesis

Although it is alive, a cell is governed by the same physical principles that shape inanimate objects. Cells and tissues possess specific mechanical properties, and their shapes reflect the forces exerted upon them. At rest, a cell is in a mechanical equilibrium in which all forces are balanced. Morphogenesis entails a shift in that balance, as new forces are generated in an oriented way. These organized force changes then play out across increasing distances. For example, the activity of myosin motor proteins on the actin cytoskeleton may change the shape of a cell, which in turn affects its interactions with neighboring cells and thus the shape of a tissue. Such coordinated changes act to sculpt entire organs and animal bodies.

While some proteins, such as actin and myosin, generate forces within cells, others act to sense the applied forces. Such proteins, called *mechanotransducers*, initiate intracellular signaling in response to mechanical stimuli, just as the molecules described in Chapter 15 trigger signaling pathways in response to chemical cues. α -Catenin and Talin are mechanotransducers that bind to the cytosolic tail of cell adhesion molecules (see Chapter 19); these unfold in response to tension exerted by neighboring cells, and this conformational change sends biochemical signals that can elicit rapid responses in cellular mechanics as well as long-term effects on gene expression. Piezo (see p. 660) is a mechanosensitive ion channel that spans the cell membrane; it opens in response to shear stress or stretching of the plasma membrane, passing ions that then initiate signaling cascades. Through such mechanosensitive molecules, cells and tissues can read the physical environment within which they exist and respond accordingly.

Tension and Adhesion Determine Cell Packing Within Epithelial Sheets

The two-dimensional epithelial sheet found in many tissues is a simple system to illustrate principles of morphogenesis. Across animal species, stable epithelial sheets display a remarkably consistent pattern of cell shapes, in which most

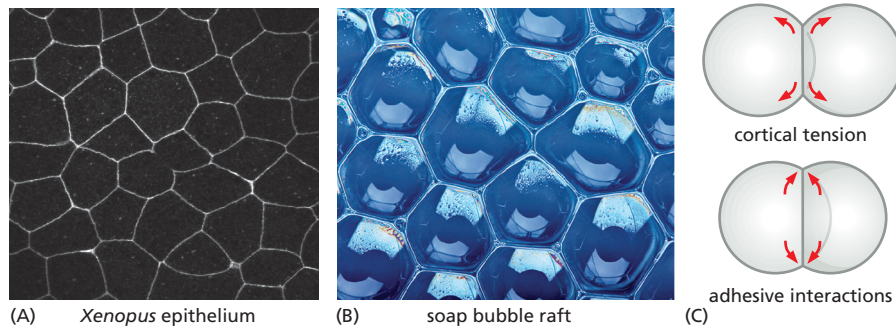


Figure 21-46 Regular distribution of cell shapes in epithelia. (A) Fluorescence micrograph of cell membranes in an early *Xenopus* embryo shows that most epithelial cells have hexagonal profiles, with six neighbors forming three-way junctions where cells meet. (B) A two-dimensional “raft” of soap bubbles on the surface of water shows a similar organization. (C) The length and geometry of cell-cell contacts are set by opposing activities. The contractile network of the cortical actomyosin cytoskeleton within each cell tends to pull cells apart, while adhesive interactions mediated by cadherins and other molecules tend to hold cell surfaces together. (A, courtesy of Saranyaraajan Varadarajan and Ann Miller; B, Shebeko/Shutterstock.)

cells possess a hexagonal profile at their apical surface (Figure 21-46A). This arrangement is determined by the mechanical properties of the cells and their connections with neighbors, which are mediated primarily by *adherens junctions* (see Chapter 19). Contractility of an actomyosin ring underlying the plasma membrane adjacent to the junctions generates cortical tension within each cell, which tends to pull it away from its neighbors. However, this force is opposed by cell-cell adhesion, mediated by molecules such as cadherins within the junctions. The balance between these inward and outward forces generates a predictable configuration of cells, which can be mathematically modeled and is analogous to the packing arrangement of inorganic entities such as soap bubbles (Figure 21-46B and C). Although the physical mechanisms that balance surface-tension minimization and contact interfaces in bubbles are very different from those that shape cells, the mathematical solution representing a minimal energy of arrangement is similar.

Changing Patterns of Cell Adhesion Molecules Force Cells into New Arrangements

The shapes of tissues are ultimately determined by patterns of gene expression, which produce molecules that change the physical properties of cells in which they are expressed. One important class of genes that contributes to these properties codes for the adhesion molecules that cells display on their surface. Through changes in its surface molecules, a cell can break old attachments and make new ones. For example, cells in one region of a tissue may develop surface properties that make them cohere with one another and become segregated from a neighboring group of cells that possess different surface molecules. Cadherins and other cell-cell adhesion proteins that are differentially expressed in the various tissues of a developing animal play important roles. In Chapter 19, we saw how selective cell-cell adhesion is sufficient to drive the sorting of cells dissociated from an embryo into groups of a similar type (see Figure 19-7); treatment with antibodies that bind cadherins interferes with this process. Changes in expression of the various cadherin genes correlate closely with the different modes of association among cells during embryogenesis, as occurs, for example, during gastrulation, neural tube formation, and somitogenesis. Cadherins also control the formation and dissolution of epithelial sheets and clusters of cells (see Movie 19.1). Finally, because adhesion molecules not only glue one cell to another but also provide anchorage for intracellular actin filaments, the sites of cell-cell adhesion influence tension and cell movements in the developing tissue.

Repulsive Interactions Help Maintain Tissue Boundaries

The different types of cadherins enable different types of cells to cohere selectively: cells expressing one type of cadherin will maximize their contact with cells expressing the same cadherin and thereby segregate from other cells, creating specific tissue boundaries. Cell mixing can be inhibited and boundaries created and maintained in another way as well: cells of different types

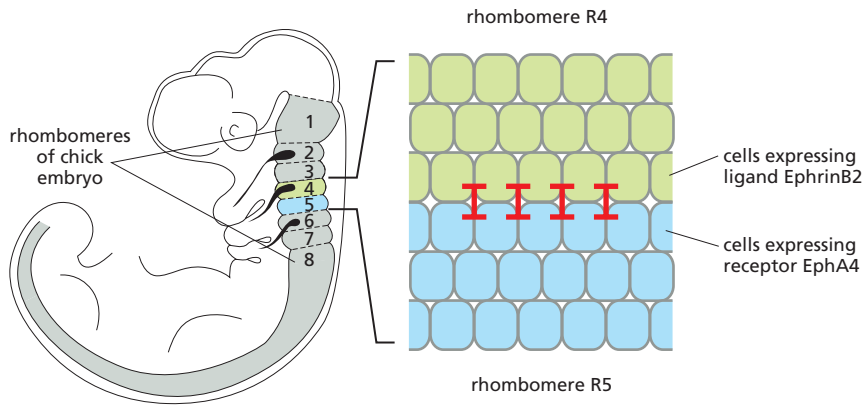


Figure 21-47 Sorting out by repulsion.

Ephrin–Eph signaling in hindbrain segmentation in a chick embryo. Each pair of rhombomeres (segments in the hindbrain) is associated with a branchial arch (a modified gill rudiment) to which the pair sends innervation. Rhombomeres are distinguished from one another by expression of different *Hox* genes (see Figure 21-30). Differences in surface tension at the interface between cells that express EphrinB2 in rhombomere R4 and EphA4 in rhombomere R5 lead to mutual repulsion (red bars), creating a sharp boundary.

can sometimes actively repel one another. The bidirectional activation of Eph receptors and ephrins discussed in Chapter 15 often mediates such repulsion, acting at interfaces between different groups of cells to keep the groups from mixing, and repelling invasion by inappropriate visitors. Ephrin–Eph signaling operates at the boundaries of the somites in the vertebrate embryo discussed earlier (see Figure 21-38). By modulating the cortical actomyosin cytoskeleton, ephrin–Eph signaling is thought to induce differences in cortical tension between cells, thereby decreasing the force of the cell–cell contact. Another example is in rhombomeres, which are segmented regions of the developing vertebrate brain. Neighboring rhombomeres express complementary combinations of ephrins and Eph receptors, and this keeps the cells in adjacent rhombomeres strictly segregated, with a boundary between them that is sharply defined (Figure 21-47).

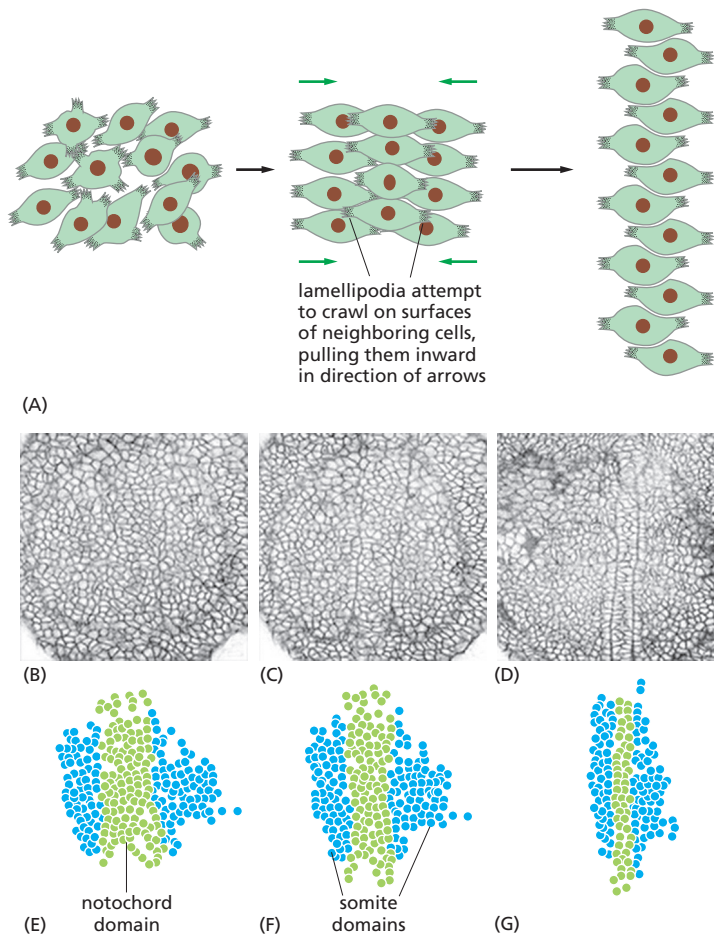


Figure 21-48 Convergent extension

by mesenchymal cell migration. (A) A population of cells form lamellipodia, with which they attempt to crawl over one another. Alignment of the lamellipodial movements along a common axis, controlled by planar-polarity signaling (discussed shortly), leads to convergent extension. (B–G) The pattern of convergent extension of dorsal mesoderm during zebrafish gastrulation at 8.8 (B, E), 9.3 (C, F), and 11.3 (D, G) hours after fertilization. Cells that will give rise to the notochord are labeled in green, and cells that will give rise to somites and muscle are labeled in blue. The notochord and somite domains are spatially separate from the start of the recording (B, E), but their boundaries are at first barely visible and only a little later become obvious. Convergence narrows the notochord domain to a width of about two cells at the last time point (D, G). (A, after J. Shih and R. Keller, *Development* 116:901–914, 1992. With permission from the Company of Biologists; B–G, after N.S. Glickman et al., *Development* 130:873–887, 2003. With permission from the Company of Biologists.)

Groups of Similar Cells Can Perform Dramatic Collective Rearrangements

Cell sorting mediated by cadherin and repulsion mediated by ephrin-Eph exemplify how differences in cell-surface properties can drive tissue arrangements, causing cells that express different sets of genes to separate from one another. However, groups of cells that are all similar can also undergo dramatic rearrangements. Many animal embryos extend their major body axis through such rearrangements, in a process called **convergent extension**. Convergent extension occurs when a population of cells moves coordinately toward a specific boundary, intercalating between each other as they do so. This causes the tissue to narrow along one axis (converge) and elongate along another (extend). The overall change in tissue shape results from cell displacement, rather than a change in shape of individual cells, and is largely autonomous to the group of cells involved. A striking demonstration is seen when small, square fragments of tissue from the appropriate region of a frog embryo are isolated in culture. These will spontaneously narrow and elongate, just as they would in the developing animal.

Convergent extension is a common movement that elongates many organs, although the cell behaviors that underlie it can differ. During frog gastrulation, for example, cells in one region of the surface epithelium lose their epithelial character as they migrate into the interior of the embryo. The loosely connected cells, called *mesenchymal cells*, form actin-based protrusions called lamellipodia that are oriented so that they crawl over one another and interdigitate as they converge toward the embryonic midline (**Figure 21–48**). In contrast, when the *Drosophila* embryo extends its anteroposterior axis, the cells involved remain epithelial. These cells maintain tight adhesive junctions but rearrange the relationships to their neighbors by shrinking cell junctions along one axis while extending them in another axis to accomplish cell intercalation (**Figure 21–49**; see also **Figure 19–15**). Remodeling of the junctions involves differential contractile tension on them that is generated by localized myosin activity. In both cases, whether mesenchymal migration or epithelial junction remodeling, the coordination of cell behaviors depends on a signaling pathway that can generate *planar cell polarity*, as we discuss next.

Planar Cell Polarity Orients Cell Behaviors Within an Embryo

Just as a compass allows us to orient ourselves in a landscape, cells undergoing morphogenetic movements must have a system to orient themselves with respect to the different body axes of the animal. We discussed in Chapter 19 how all

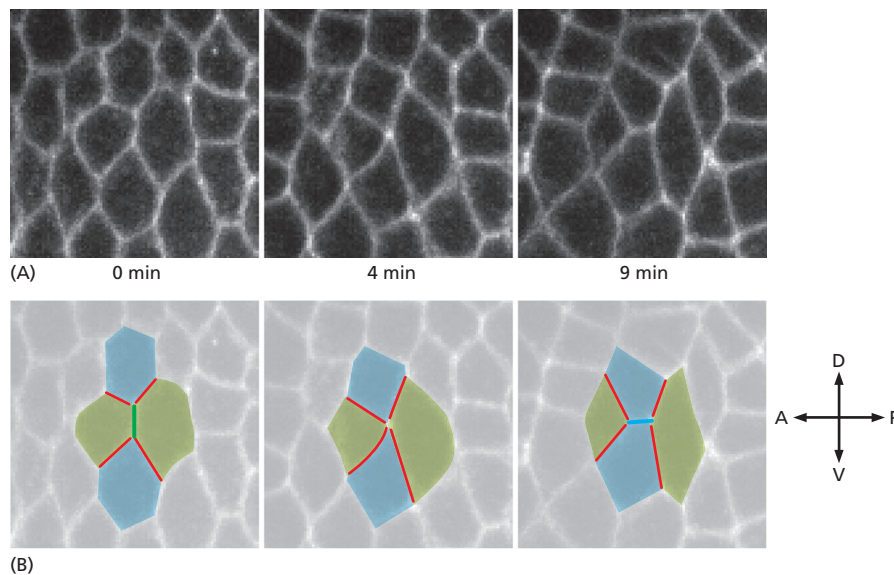


Figure 21–49 Convergent extension by rearrangement of epithelial junctions.

Shown in (A) are fluorescence images of the *Drosophila* embryonic epidermis over a 9-minute period, with drawings in (B) that colorize specific cells and cell junctions (**Movie 21.6**). As in other epithelia, most cells are initially hexagonal, and their junctions are shared between three neighbors. To elongate the tissue along the anteroposterior axis, some of the dorsoventral-oriented junctions (*green*) shrink, forming a transient arrangement in which cells are pentagonal or quadrilateral, and a junction is shared between four neighbors. A new junction (*blue*) then forms along the anteroposterior axis, now shared between two cells that did not previously contact each other and separating two cells that previously did contact each other. This process of shrinking and elongating junctions in a planar-polarized fashion, occurring all along the epidermis, narrows and extends the tissue while maintaining epithelial integrity throughout. (Images courtesy of Huapeng Yu and Jennifer Zallen.)

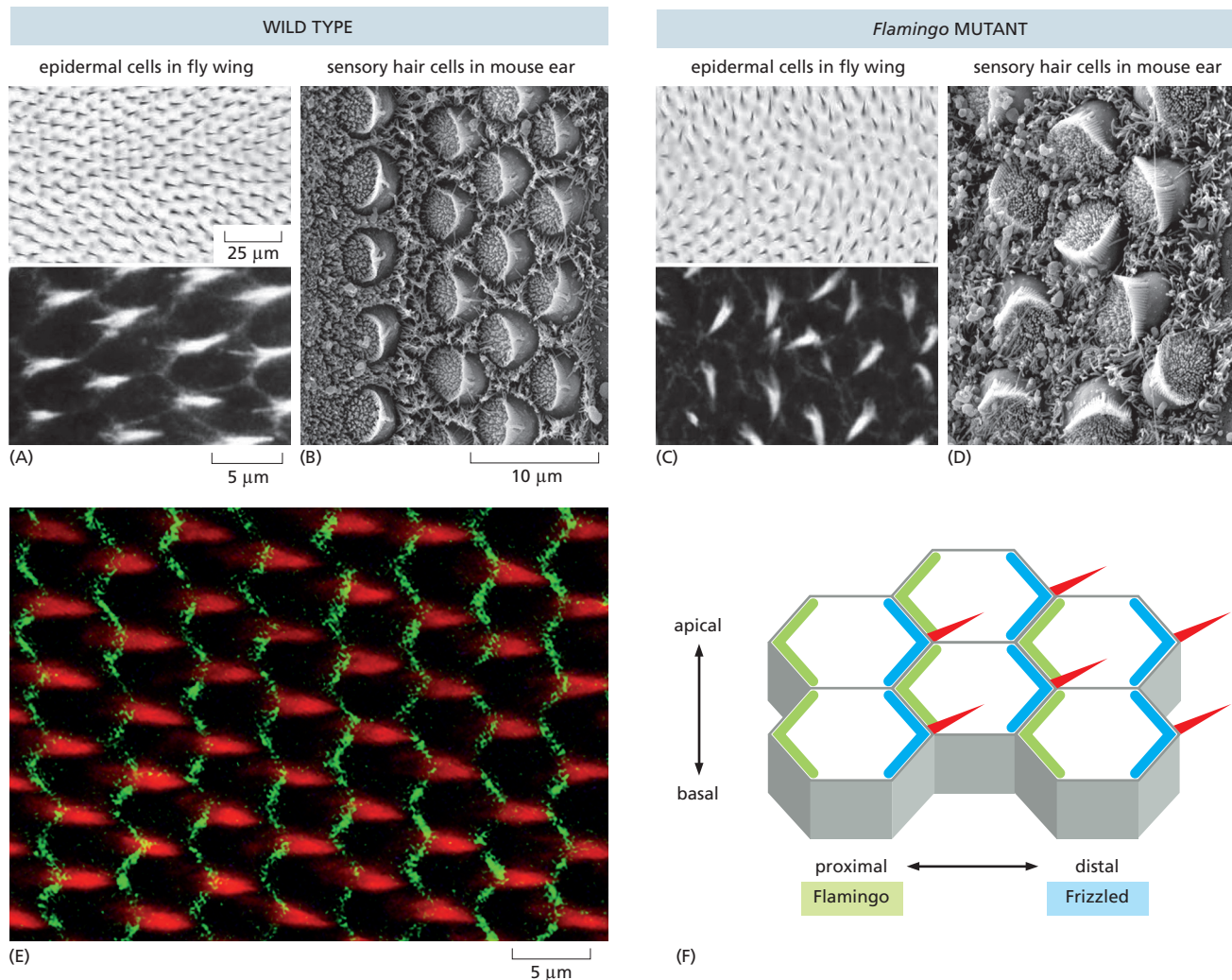


Figure 21-50 Planar cell polarity. (A) Wing hairs on the wing of a fly. Each cell in the wing epithelium forms a small, spiky protrusion or “hair” at its apex, and all the hairs point the same way, toward the tip of the wing. This reflects a planar polarity in the structure of each cell. (B) Sensory hair cells in the inner ear of a mouse similarly have a well-defined planar polarity, manifest in the oriented pattern of stereocilia (actin-filled protrusions) on their surface. The detection of sound depends on the correct, coordinated orientation of the hair cells. (C) A mutation in the gene *Flamingo* in the fly, coding for a nonclassical cadherin, disrupts the pattern of planar cell polarity in the wing. (D) A mutation in a homologous *Flamingo* gene in the mouse randomizes the orientation of the planar cell polarity vector of the hair cells in the ear. The mutant mice are deaf. (E) Micrograph showing localization of Flamingo (green) in a pupal fly wing, in relation to the hair growing from each cell (red). (F) Diagram illustrating planar-polarized localization of Flamingo to one side of each epithelial cell, and localization of another planar polarity-regulating protein called Frizzled to the opposite side, from which the hair will form. (A and C, from J. Chae et al., *Development* 126:5421–5429, 1999. With permission from the Company of Biologists; B and D, from J.A. Curtin et al., *Curr. Biol.* 13:1129–1133, 2003. With permission from Elsevier; E, courtesy of Paul Adler.)

epithelial cells have apical–basal polarity, but the cells of many epithelia show an additional polarity at right angles to this axis: the cells are arranged as if they had an arrow written on them, all pointing in a common direction in the plane of the epithelium. This type of polarity is called **planar cell polarity**. In the wing of a fly, for example, each epithelial cell has a tiny asymmetric projection, called a wing hair, on its surface, and the hairs all point toward the tip of the wing. Similarly, in the inner ear of a vertebrate, each mechanosensory hair cell has a precisely oriented asymmetric bundle of actin-filled, rodlike protrusions called stereocilia sticking up from its apical plasma membrane as a detector of sound and of forces such as gravity. Tilting the bundle in one direction causes ion channels in the membrane to open, electrically activating the cell; tilting in the opposite direction has the opposite effect. For the ear to function properly, the hair cells

must be oriented correctly. Planar cell polarity is also important in the respiratory tract, where every ciliated cell must orient the beating of its cilia so as to sweep mucus upward, away from the lungs.

Screens for mutants with misoriented wing hairs in *Drosophila* have identified a set of genes that control planar cell polarity in many tissues. Some of these genes code for components of the Wnt signaling pathway, and others code for transmembrane proteins such as specialized members of the cadherin superfamily that interact across neighboring cell-cell junctions. Together with intracellular proteins that influence the localization of the transmembrane proteins, two complementary, mutually exclusive domains are defined along the apical surface in such a way as to exert a polarizing influence that propagates from cell to cell, leading to polarization of the entire epithelial tissue. Essentially the same system of proteins controls planar cell polarity in vertebrates; mice deficient in homologs of the *Drosophila* planar-polarity genes have a variety of defects, including incorrectly oriented hair cells in the inner ear, making them deaf (Figure 21–50).

An Epithelium Can Bend During Development to Form a Tube

Convergent extension is an example of morphogenesis in two dimensions. When cells need to leave the plane of a tissue and produce three-dimensional structures, they engage in additional cell behaviors. The universal process of gastrulation involves future endodermal and/or mesodermal cells migrating into the interior of the embryo, and this is often initiated by a localized bending of the blastula epithelium. Similar bending can generate biological tubes from flat epithelial sheets, including the neural tube characteristic of chordates, although there are other routes to tube formation as sketched in Figure 21–51.

Tissues can be bent by a widely used cell behavior called *apical constriction*, which occurs when cells use myosin motor proteins to contract bundles of actin filaments anchored at adherens junctions near their apical surface. Shortening of these filament bundles causes the epithelial cells to narrow at their apex, transforming the cell from a columnar shape into a distinctive wedge or bottle shape. Because the cells are connected to each other by adherens junctions, coordinated contraction in a group can result in the entire sheet bending or even rolling up into a tube (Figure 21–52). The cell groups that undergo apical constriction are selected by localized expression of patterning genes that trigger the activity of the contractile actomyosin networks. In *Drosophila* embryos, the ventrally restricted transcription regulator Twist (Figure 21–26) controls myosin behavior in presumptive mesodermal cells, which then gastrulate by creating a transient tube-like invagination. In vertebrate embryos, patterns of Sonic hedgehog (Shh) and BMP

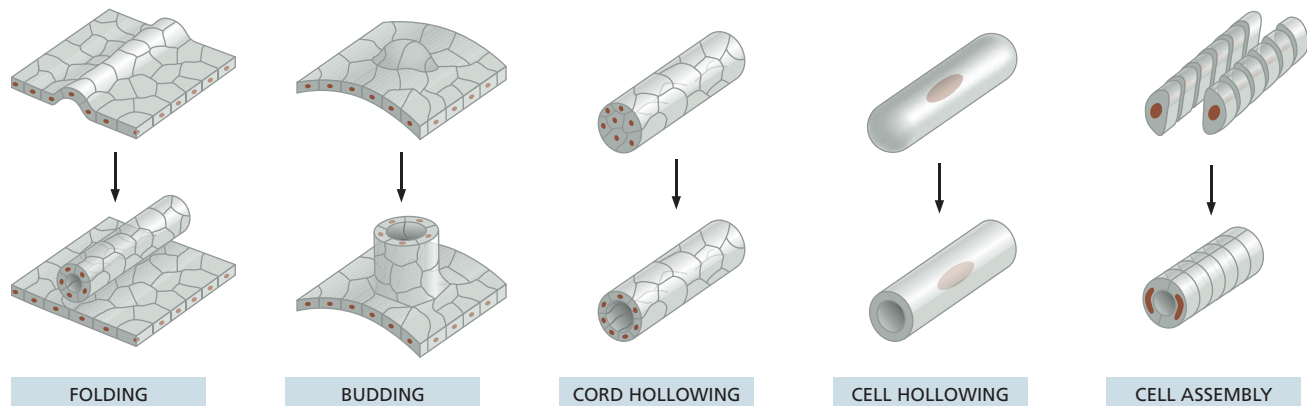


Figure 21–51 Cell behaviors involved in tube formation. Folding generates the neural tube, budding underlies the formation of lungs and trachea, cord hollowing occurs during the formation of mammalian salivary glands, cell hollowing is involved in the formation of tracheal terminal cell tubes, and cell assembly generates the heart tube that forms at the earliest stage of heart development.

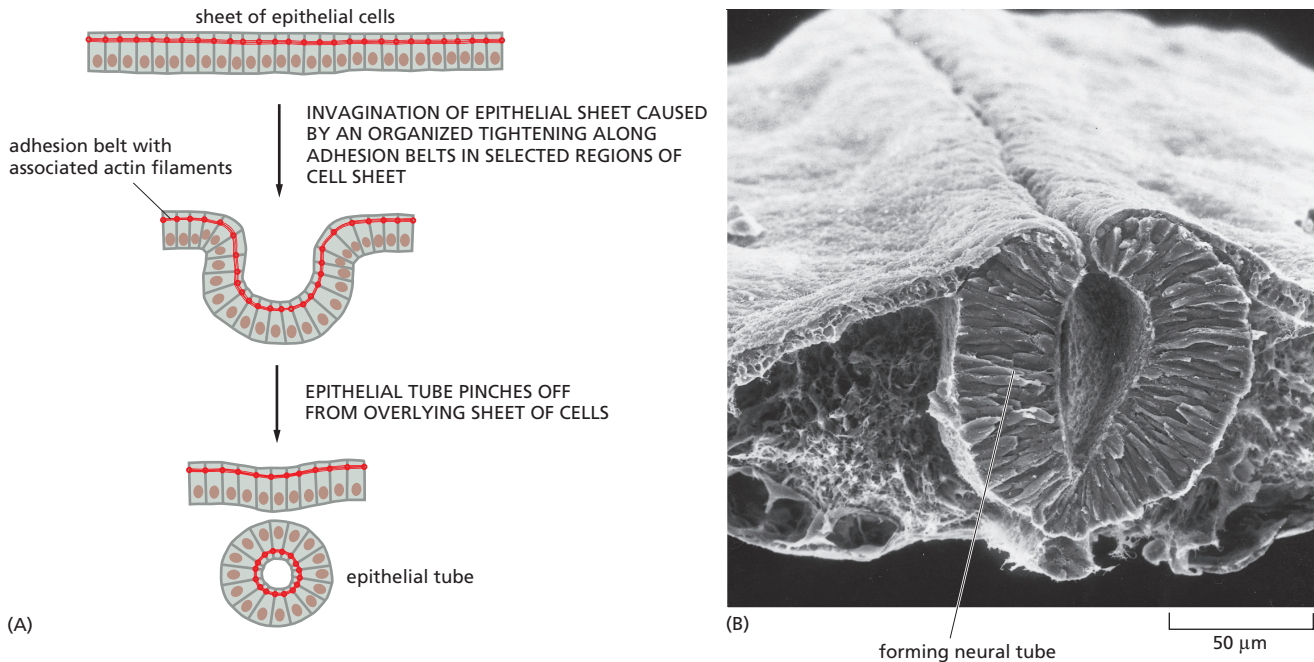


Figure 21–52 Bending of an epithelial sheet to form a tube. Contraction of apical bundles of actin filaments linked from cell to cell by adherens junctions causes the epithelial cells to narrow at their apex. Depending on the number and arrangement of cells in which the contraction occurs, the invaginated epithelium can remain connected to the original sheet or alternatively can roll up into a separated tube or a hollow sphere. (A) Diagram showing how an apical contraction along one axis of an epithelial sheet can cause the sheet to form a tube. (B) Scanning electron micrograph of a cross section through the trunk of a 2-day chick embryo, showing the formation of the neural tube by the process diagrammed in panel A. The vertebrate neural tube gives rise to the brain and spinal cord. (B, courtesy of Jean-Paul Revel.)

signaling induce apical constriction at three hinge points that transform the neural plate into a closed neural tube.

Interactions Between an Epithelium and Mesenchyme Generate Branching Tubular Structures

Embryos often develop branching tubular structures when large surfaces are required for functions such as excretion, absorption of nutrients, and gas exchange. The lungs are an example. They originate from epithelial buds that grow out from the floor of the foregut and penetrate neighboring mesenchyme to form the bronchial trees, systems of tubes that branch repeatedly as they extend, eventually making an estimated 17 million branches in humans. Endothelial cells that form the lining of blood vessels invade the same mesenchyme, thereby creating a system of closely apposed airways and blood vessels, as required for gas exchange in the lung (**Figure 21–53**). This whole process of *branching morphogenesis* depends on signals that pass in both directions between the growing epithelial buds and the mesenchyme. Genetic studies in mice indicate that fibroblast growth factor (FGF) proteins and their receptor tyrosine kinases play a central part in these signaling processes. FGF signaling has various roles in development, but it is especially important in the many interactions that occur between a developing epithelium and mesenchyme.

In the case of lung development, FGF10 is expressed in clusters of mesenchyme cells that lie near the tips of the growing epithelial tubes, and its receptor is expressed in the invading epithelial cells. In FGF10-deficient mutant mice, primary buds of the lung epithelium form, but fail to grow out of the mesenchyme to create branching bronchial trees. Conversely, a microscopic bead soaked in FGF10 and placed near embryonic lung epithelium in culture will induce a bud to form and grow out from the epithelium toward the bead. Evidently, the epithelium enters the mesenchyme only by invitation, in response to FGF10.

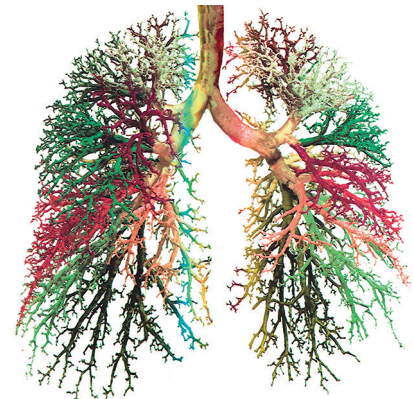
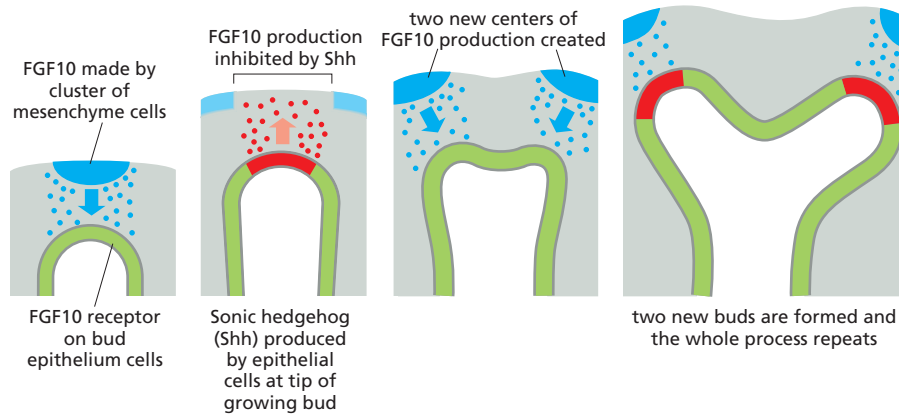


Figure 21–53 The airways of the lungs, shown in a cast of the adult human bronchial trees. Resins of different colors have been injected into different large branches of the trees. (James Cavallini/ Science Source.)



But what makes the growing epithelial tubes of the lung branch repeatedly as they invade the mesenchyme? This depends on a Sonic hedgehog signal that is sent in the opposite direction, from the epithelial cells at the tips of the buds back to the mesenchyme, where it inhibits FGF10 production to promote bifurcation as shown in [Figure 21-54](#). In mice lacking Sonic hedgehog, the lung epithelium grows and differentiates, but it forms a sac instead of a branching tree of tubules.

FGF is also the major signal in formation of the air-exchange system of insects, which consists of a pattern of fine, air-filled channels called *tracheae* and *tracheoles*. These originate from the epidermis covering the surface of the body and extend inward to invade the underlying tissues, branching and narrowing as they go ([Figure 21-55](#)). The FGF acts on cells at the tips of the advancing tracheae, causing them to extend filopodia and migrate toward the source of the FGF signal. Because the tip cells remain connected to the remainder of the tracheal epithelium, the pulling force that they generate elongates the tracheal tube.

Initially, the pattern of FGF production in fly embryos is defined by the D-V and A-P patterning systems discussed earlier. In later stages of development, however, FGF expression is induced by transcription regulators called *hypoxia-inducible factors* (HIFs) that are activated by hypoxia (low oxygen levels). In this way, hypoxia stimulates the formation of finer and finer and more extensively branched trachea, until the oxygen supply is sufficient to stop the process.

The Extracellular Matrix Also Influences Tissue Shape

So far, we have considered only forces generated by cells during morphogenesis. However, we must not ignore another important constituent of organs: the extracellular matrix. As detailed in Chapter 19, this network of insoluble molecules is found in most animal tissues, and a specialized matrix called the basal lamina underlies all epithelial sheets. Extracellular matrices provide substrates for morphogenetic cell-migration events, as we will discuss shortly. Furthermore, the mechanical stiffness of the matrix can provide instructive signals for shaping organs by modifying the resistance against which cellular forces are generated. During branching morphogenesis of the mammalian salivary gland, for

Figure 21-54 Branching morphogenesis of the lung. How FGF10 and Sonic hedgehog are thought to induce the growth and branching of the buds of the bronchial tree. Many other signal molecules, such as BMP4, are also expressed in this system, and the illustrated branching mechanism is only one of several possibilities. As indicated, FGF10 protein is expressed in clusters of mesenchyme cells near the tips of the growing epithelial tubes, and its receptor is expressed in the epithelial cells themselves. The Sonic hedgehog signal is sent in the opposite direction, from the epithelial cells at the tips of the buds back to the mesenchyme. The patterns of gene expression and their timing suggest that the Sonic hedgehog signal may serve to shut off FGF10 expression in the mesenchyme cells closest to the growing tip of a bud, splitting the FGF10-secreting cluster into two separate clusters, which in turn cause the bud to branch into two.

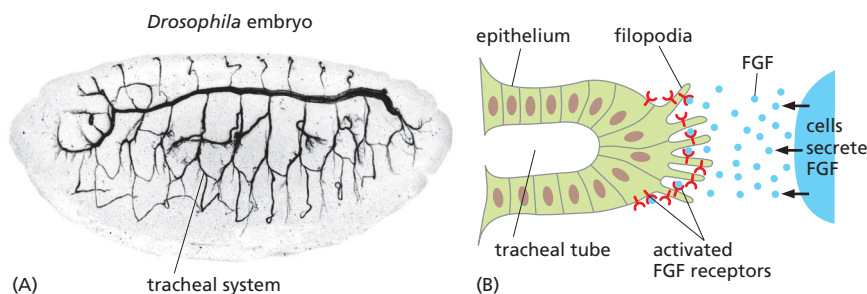


Figure 21-55 Branching morphogenesis of airways in a fly. (A) *Drosophila* embryonic tracheal system. (B) FGF (produced in *Drosophila* by the *Branchless* gene) signals from surrounding cells to the tracheal epithelium and activates its FGF receptors, leading to filopodia formation and tube elongation. [A, from G. Manning and M.A. Krasnow, in *The Development of Drosophila melanogaster* (A. Martinez-Arias and M. Bate, eds.), Vol. 1, pp. 609–685. New York: Cold Spring Harbor Laboratory Press, 1993. With permission from Cold Spring Harbor Laboratory Press.]

example, the basal lamina surrounding the growing bud is perforated by matrix-degrading enzymes. This localized softening of the matrix encourages preferential outgrowth of the epithelium into the more pliable regions. Changes in extracellular resistance, as well as dynamic intracellular forces, can therefore create force imbalances that drive morphogenetic events.

Cell Migration Is Guided by Environmental Signals

The birthplace of cells is often far from their ultimate location in the body. Our skeletal muscles, for example, derive from muscle-cell precursors, or *myoblasts*, in somites from which they migrate into the limbs and other regions. The routes that the migrant cells follow and the selection of sites that they colonize determine the eventual pattern of muscles in the body. The embryonic connective tissues form the framework through which the myoblasts travel, and these tissues provide the cues that guide myoblast distribution. No matter which somite they come from, the myoblasts that migrate into a forelimb bud will form the pattern of muscles appropriate to a forelimb, and those that migrate into a hindlimb bud will form the pattern appropriate to a hindlimb. It is the connective tissue that provides the patterning information.

As a migrant cell travels through the embryonic tissues, it repeatedly extends surface projections that probe its immediate surroundings, testing for cues to which it is particularly sensitive by virtue of its specific assortment of cell-surface receptor proteins. Inside the cell, these receptors are connected to the cortical actin and myosin cytoskeleton, which moves the cell along (see Chapter 16). Some extracellular matrix molecules, such as the protein fibronectin, provide adhesive sites that help the cell advance; others, such as chondroitin sulfate proteoglycan, inhibit locomotion and repel immigration. The nonmigrant cells along the migration pathway may likewise have inviting or repellent macromolecules on their surface; some may even extend filopodia to make their presence known.

In addition to general adhesive molecules, migratory cells can also be guided by specific chemical or mechanical signals that are sensed by dedicated receptors. Among many such influences, a few stand out as especially important. In particular, many types of migrating cells are guided by chemotaxis that depends on a G-protein-coupled receptor, called CXCR4, which is activated by an extracellular ligand, CXCL12. Cells expressing this receptor can shuffle their way along tracks marked out by CXCL12 (Figure 21–56). Chemotaxis toward sources of CXCL12 plays a major part in guiding the migrations of lymphocytes and various other white blood cells; of neurons in the developing brain; of myoblasts entering limb

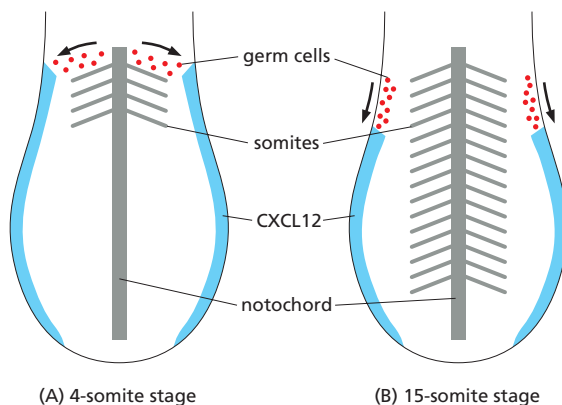


Figure 21–56 CXCL12 guides migrating germ cells. Zebrafish germ cells migrate to domains that express CXCL12. As the sites of CXCL12 expression change, cells follow the CXCL12 track and are guided to the region where the gonad develops at a later developmental stage. (A) At the 4-somite stage, germ cells move from a position that is close to the midline to more lateral regions where CXCL12 is expressed. (B) As the CXCL12 expression retracts, germ cells are guided to more posterior positions, where the gonads are developing.

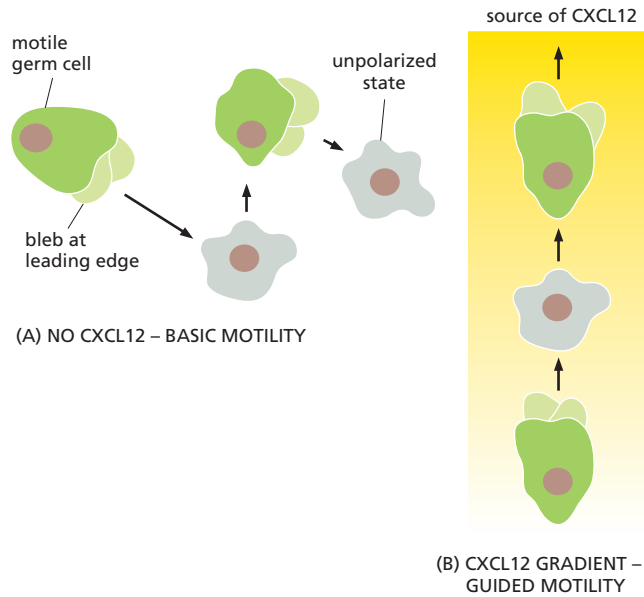


Figure 21–57 Directional migration by local blebbing. (A) Germ cells migrate via protrusions at the leading edge of the cell called blebs, where the plasma membrane detaches locally from the underlying actin cortex and is pushed outward (see Figure 16–18). (B) The persistence and site of the protrusions are biased toward higher levels of CXCL12. Thus, germ cells migrate up the CXCL12 gradient. Note that this form of cell migration is distinct from that based on actin-rich lamellipodia (see Figures 21–48 and 21–59).

buds; of primordial germ cells as they travel toward the developing gonads; and of cancer cells when they metastasize.

Detailed studies of primordial-germ-cell migration have shown that CXCL12 signaling does not induce cell migration per se but rather serves to control its direction. In the absence of CXCL12 signaling, germ cells still display the membrane blebbing associated with cell migration, but the position of the cell front where blebs form is randomly chosen (Figure 21–57), and migration is more of a random walk; if CXCL12 signaling is intact, blebbing is more frequent on the side of the cell that faces the source of CXCL12, resulting in directional migration.

The Distribution of Migrant Cells Depends on Survival Factors

The final distribution of migrant cells depends not only on the routes they take, but also on whether they survive the journey and thrive in the environment they find at their journey's end. Specific sites provide survival factors needed for specific types of migrant cells to survive.

Among the most important sets of migrant cells in the vertebrate embryo is that of the **neural crest**. They arise from the border region between the part of the ectoderm that will form epidermis and the part that will form the central nervous system. As the neural ectoderm rolls up to form the neural tube, the neural crest cells break loose from the epithelial sheet along this border region and set out on their long migrations (see Figure 19–8 and Movie 21.7). They settle ultimately in many sites and give rise to a surprising diversity of cell types. Some will differentiate into the neurons and glial cells of the peripheral nervous system—not only in the sensory ganglia that lie close to the spinal cord, but also, following a much longer migration, in the wall of the gut. Others will form skeletal tissue in the face; still others will lodge in the skin and specialize as pigment cells.

A signaling pathway required for neural crest cell survival was identified by studying animals with defective pigmentation. Many naturally occurring mouse mutants that show white spots on their normally black coat of hair are defective in a gene that encodes a secreted peptide called Stem cell factor. Stem cell factor is produced by tissues along the migration pathways and acts as a survival factor for the migrating neural crest cells. In animals defective for Stem cell factor or its receptor, a transmembrane tyrosine kinase called Kit, many of the migrating neural crest cells die by apoptosis. As a result, the mutant individuals have nonpigmented (albino) patches of skin (Figure 21–58). Stem cell factor is an important survival signal for other types of migratory cells as well, including

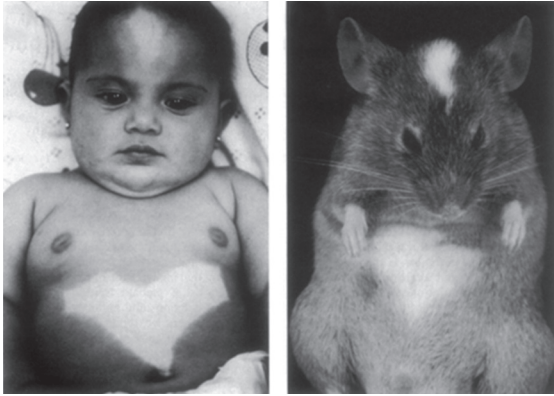


Figure 21–58 Effect of mutations in the *Kit* gene. Both the baby and the mouse are heterozygous for a loss-of-function mutation that leaves them with only half the normal quantity of *Kit* gene product, which is the receptor for stem cell factor (SCF). In both cases, pigmentation is defective because pigment cells depend on SCF for their survival. (Courtesy of R.A. Fleischman, from R.A. Fleischman et al., *Proc. Natl. Acad. Sci. USA* 88:10885–10889, 1991.)

primordial germ cells and blood-cell precursors. Mice completely lacking Stem cell factor or *Kit* are sterile and eventually die of anemia.

Cells Migrate in Groups to Achieve Large-Scale Morphogenetic Movements

In addition to traveling solo, cells can migrate in groups, maintaining contact with one another and exhibiting coordinated behaviors that differ from those of independently migrating cells. These movements are termed *collective cell migration*, and they come in several forms distinguished by how the cells are organized. For example, neural crest cells undergo *chain migration*, in which cells follow one another in streamlike files (**Figure 21–59A**). Although their association is loose, contact between the cells is required for directed migration along paths determined by chemotactic factors such as CXCL12. Whereas individual neural crest cells respond poorly to this signal, head-to-tail cell contact induces polarity within the group that drives protrusion at the leading edge of each cell in the migrating chain.

Zebrafish development provides a good example of a second form of collective cell movement, called *clustered migration* (**Figure 21–59B**). Such migration occurs in groups of cohesive cells that are usually mesenchymal in origin, with leader cells of the group exhibiting the most protrusive behavior. In the zebrafish embryo, a cell cluster called the lateral line primordium migrates from a position just behind the ear and crosses the trunk musculature toward the tail. Cells in the migrating primordium continue to divide, and some are left behind to form sensory structures in its wake. The specific migratory route is laid out by a prepatterned stripe of CXCL12, but interestingly this ligand is not expressed in a gradient. Instead, the cell cluster itself generates a local gradient. Leader and follower cells both express the CXCR4 receptor, but in addition the followers express a nonfunctional CXCL12 receptor called CXCL7, which acts as a “sink” to diminish the effect of the CXCL12 signal at the rear of the cluster. The result is differential signaling within the cell cluster that orients the direction of travel.

Entire epithelia can move in a third form of collective migration called *sheet migration* (see **Figure 21–59C**). Here, strong adhesion and apical–basal polarity are maintained in the group; cells at the leading edge of the sheet extend robust protrusions that explore the environment, while follower cells create short basal protrusions that are also oriented in the direction of migration. Collective sheet migration can fuse flanking epithelia together, sealing gaps that have formed during development, as seen at the end of *Drosophila* and *C. elegans* embryogenesis or during wound healing. It can also create single organs from bilateral primordia, such as through the fusion that forms the mammalian palate. Interestingly, epithelial migrations can occur in the absence of a free leading edge, as seen in the spherical epithelium of the developing *Drosophila* egg chamber. In this case, collective epithelial migration driven by oriented basal protrusions drives repeated rounds of rotation of the entire tissue, a process required to shape the mature egg. Migration of cells to the top of mammalian gut villi, which

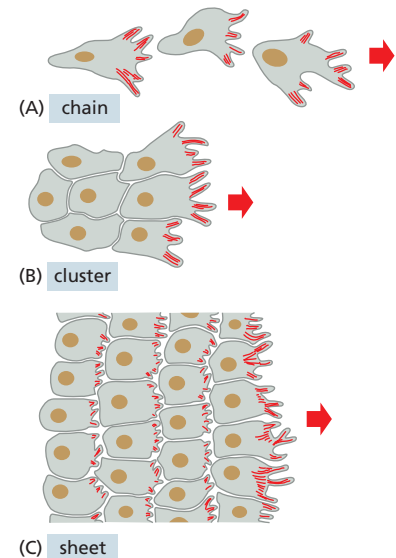


Figure 21–59 Collective cell migration. Cell protrusions are shown in red. (A) Cells undergoing chain migration are connected only loosely, but transient contact with the cell ahead polarizes the protrusions (red) of the cell behind, promoting a shared trajectory. (B) Clustered migration involves groups of adherent cells in which a few “leader” cells extend protrusions in the direction of travel. (C) In sheet migration, large groups of tightly adherent epithelial cells move coherently together, usually with a free edge where cells take on a partially mesenchymal morphology with robust protrusions that lead to migration. Smaller oriented protrusions are also seen in cells behind the leading edge.

preserves intestinal homeostasis (see Figure 22–4), appears to involve a similar movement of a continuous epithelial sheet.

Summary

Animal tissues are sculpted by dramatic changes in cell shape, arrangement, or position. These changes result from shifts in the balance of forces on cells, often triggered by the expression of genes that regulate actomyosin contractility or cell adhesion. Cells that have similar adhesion molecules on their surfaces cohere and tend to segregate from other cell groups with different surface properties. Selective cell–cell adhesion is often mediated by cadherins; repulsion is often driven by ephrin–Eph signaling. Within a tissue, cells can rearrange themselves to drive convergence and extension movements that result in elongation of a tissue along a body axis. Many movements are coordinated through a planar-polarity signaling pathway that is also responsible for orienting cells correctly in various types of epithelium. Epithelial tubes can originate in various ways, most simply by the rolling up and pinching off of a segment of epithelium due to constriction of the apical surface. Elaborate branched tubular structures, such as the airways of the lung, are generated through bidirectional signaling between an epithelial bud and the mesenchyme that it invades in a process called branching morphogenesis. Branching and other morphogenetic events are regulated by changes both in the extracellular matrix and within the cells themselves. Long-range migrating cells, such as those of the neural crest, break loose from their original neighbors and travel through the embryo to colonize new sites. Many migrant cells, including primordial germ cells, are guided by chemotaxis dependent on the receptor CXCR4 and its ligand CXCL12. Interconnected groups of cells can also migrate, with leader and follower cells showing distinctive behaviors. Such collective cell migrations accomplish large-scale tissue shaping.

GROWTH

One of the most fundamental aspects of development is one we know surprisingly little about—how the size of an animal or an organ is determined. Why, for example, do we grow to be so much larger than a mouse? All mammals develop from eggs of a similar size, but there is a 100-million-fold difference in adult weight between the smallest (a shrew) and the largest (a blue whale). Even within a species, size can vary greatly; a Great Dane, for instance, can weigh over 40 times more than a Chihuahua (Figure 21–60). In addition to hereditary factors, the environment also plays a role—a goldfish released into the wild may grow 10 times the size of one kept in a home aquarium.

Three variables define the size of an organ or organism: the number of cells, the size of the cells, and the quantity of extracellular material per cell. Size differences can arise from changes in any of these factors (Figure 21–61). If we compare a mouse with a human, for example, we find that the difference lies chiefly in the number of cells, there being roughly 3000 times more cells in a human, corresponding to a body that is roughly 3000 times more massive. Wild and cultivated species of food plants, on the other hand, often differ in body size chiefly because of differences in cell size.



Figure 21–60 Members of the same species can have dramatically different sizes. The Chihuahua weighs 2–5 kg, whereas a Great Dane weighs 45–90 kg. (Courtesy of Deanne Fitzmaurice.)

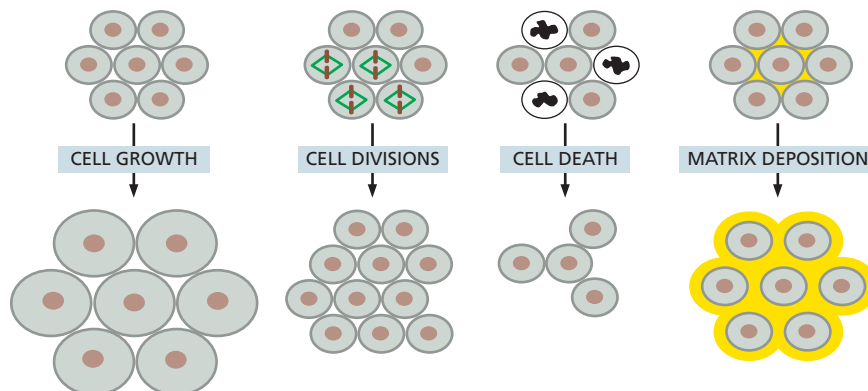


Figure 21–61 Determinants of organ size. A tissue may tune cell growth, proliferation, death, or a combination of these to regulate size. Extracellular matrix deposition also contributes, especially in animal tissues such as bone and cartilage, and in plants.

The challenge, therefore, is to understand how cell numbers, cell size, and extracellular matrix production are regulated. First, we need to identify the signals that drive or inhibit growth. Then we need to discover how the signals themselves are regulated. In some cases, organ growth is dictated by developmental programs that do not monitor the size the structure has attained. In other cases, the growth of an organ or of the body as a whole is controlled homeostatically, so that a “target size” is reached even in the face of drastic disturbances. This suggests the developing structure somehow senses its own size, through local or systemic signals, and uses this information to regulate its growth or shrinkage.

The variation in control strategies is nicely illustrated by some classic transplantation experiments. If several fetal thymus glands are transplanted into a developing mouse, each grows to its characteristic adult size. In contrast, if multiple fetal spleens are transplanted, each ends up smaller than normal, but collectively they grow to the size of one adult spleen. Thus, thymus growth is regulated by local mechanisms intrinsic to the individual organ, whereas spleen growth is controlled by a systemic feedback mechanism that senses the quantity of spleen tissue in the body as a whole; in neither case is the mechanism known. Very often, the sizes and proportions of body parts depend on combinations of size-measuring feedback controls and intracellular programs, as well as on environmental influences such as nutrition and habitat.

The Proliferation, Death, and Size of Cells Determine Organ and Organism Size

The worm *C. elegans* demonstrates the ways in which size differences can arise. As we discussed earlier, each individual of a given sex is generated by almost exactly the same sequences of cell divisions and cell deaths, and consequently has precisely the same number of somatic cells—959 in the adult hermaphrodite (see Figure 21–42). More than 1000 cell divisions generate 1090 somatic cells during hermaphrodite development, but 131 of these cells undergo apoptotic cell death. Thus, precise regulation of both cell division and cell death determines the final numbers of somatic cells in the worm.

It takes only 3 days for *C. elegans* to generate all of its adult somatic cells, after which cell divisions occur only in the germ line. Yet the worm continues to grow, doubling in size between sexual maturity and death 2–3 weeks later. This doubling results from somatic-cell growth: although the cells no longer divide, they continue to go through rounds of DNA synthesis; this *endoreplication* of the genome makes the cells *polyploid*. As in all organisms, the size of a cell is proportional to its ploidy (that is, the number of genome copies that it contains): a doubling of ploidy roughly doubles cell volume. Thus the worm’s final size is set by a combination of programmed cell divisions and cell deaths, along with regulation of the sizes of individual cells through changes in ploidy.

Endoreplication of the genome as a means to increase cell size is not peculiar to *C. elegans* and occurs in a developmentally regulated manner in specific tissues of many animals. In one extreme case, neurons that innervate large areas of the mollusk *Aplysia* may contain hundreds of thousands more copies of the genome than does a diploid cell and grow up to 1 mm in diameter (Figure 21–62).

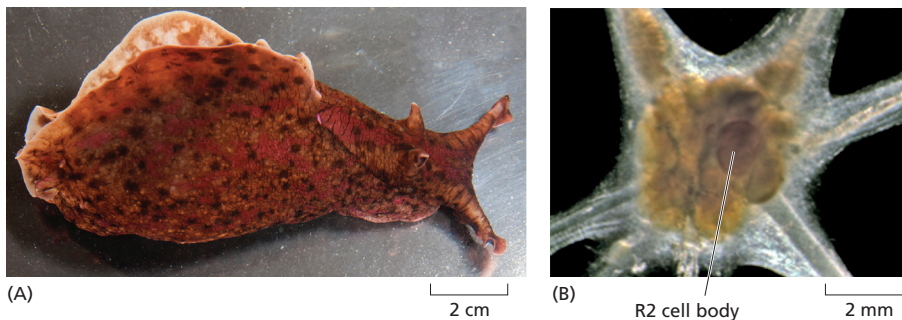


Figure 21–62 Giant neurons of the sea slug are highly polyploid. (A) The mollusk species *Aplysia californica*, commonly called a sea hare because its chemosensory tentacles stick up like ears, can weigh more than 2 kg. The *Aplysia* central nervous system contains groups of neural cell bodies called ganglia. (B) A dissected abdominal ganglion contains massive neurons such as R2, which has undergone endoreplication to produce ~300,000 copies of the diploid genome and is thought to be the largest somatic cell found in nature. *Aplysia* and its giant neurons enable physiological, biochemical, and genomic studies at the level of single cells, providing an accessible model for studies of how neural circuits control behavior. (A and B, courtesy of Lynne Fieber and Michael Schmale.)

Figure 21–63 Two species of the frog genus *Xenopus*. *X. tropicalis* (top) has an ordinary diploid genome. *X. laevis* (bottom) has roughly twice as much DNA per cell and evolved through hybridization of two *X. tropicalis*-like species followed by a whole-genome duplication event roughly 30 million years ago. (From E. Amaya, M. Offield, and R. Grainger, *Trends Genet.* 14:253–255, 1998. With permission from Elsevier.)

Changes in ploidy can also arise during the evolution of new species, where they can drive size changes of the entire animal. The frog *Xenopus laevis* possesses a tetraploid genome and has a mass of about twice that of diploid relatives such as *Xenopus tropicalis* (Figure 21–63). These two species show robust size-scaling relationships at the organism, cellular, and subcellular levels, with the smaller frog possessing smaller cells, and the smaller cells, in turn, containing smaller nuclei. In plants, as in animals, cell size increases as ploidy increases (Figure 21–64). This effect has been exploited in the agricultural breeding of plants for large size: most of the major fruits and vegetables that we consume are polyploids.

Changes in Cell Size Usually Result from Modified Cell Cycles

Although fundamental to understanding how organ and organism size are determined, how cell size itself is set and maintained remains an open question. As discussed in Chapter 17, increases in cell size (growth) are usually coordinated with progression through the cell cycle (division) to maintain cell-size homeostasis. However, cells can uncouple the two processes, for example during growth of an egg, which enlarges dramatically without replicating its genome or dividing. After fertilization, this phenomenon is reversed when rounds of DNA replication and cell division occur in the absence of growth to form many small cells.

Endoreplication, which reduplicates the genome to enlarge a somatic cell, results from another variation on the cell cycle. Although S phase proceeds,

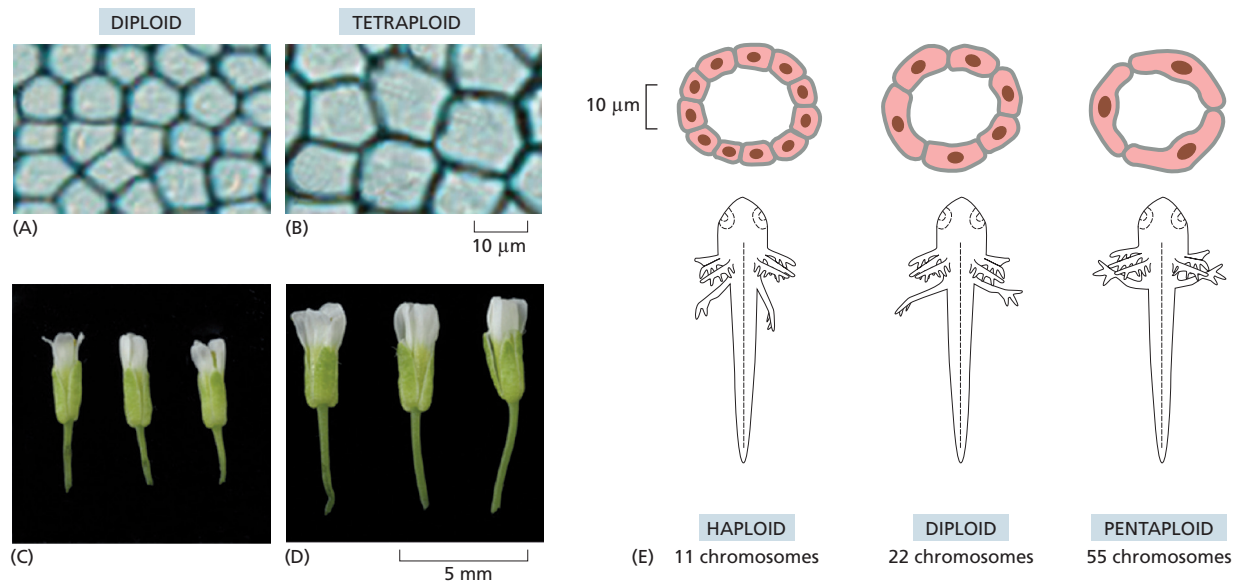


Figure 21–64 Effects of ploidy on cell size and organ size. In all organisms, from bacteria to humans, cell size is proportional to ploidy—the number of copies of the genome per cell. This is illustrated for (A–D) *Arabidopsis* flowers and (E) for salamanders. In each case, the upper panels show cells in a specific tissue [a petal for *Arabidopsis*, a pronephric (kidney) tubule for the salamander]; the lower panels show the gross anatomy—flowers for *Arabidopsis*, the whole body for the salamander. In the case of *Arabidopsis* flowers, increase in cell size increases organ size. By contrast, the salamander and its individual organs attain their normal standard size regardless of ploidy, because large cell size is compensated for by fewer cells. This indicates that the size of an organism or organ in this species is not controlled simply by counting cell divisions or cell numbers; size must somehow be regulated at the level of total cell mass. [A–D, from C. Breuer et al., *Plant Cell* 19:3655–3668, 2007. With permission from the American Society of Plant Biologists; E, adapted from G. Fankhauser, in *Analysis of Development* (B.H. Willier et al., eds.), pp. 126–150. Philadelphia: Saunders, 1955.]

mitotic cyclin levels are too low to induce M phase, and the polyploid cells usually cease dividing completely, entering a *post-mitotic* state. Additional rounds of DNA replication in the absence of cell division lead to an increase in cell size. In some tissues, such as the liver and cardiac muscle, DNA replication is followed by nuclear division, but not cytokinesis, resulting in polyploid cells with multiple nuclei, through a process called *endomitosis*. Whether or not nuclear division occurs, ploidy increases are accompanied by cell growth and contribute to tissue function. For example, the trophoblast cells of the mammalian placenta undergo eight rounds of endoreplication, producing giant cells crucial for formation of the tissue barrier between mother and fetus.

How do increases in ploidy increase the size of cells? One idea is that elevated gene-copy numbers increase the biosynthetic capacity by allowing production of more gene products. However, a strong positive correlation between genome size and cell size is observed across all species, independent of the fraction of the genome that codes for proteins. For example, although all vertebrates have a similar number of coding genes, axolotl salamanders possess a diploid genome that is 10 times the size of the human genome (containing 10 times more noncoding DNA). Remarkably, axolotl cells are also 10 times larger than human cells. The molecular basis of the scaling relationship between genome size and cell size is one of the many outstanding mysteries in biology.

Animals and Organs Can Assess and Regulate Total Cell Mass

The size of an animal or organ depends on both cell size and cell number; that is, on total cell mass. Remarkably, many animals and organs can somehow assess their total cell mass and regulate it, providing evidence for feedback controls of the sort highlighted earlier in our introductory account of the general principles of growth control. In contrast with *C. elegans*, if cell size is artificially increased or decreased in these cases, cell numbers adjust to maintain the same total cell mass. This has been beautifully illustrated by experiments done long ago in salamanders, where cell size can be manipulated by altering the animal's ploidy. As shown in Figure 21–64E, salamanders of different ploidies end up being the same size with very different numbers of cells. The individual cells in a pentaploid salamander, for example, are about five times the size of those in a haploid salamander, but there are only one-fifth as many cells. This scaling operates not only in the body as a whole, but also in its individual organs. Pentaploid salamanders possess the same size brain as their diploid counterparts, but with many fewer neurons and a simplified structure.

The **imaginal discs** of *Drosophila* provide another striking example of homeostatic size control. The discs are epithelial pouches that grow by cell proliferation during the larval period and, during the pupal stage, form the organs and extremities of the adult fly (Figure 21–65). Experiments have been chiefly done on the wing imaginal disc. Mutations in components of the cell-cycle control machinery can be used to speed up or slow down the rate of cell division in the disc. Remarkably, such mutations result in an excessive number of abnormally small cells or a reduced number of abnormally large cells, respectively, leaving the size (area) and patterning of the adult wing practically unchanged. Thus, the size of

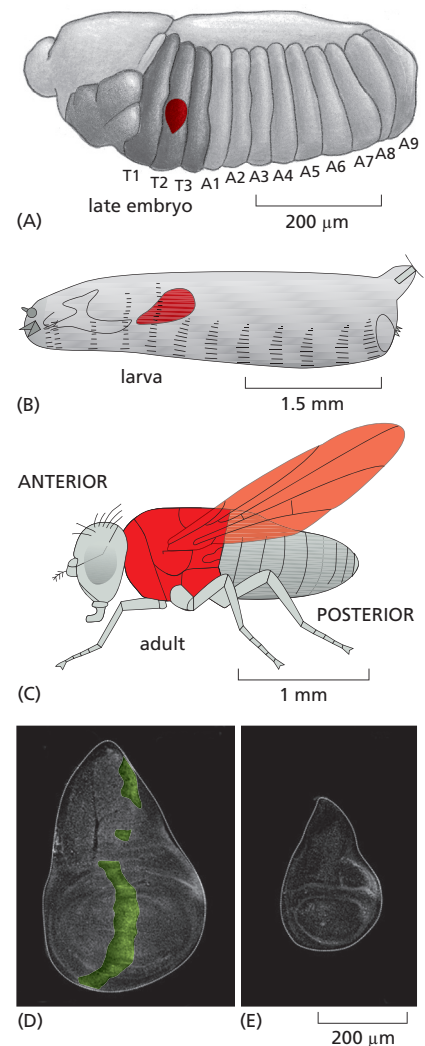


Figure 21–65 Growth of the *Drosophila* wing imaginal disc. Drawings in panels A–C show the locations of cells that give rise to the adult wing and dorsal thorax after metamorphosis. (A) Initially, ~30 cells are specified in the second thoracic segment of the embryo as precursors to the wing and dorsal thorax. These cells divide rapidly during the 4 days of larval life. (B) In the larva before metamorphosis, these cells are organized into a simple pouch-like epithelial organ called the wing imaginal disc, which contains ~30,000 cells and is ~400 μm in diameter. Other imaginal discs (not shown) give rise to most of the other external structures of the fly. (C) The final adult wing is ~2.25 mm long. (D, E) The morphogen Dpp is expressed in a stripe down the center of the wing imaginal disc (green), diffusing out toward the edges and influencing disc growth as well as cell fates. When Dpp expression is removed from the disc late during larval development, the wing disc and resultant wing is proportionally smaller. (D and E, from S. Matsuda and M. Affolter, *eLife* 6:e22319, 2017, doi 10.7554/eLife.22319.)

the disc is not determined by counting the number of cell divisions so as to produce a set number of cells. Instead, there must be a regulatory mechanism that halts growth when the disc's total cell mass reaches the appropriate value, so that the adult wing that subsequently develops is normal. Moreover, if young discs or even disc fragments are taken out of their normal context and transplanted into the growth-permissive abdominal cavity of an adult female, they will grow until they reach their normal size and then stop. Clearly, the mechanisms that regulate organ size are intrinsic to the disc.

We know little about how organisms or organs assess total cell mass or monitor their own growth. Nevertheless, we are beginning to understand some of the signal molecules that drive or halt growth in response to the mysterious cues that convey information about the size attained.

Various Extracellular Signals Stimulate or Inhibit Growth

To produce an organ with the correct morphology and proportions, cell proliferation and cell growth must be coordinated within a tissue. While some control occurs systemically through growth-regulatory hormones, discussed later, there are also interesting examples in which tissues regulate their own growth. One example of how this may be achieved has been investigated in the *Drosophila* wing imaginal disc (see Figure 21–65), where the TGF β family member Dpp plays a role. We encountered Dpp before as a morphogen involved in patterning the dorsoventral axis of the fly embryo; in the disc, it works analogously, diffusing from a narrow stripe down the middle of the disc to dictate different cell fates at different distances. Some experiments indicate that the gradient of Dpp emanating from this central source also influences tissue size, perhaps because growth of the disc leads cells farther from the source to experience lower levels of Dpp signaling (see Figure 21–65D and E). Linking tissue growth to pattern formation would ensure that changes in organ size are accompanied by appropriate changes in cell differentiation and distribution.

Not all growth-regulating extracellular signals stimulate growth; some inhibit it by promoting cell death or inhibiting cell growth, cell division, or both. *Myostatin* is another TGF β family member that specifically inhibits the growth and proliferation of myoblasts—the precursor cells that fuse to form the huge, multinucleated cells of skeletal muscle. In this case, the tissue itself produces the signal, so as more muscle forms, myostatin levels increase and eventually turn off muscle growth. The importance of this simple organ-size feedback mechanism can be seen when the *Myostatin* gene is deleted: the muscles of mutant mice grow to be several times larger than normal. Remarkably, two breeds of cattle that were bred long ago for large muscles both have mutations in the *Myostatin* gene, as does a breed of dogs (Figure 21–66).

The Hippo Pathway Relays Mechanical Signals Regulating Growth

Organ growth is governed in many tissues by an intracellular signaling system called the Hippo pathway. It was discovered in *Drosophila*, but it operates in vertebrates as well. Named after the overgrown, buckled tissues that result when

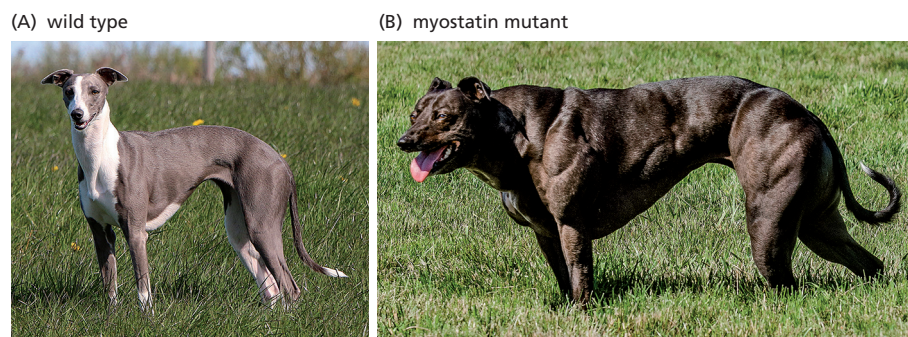
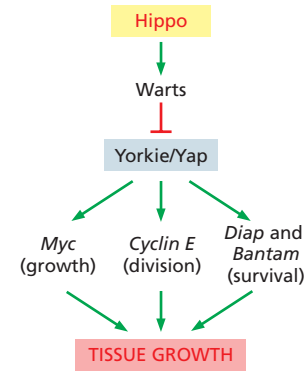


Figure 21–66 Myostatin limits muscle growth. (A) A standard whippet and (B) a bully whippet that lacks myostatin. (A, Bianca Grueneberg/Getty Images; B, ©2020 Stuart Isett. All rights reserved.)

Figure 21–67 The Hippo pathway in *Drosophila*. Hippo, a protein kinase, limits growth by phosphorylation and activation of the kinase Warts, which in turn phosphorylates and inactivates the transcriptional coactivator Yorkie (Yap and Taz in vertebrates). When unphosphorylated, Yorkie/Yap drives tissue growth: it activates the transcription of the growth-promoting gene *Myc*, the cell-cycle progression gene *Cyclin E*, the anti-apoptotic gene *Diap*, and the microRNA *Bantam*. Hippo-induced phosphorylation of Yorkie/Yap blocks this effect.



the pathway is inactive, Hippo signaling in wild-type animals inhibits growth both by promoting cell death (by blocking an apoptosis inhibitor) and by inhibiting cell-cycle progression (by inhibiting the expression of the cell-cycle gene *Cyclin E*). Some components of the pathway in *Drosophila* are shown in **Figure 21–67**. The organs of animals that are abnormally resistant to Hippo repression can grow to a monstrous size (**Figure 21–68**).

Unlike many signaling pathways that involve binding of a signal molecule to a dedicated receptor, the Hippo pathway instead appears to be controlled primarily by mechanical forces and cell architecture. Cell-cell and cell-matrix contact, cytoskeletal tension, and the apical-basal polarity of the cell can each influence Hippo activation. Through this mechanism, tissues can sense physical as well as chemical cues in their environment and regulate their growth accordingly.

Hormones Coordinate Growth Throughout the Body

We have already seen how some signals act systemically as hormones to regulate the development of the animal as a whole. Some of these serve to regulate growth. In mammals, for instance, growth hormone (GH) is secreted by the pituitary gland into the bloodstream and stimulates growth throughout the body: excessive production of GH leads to pituitary gigantism, and too little leads to dwarfism (**Figure 21–69**). Pituitary dwarfs with lower GH levels have bodies and organs that are proportionately small, and stand in contrast to achondroplastic dwarfs, whose limbs are disproportionately short usually because of a mutation in a gene encoding an FGF receptor that disrupts normal cartilage and bone development (**Figure 21–70**).

Growth hormone stimulates growth primarily by inducing the liver and other organs to produce insulin-like growth factor 1 (IGF1), which acts mainly as a local signal within many tissues. IGF1 and related factors are potent growth promoters that increase cell survival, cell growth, cell proliferation, or some combination of these, depending on the cell type. Large breeds of dogs such as Great Danes owe their great size to high levels of IGF1, while miniature breeds such as Chihuahuas have low levels (see **Figure 21–60**).

It is important to note that in all species, nutritional conditions also play a fundamental part in regulating the pace and extent of growth, and in animals they do so through hormonal signal networks that are highly conserved between

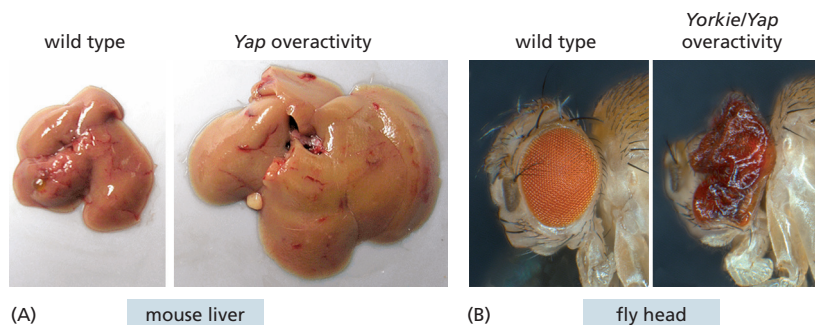


Figure 21–68 Overcoming Hippo repression increases organ size. (A) When *Yap* is overexpressed in the mouse liver, Hippo signaling cannot repress its activity and excess growth results. (B) Similar overgrowth is seen when fly eyes express a form of Yorkie/Yap that cannot be phosphorylated and inhibited by Hippo signaling. (A, from J. Dong et al., *Cell* 130:1120–1133, 2007. With permission from Elsevier; B, courtesy of Jung Kim.)

Figure 21–69 Pituitary dwarf and pituitary giant. The “giant” on the right is Robert Wadlow (1914–1940), the tallest recorded man at 8 feet 11 inches (2.72 m), together with his father, who was almost 6 feet tall (1.82 m). The dwarf on the left is General Tom Thumb, which was the stage name of Charles Sherwood Stratton (1838–1883). On his 18th birthday, he was measured at 2 feet 8.5 inches (82.6 cm) tall, and at his death, he was 3 feet 4 inches (102 cm). (Left, Bettman/Getty; right, History and Art Collection/Alamy Stock Photo.)

vertebrates and invertebrates. Genetic experiments, especially in *Drosophila*, have begun to unravel the logic of these controls and to indicate how they may operate alongside other machinery, such as the Hippo pathway, to determine final size.

The Duration of Growth Influences Organism Size

In addition to rate of growth, the final size of an organism will be determined by the time over which growth occurs. Some animals have a defined period of the life span during which growth is permitted; these species are said to show *determinate growth*. In humans, the growth period terminates at the end of puberty, while in many insects it terminates with the onset of metamorphosis. In both cases the change is triggered by the release of hormones that transition the organism into sexual maturity. Thereafter, cell proliferation is restricted to supporting tissue homeostasis via the action of adult stem cells (see Chapter 22), although significant cell growth can continue in cells such as fat cells. The timing of hormone release is set by many influences, among which both nutritional status and juvenile body size provide important inputs to set final body size.

Other species show *indeterminate growth* and are capable of size increases throughout their life as long as environmental conditions are amenable. Lobsters and many fish are familiar examples, but indeterminate growth is found along numerous branches of the tree of life. Plants, in which cell proliferation takes place in specialized regions called meristems (see Chapter 19), also show both determinate and indeterminate growth patterns. Giant sequoias and other redwood trees are an example of the latter, whereas the former include agricultural crops selected for maximal and synchronous fruit production. Some species, including tomatoes, exist in both determinate and indeterminate varieties. The mechanisms that assign a terminal growth point to one organism while enabling a closely related species to continue growth throughout life remain unknown.

Summary

The sizes of animal species and their organs vary widely and depend mainly on the size and number of cells, which are increased through cell growth and cell division, respectively. Cell numbers are reduced by cell death—mainly by apoptosis. The mystery is how decisions about growth, division, and death within individual cells are regulated and coordinated to produce the characteristic final size of the adult animal.

Some signals such as growth factors, mitogens, and survival factors act in organs to stimulate growth, while other signal molecules do the opposite. These signals are deployed under the control of developmental patterning programs and influence the cell cycle or apoptotic machinery to produce tissue-specific growth outcomes. Many animals and organs can assess their own total cell mass and regulate it, reaching a consistent target size in the face of artificial changes to cell growth or cell numbers. The compensatory mechanisms are not known, but signaling pathways that ensure that organs do not grow beyond a certain size have been identified.

Although most of these signals operate locally to help sculpt the size of organs and appendages, others act as hormones to regulate the growth of the animal as a whole. Nutrients can regulate growth through stimulating the release of hormonal signals throughout the body. The period of life during which growth is permitted can also be controlled by hormones. Final organism size reflects the integrated output of all the above mechanisms.

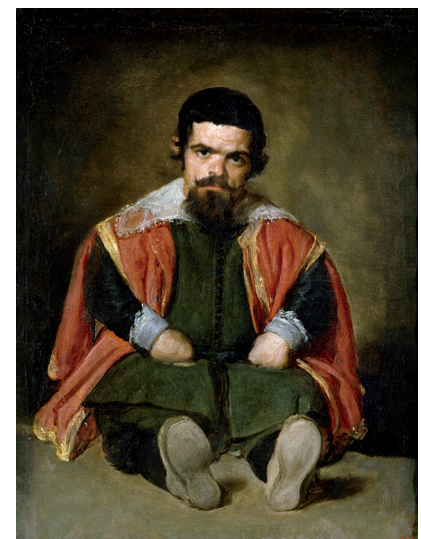
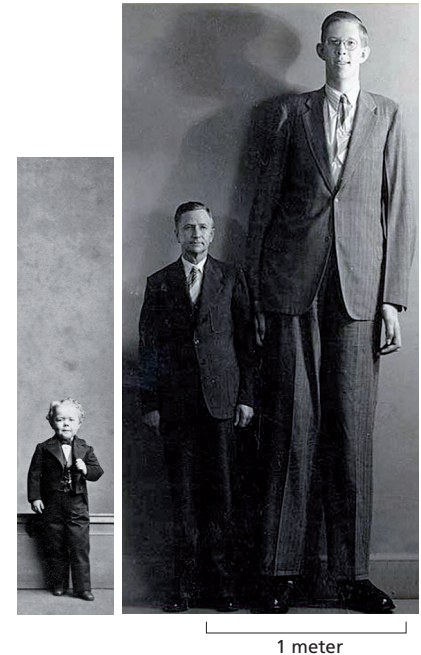


Figure 21–70 Achondroplasia. This type of dwarfism, shown in Velázquez's painting, occurs in one of 10,000–100,000 births; in more than 99% of cases it results from a mutation at an identical site in the genome, corresponding to amino acid 380 in the FGF receptor FGFR3 (a glycine in the transmembrane domain). The mutation is dominant, and almost all cases are due to new, independently occurring mutations, implying an extraordinarily high mutation rate at this particular site in the genome. The defect in FGF signaling causes dwarfism by interfering with the growth of cartilage in developing long bones. (The Picture Art Collection/Alamy Stock Photo.)

PROBLEMS

Which statements are true? Explain why or why not.

21-1 In the early cleavage stages, when the embryo cannot yet feed, the developmental program is driven and controlled entirely by the material deposited in the egg by the mother.

21-2 Because of the many later developmental transformations that produce the elaborately structured organs, the body plan set up during gastrulation bears little resemblance to the body plan in the adult.

21-3 As development progresses, individual cells within a lineage become more and more restricted in the range of cell types they can give rise to.

21-4 At different stages of embryonic development, the same signals are used over and over again by different cells, but with different biological outcomes.

21-5 Changes in the coding regions of genes involved in development are primarily responsible for the differences between species.

21-6 The cell cycle is the ticking clock that sets the tempo of developmental processes, with maturational changes in gene expression being dependent on cell-cycle progression.

Discuss the following problems.

21-7 Name the three processes that are fundamental to animal development, and describe each of them in a single sentence.

21-8 Name the three germ layers of the early embryo that are formed during gastrulation, and list the principal structures each gives rise to in the adult.

21-9 In the early *Drosophila* embryo, there seems to be no requirement for the usual forms of cell-cell signaling; instead, transcriptional regulators and mRNA molecules move freely between nuclei. How can that be?

21-10 Morphogens play a key role in development, creating concentration gradients that inform cells of where they are and how to behave. Examine the simple patterns represented by the flags in [Figure Q21-1](#). Which do you suppose could be created by a gradient of a single morphogen? Which would require gradients of two morphogens? Assuming that such patterns were present in a sheet of cells, explain how they could be created by morphogens.



Figure Q21-1 National flags from three countries (Problem 21-10). (Left, railway fx/Shutterstock; center, Creative Photo Corner/Shutterstock; right, Derek Brumby/Shutterstock.)

21-11 Two adjacent cells in the nematode worm normally differentiate into an anchor cell (AC) and a ventral uterine precursor (VU) cell, but which of the two becomes the AC and which becomes the VU cell is completely random. The cells have an equal chance of adopting either fate, but they always adopt opposite fates. Mutations of the *Lin12* gene alter these fates. In hyperactive *Lin12* mutants, both cells become VU cells, while in inactive *Lin12* mutants, both cells become ACs. Thus, *Lin12* is central to the decision-making process. In genetic mosaics in which one precursor cell has the hyperactive *Lin12* and the other precursor has the inactive *Lin12*, the cell with the hyperactive *Lin12* gene always becomes the VU cell, and the cell with the inactive *Lin12* gene always becomes the AC. Assuming that one cell sends a signal and the other cell receives it, explain how these results suggest that the *Lin12* gene encodes a protein required to receive the signal. Offer a suggestion for how the fates of these two precursor cells are normally decided in wild-type worms.

21-12 It was clear from the early days of studying development that certain “morphogenetic” substances were present in the egg and segregated asymmetrically into cells of the developing embryo. One such investigation in ascidian (sea squirt) embryos examined endodermal alkaline phosphatase, which could be visualized using a histochemical stain. Treatment of embryos with cytochalasin B stopped cell division but did not block expression of alkaline phosphatase at the appropriate time. Treatment with actinomycin D, which blocks transcription, did not interfere with expression of alkaline phosphatase. Treatment with puromycin, which blocks translation, eliminated expression of alkaline phosphatase. What is the likely nature of the morphogenetic substance that gives rise to alkaline phosphatase?

21-13 The mouse *HoxA3* and *HoxD3* genes are paralogs that occupy equivalent positions in their respective *Hox* gene clusters and share roughly 50% identity in their protein-coding sequences. Mice with defects in *HoxA3* have deficiencies in pharyngeal tissues, whereas mice with defects in *HoxD3* have deficiencies in the axial skeleton, suggesting quite different functions for the paralogs. Thus, it came as a surprise when it was found that replacing a defective *HoxD3* gene with the normal *HoxA3* gene corrected the deficiency, as did the reciprocal experiment of replacing a mutant *HoxA3* gene with a normal *HoxD3* gene. Neither transplanted gene, however, could supply its normal function; that is, a normal *HoxA3* gene at the *HoxD3* locus could not correct the deficiency caused by a mutant *HoxA3* gene at the *HoxA3* locus. The same was true for the *HoxD3* gene. If the *HoxA3* and *HoxD3* genes are equivalent, how do you suppose they can play such distinct roles in development? Why do you suppose they cannot perform their normal function in a new location?

21-14 The segmentation of somites in vertebrate embryos is thought to depend on oscillations in the expression of the *Hes7* gene. Mathematical modeling explains

these oscillations in terms of the delays in production of the unstable Hes7 protein, which acts as a transcription regulator to shut off its own expression. Once Hes7 decays, with a half-life of about 20 minutes, its transcription resumes. To test this model, you decide to reduce the total delay by removing one, two, or all three of the introns from the *Hes7* gene in mice. Why do you expect that intron removal would reduce the delay? What would you predict would happen to the oscillation time, and somitogenesis, if the model were correct?

21–15 The oscillatory clock that drives somitogenesis in vertebrates involves three essential components: *Her7* (an unstable repressor of its own synthesis), *Delta* (a transmembrane signaling molecule), and *Notch* (a transmembrane receptor for *Delta*). *Notch* is bound by *Delta* on neighboring cells, activating the *Notch* signaling pathway, which then activates *Her7* transcription. Normally, this system works flawlessly to create sharply defined somites (**Figure Q21–2A**). In the absence of *Delta*, however, only the first five somites form normally, and the rest are poorly defined (**Figure Q21–2B**). If a pulse of *Delta* is supplied later, somitogenesis returns to normal in the regions where *Delta* was present (**Figure Q21–2C**). A diagram of the connections between the components of the clock and how they interact in adjacent cells is shown in **Figure Q21–2D**. In the absence of *Delta*, why do the cells become unsynchronized?

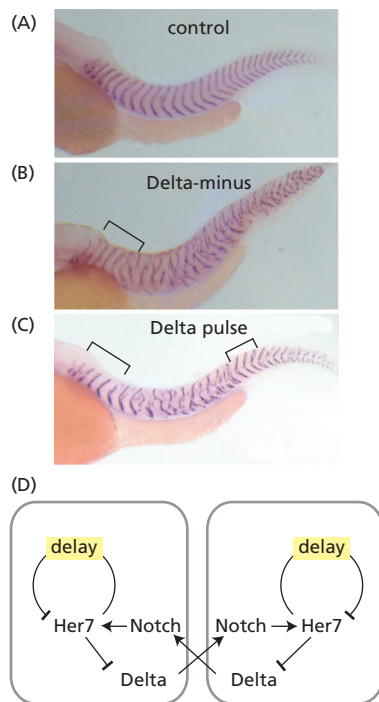


Figure Q21–2 Somite formation in zebrafish embryos (Problem 21–15). (A) Wild-type embryos with normal somites. (B) Somite formation in embryos lacking *Delta*. The bracket indicates normal-looking somites where they initially form. (C) Somite formation in embryos lacking *Delta* but receiving a pulse of *Delta* expression at the time indicated by the right-hand bracket. (D) Interactions among components of the oscillatory clock in adjacent cells. (Adapted from C. Soza-Ried et al., *Development* 141:1780–1788, 2014. With permission from the Company of Biologists.)

What is it about the presence of *Delta* that keeps adjacent cells oscillating in synchrony?

21–16 The extracellular protein factor Decapentaplegic (*Dpp*) is critical for proper wing development in *Drosophila* (**Figure Q21–3A**). It is normally expressed in a narrow stripe in the middle of the wing, along the anterior–posterior boundary. Flies that are defective for *Dpp* form stunted “wings” (**Figure Q21–3B**). If an additional copy of the gene is placed under control of a promoter that is active in the anterior part of the wing or in the posterior part of the wing, a large mass of wing tissue composed of normal-looking cells is produced at the site of *Dpp* expression (**Figure Q21–3C and D**). Does *Dpp* stimulate cell division, cell growth, or both? How can you tell?

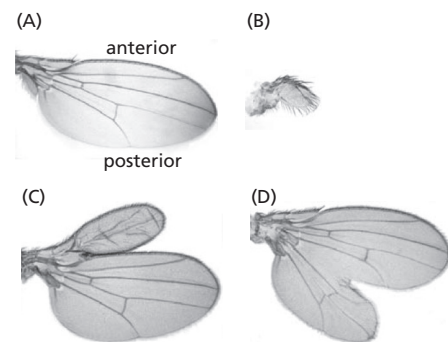


Figure Q21–3 Effects of *Dpp* expression on wing development in *Drosophila* (Problem 21–16). (A) Normal *Dpp* expression. (B) Absence of *Dpp* expression. (C) Additional anterior *Dpp* expression. (D) Additional posterior *Dpp* expression. (From M. Zecca et al., *Development* 121:2265–2278, 1995. With permission from the Company of Biologists.)

REFERENCES

General

- Baressi MJ & Gilbert SF (2019) *Developmental Biology*, 12th ed. Sunderland, MA: Sinauer Associates.
- Carroll SB (2006) *Endless Forms Most Beautiful: The New Science of Evo Devo*. New York: Norton.
- Wallingford JD (2019) We Are All Developmental Biologists. *Dev. Cell* 50(2), 132–137.
- Wolpert L, Tickle C & Martinez-Arias A (2019) *Principles of Development*, 6th ed. Oxford: Oxford University Press.

Overview of Development

- Gurdon JB (2013) The egg and the nucleus: a battle for supremacy (Nobel Lecture). *Angew. Chem. Int. Ed. Engl.* 52, 13890–13899.
- Istrail S & Davidson EH (2005) Logic functions of the genomic cis-regulatory code. *Proc. Natl. Acad. Sci. USA* 102, 4954–4959.
- Levine M (2010) Transcriptional enhancers in animal development and evolution. *Curr. Biol.* 20, R754–R763.
- Lewis J (2008) From signals to patterns: space, time, and mathematics in developmental biology. *Science* 322, 399–403.
- Meinhardt H & Gierer A (2000) Pattern formation by local self-activation and lateral inhibition. *Bioessays* 22, 753–760.
- Rogers KW & Schier AF (2011) Morphogen gradients: from generation to interpretation. *Annu. Rev. Cell Dev. Biol.* 27, 377–407.
- Shubin N, Tabin C & Carroll S (2009) Deep homology and the origins of evolutionary novelty. *Nature* 457, 818–823.

Mechanisms of Pattern Formation

- Andrey G & Duboule D (2014) SnapShot: Hox gene regulation. *Cell* 156, 856–856.e1.
- Chan YF, Marks ME, Jones FC . . . Kingsley DM (2010) Adaptive evolution of pelvic reduction in sticklebacks by recurrent deletion of a Ptx1 enhancer. *Science* 327, 302–305.
- Davis RL, Weintraub H & Lassar AB (1987) Expression of a single transfected cDNA converts fibroblasts to myoblasts. *Cell* 51, 987–1000.
- De Robertis EM (2006) Spemann's organizer and self-regulation in amphibian embryos. *Nat. Rev. Mol. Cell Biol.* 4, 296–302.
- Driever W & Nüsslein-Volhard C (1988) A gradient of bicoid protein in *Drosophila* embryos. *Cell* 54, 83–93.
- Halder G, Callaerts P & Gehring WJ (1995) Induction of ectopic eyes by targeted expression of the eyeless gene in *Drosophila*. *Science* 267, 1788–1792.
- Jaeger J (2011) The gap gene network. *Cell. Mol. Life Sci.* 68, 243–274.
- Knoblich JA (2010) Asymmetric cell division: recent developments and their implications for tumour biology. *Nat. Rev. Mol. Cell Biol.* 11, 849–860.
- Lander AD (2013) How cells know where they are. *Science* 339, 923–927.
- Lawrence PA (1992) *The Making of a Fly: The Genetics of Animal Design*. Oxford: Wiley-Blackwell.
- Lewis EB (1978) A gene complex controlling segmentation in *Drosophila*. *Nature* 276, 565–570.
- Müller P, Rogers KW, Jordan BM . . . Schier AF (2012) Differential diffusivity of Nodal and Lefty underlies a reaction-diffusion patterning system. *Science* 336, 721–724.
- Nüsslein-Volhard C & Wieschaus E (1980) Mutations affecting segment number and polarity in *Drosophila*. *Nature* 287, 795–801.
- Schuettengruber B, Bourbon HM, Di Croce L & Cavalli G (2017) Genome regulation by polycomb and trithorax: 70 years and counting. *Cell* 171, 34–57.
- Sjöqvist M & Andersson ER (2019) Do as I say, Not(ch) as I do: lateral control of cell fate. *Dev. Biol.* 447(1), 58–70.
- von Dassow G, Meir E, Munro EM & Odell GM (2000) The segment polarity network is a robust developmental module. *Nature* 406, 188–192.

Developmental Timing

- Brown DD & Cai L (2007) Amphibian metamorphosis. *Dev. Biol.* 306, 20–33.
- Giraldez AJ, Mishima Y, Rihel J . . . Schier AF (2006) Zebrafish MiR-430 promotes deadenylation and clearance of maternal mRNAs. *Science* 312, 75–79.
- Holguera I & Desplan C (2018) Neuronal specification in space and time. *Science* 362, 176–180.
- Hubaud A & Pourqu   O (2014) Signalling dynamics in vertebrate segmentation. *Nat. Rev. Mol. Cell Biol.* 15, 709–721.
- Isshiki T, Pearson B, Holbrook S & Doe CQ (2001) *Drosophila* neuroblasts sequentially express transcription factors which specify the temporal identity of their neuronal progeny. *Cell* 106, 511–521.
- Jukam D, Shariati SAM & Skotheim JM (2017) Zygotic genome activation in vertebrates. *Dev. Cell* 42, 316–332.
- Lee RC, Feinbaum RL & Ambros V (1993) The *C. elegans* heterochronic gene *lin-4* encodes small RNAs with antisense complementarity to *lin-14*. *Cell* 75, 843–854.
- Oates AC (2020) Waiting on the Fringe: cell autonomy and signaling delays in segmentation clocks. *Curr. Opin. Genet. Dev.* 63, 61–70.
- Song J, Irwin J & Dean C (2013) Remembering the prolonged cold of winter. *Curr. Biol.* 23, R807–R811.
- Wightman B, Ha I & Ruvkun G (1993) Posttranscriptional regulation of the heterochronic gene *lin-14* by *lin-4* mediates temporal pattern formation in *C. elegans*. *Cell* 75, 855–862.

Morphogenesis

- Bussmann J & Raz E (2015) Chemokine-guided cell migration and motility in zebrafish development. *EMBO J.* 34, 1309–1318.
- Butler MT & Wallingford JB (2017) Planar cell polarity in development and disease. *Nat. Rev. Mol. Cell Biol.* 18, 375–388.
- Glimour D, Rembold M & Leptin M (2017) From morphogen to morphogenesis and back. *Nature* 541, 311–320.
- Goodwin K & Nelson CM (2020) Branching morphogenesis. *Development* 147, dev184499.
- Lecuit T & Lenne PF (2007) Cell surface mechanics and the control of cell shape, tissue patterns and morphogenesis. *Nat. Rev. Mol. Cell Biol.* 8, 633–644.
- Le Douarin NM & Kalcheim C (1999) *The Neural Crest*, 2nd ed. Cambridge: Cambridge University Press.
- Solnica-Krezel L & Sepich DS (2012) Gastrulation: making and shaping germ layers. *Annu. Rev. Cell Dev. Biol.* 28, 687–717.
- Takeichi M (2011) Self-organization of animal tissues: cadherin-mediated processes. *Dev. Cell* 21, 24–26.
- Te Boekhorst V, Preziosi L & Friedl P (2016) Plasticity of cell migration in vivo and in silico. *Annu. Rev. Cell Dev. Biol.* 32, 491–526.
- Walck-Shannon E & Hardin J (2014) Cell intercalation from top to bottom. *Nature* 15, 34–48.

Growth

- Andersen DS, Colombani J & L  opold P (2013) Coordination of organ growth: principles and outstanding questions from the world of insects. *Trends Cell Biol.* 23, 336–344.
- Conlon I & Raff M (1999) Size control in animal development. *Cell* 96, 235–244.
- Hariharan IK & Bilder D (2006) Regulation of imaginal disc growth by tumor-suppressor genes in *Drosophila*. *Annu. Rev. Genet.* 40, 335–361.
-   vreb   JI & Edgar BA (2018) Polyploidy in tissue homeostasis and regeneration. *Development* 145, dev156034.
- Penzo-M  ndez AI & Stanger BZ (2015) Organ-size regulation in mammals. *Cold Spring Harb. Perspect. Biol.* 7, a019240.
- Restrepo S, Zartman JJ & Basler K (2014) Coordination of patterning and growth by the morphogen DPP. *Curr. Biol.* 24, R245–R255.
- Zheng Y & Pan D (2019) The hippo signaling pathway in development and disease. *Dev. Cell* 50, 264–282.